SUMMARY OF CHANGES

NCI Protocol #: 10017
Local Protocol #: I 285416

NCI Version Date:
Protocol Date: 7/22/19

Protocol Title: A Randomized Phase 2 Trial of Atezolizumab (MPDL3280A), SGI-110 and CDX-1401 Vaccine in Recurrent Ovarian Cancer.

Summary of Change for Consent Revisions- Phase I and II

#	Page #	Revision
1.		Date updated for consent revision: **July 22 , 2019**
2.		**Phase I complete, no changes in the Phase I consent form**
3.	**1**	Title for Phase II consent revised to remove vaccine arm. **(Official study title unchanged.)**
4.	**1**	**Why is this study being done?** Group 3 removed.
5.	**1 to 3**	**What are the study groups?** Group 3 removed, schema and table updated accordingly.
6.	**6**	**What possible risks can I expect from taking part in this study?** Possible side effects of CDX-1401 and Poly-ICLC removed.
7.	**9**	**What are the costs of taking part in this study?** CDX-1401 and Poly-ICLC removed.
8.	**General**	Minor typos corrected.

Summary of Change for Protocol Revisions

1.	Page 2	**Date and revision # updated** *Revision / Version 17 / July 22, 2019*
2.	**Synopsis & Section 12** **Pages 3 and 108**	**Revised Study Schema and the statistical description with now only 2 cohorts in Phase 2. Reference #83 for "crossover" design added.** Given the constraints of a very limited time supply of CDX-1401, considering feasibility, the decision had to be made to omit the vaccine arm. 2 arms only study will allow additional patients/arm to increase the power and "crossover" will provide interesting additional secondary endpoint for concomitant vs. sequential treatment with Atezolizumab and SGI-110

		combinations.
3	**Synopsis** & **Section 5** Pages 4, 5, 37 and through out the protocol	Phase I part is now complete and the **SGI-110 20mg/m^2 has been declared as the safe dose for Phase II**, in combination with Atezolizumab 840 mg. Therefore, Phase II dose of SGI 110 stated as needed throughout the protocol.
4	**Section 5** Page 35 to 38	**Treatment plan revised** to reflect the Phase II 2 cohort design and the **table updated** accordingly.
5.	Through-out the protocol	**With the omission of cohort 3 in the Phase IIb part of the study, all irrelevant background and information materials related to the vaccine arm has been removed and sections updated throughout the protocol as needed for the new study design with only 2 cohorts.**
6.	**Study Calendar** Table 4 Page 96	Study Calendar Table 4 for cohort 3 was removed as it is no longer applicable.
7.	**TOC** & **References** Pages 6 and 117	**TOC and References updated.**

NCI Protocol #: 10017

Local Protocol #: I 285416

ClinicalTrials.gov Identifier: NCT03206047

TITLE: A Randomized Phase 2 Trial of Atezolizumab (MPDL3280A), SGI-110 and CDX 1401 Vaccine in Recurrent Ovarian Cancer.

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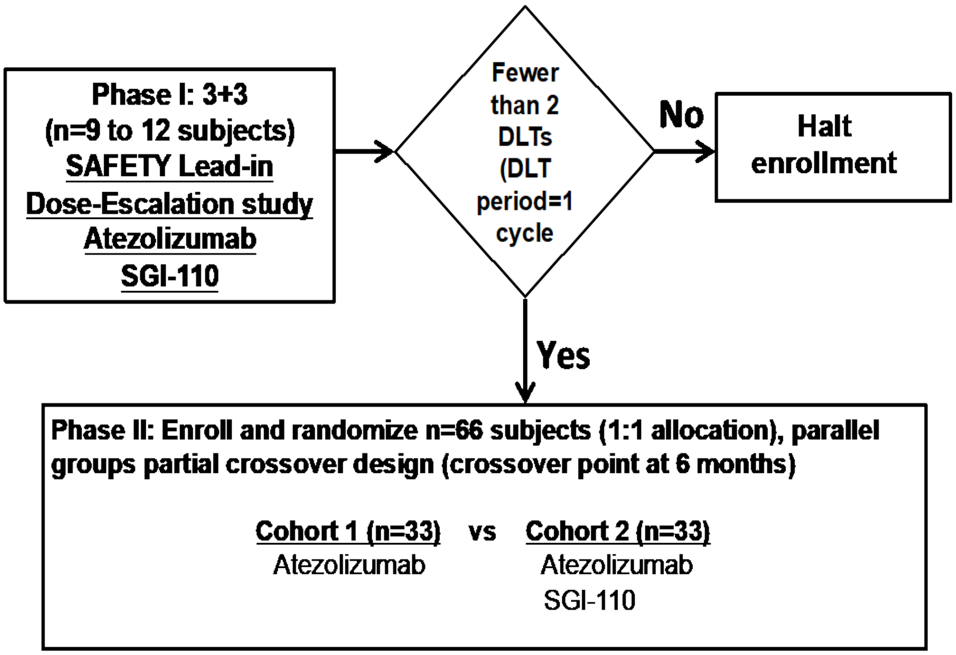
NCI-Supplied Agent(s):
Atezolizumab (MPDL3280A), NSC 783608, Genentech, Inc.
SGI-110 (guadecitabine), NSC 780463, Astex Pharmaceuticals

IND #: [redacted] (Ref: MPDL3280A (NSC 783608), SGI-110 (NSC 780463)

IND Sponsor: DCTD

Protocol Type / Version # / Version Date: *Revision / Version 17 /July 22, 2019*

STUDY SCHEMA



For the Phase I component of the study (safety lead-in dose escalation study with total of 9-12 patients), 3 eligible patients will receive SGI-110 (guadecitabine), on days 1-5 at an initial dose of 30mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles. The following decision rules will be followed: (i) If 0/3 dose limiting toxicities (DLTs) are observed, proceed to the next dose level of SGI-110 in combination with Atezolizumab (i.e. SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). (ii) If 1/3 DLTs are observed enroll 3 more patients at the same dose of SGI-110 (i.e. 30mg/m^2). If 1/6 DLTs are observed, proceed to the next dose level of SGI-110 in combination with Atezpolizumab (i.e. SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). At the next dose level of 45mg/m^2, if 0 or 1 out of 3 patients experiences a DLT, 3 more patients will be added. If 0 or 1 out of the 6 patients experiences a DLT, this will be declared as a safe dose for the Phase II. If $\geq$ 2 DLTs are experienced, the treatment will be considered too toxic and the lower dose-level will be considered the MTD.

Considering the number of subjects who experienced neutropenia on the study at the SGI-110 30mg/m^2 dose level, the FDA advised to de-escalate to SGI-110 20mg/m^2 and study 3 to 6 subjects as an expansion of the Phase 1 portion of the study (Dose Level -1 in the table below). If 1 out of 3 patients experience a DLT, 3 more patients will be added. If 0 out of 3 or 1 out of the 6 patients experience a DLT, this will be declared as a safe dose for the Phase II.
Phase I part is now complete and the SGI-110 20mg/m^2 has been declared as the safe dose for Phase II, in combination with Atezolizumab 840 mg.

Phase I Dosing

Dose Level	Dose	
Dose Level 1	SGI-110: 30 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles.
Dose Level 2	SGI-110: 45 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles.
Dose Level -1	SGI-110: 20 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles.

In patients receiving therapeutic benefit, dose reduction with continued treatment can be considered regardless of degree and type of toxicity at the discretion of the Principal Investigator.

In the randomized Phase IIb, an additional n=66 patients will be enrolled, allocated n=33 each for cohort 1 (Atezolizumab alone) and cohort 2 (Atezolizumab + SGI-110). Given the constraints of a very limited time supply of CDX-1401, considering feasibility, the decision had to be made to omit the vaccine arm proposed originally as cohort 3 (Atezolizumab + SGI-110 + CDX-1401). Eligible patients will be randomized to one of the following two treatment groups:

- **Cohort 1:** Atezolizumab 840 mg IV every 2 weeks; on days 1 and 15 of each 28 day cycle for a total of 24 cycles.
- **Cohort 2:** SGI-110 subcutaneous on days 1-5 of each cycle for a total of 6 cycles plus Atezolizumab 840 mg IV every 2 weeks, on days 8 and 22, for a total of 24 cycles.

  (1 cycle = 28 days)

Accrual into the randomized Phase IIb study (Cohort 2) will utilize the dose of SGI-110 that has been determined to be safe in the Phase I trial, which is 20 mg/m^2. The phase IIb component of this trial is designed as a randomized partial crossover design. Subjects randomized to Cohort 1 will be crossed over at the six months time point, following the primary measure of efficacy, and will be eligible to receive SGI-110 in addition to Atezolizumab.

Randomized Phase IIb Dosing

Treatment Cohorts	# of Patients	SGI-110	Atezolizumab
Cohort 1	33	None	840 mg IV Day 1 and 15 of each 28 day cycle for a total of 24 cycles.
Cohort 2	33	20 mg/m^2 SC on days 1-5 of each 28 day cycle for a total of 6 cycles.	840 mg IV Day 8 and 22 of each 28 day cycle for a total of 24 cycles.

In patients receiving therapeutic benefit, dose reduction with continued treatment can be considered regardless of degree and type of toxicity at the discretion of the Principal Investigator.

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1. OBJECTIVES

1.1. Phase I Primary Objectives

1.1.1 To determine the safety of fixed doses of atezolizumab (MPDL3280A) in combination with SGI-110.

1.1.2 To evaluate toxicity of the combination as defined by NCI Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0.

1.2 Phase IIb Primary Objective

1.2.1 To study if SGI-110 improves the benefit of atezolizumab by analyzing progression free survival (PFS), using RECIST 1.1 criteria (Section 10.1).

1.3 Phase I Secondary Objectives

1.3.1 To observe and record anti-tumor activity. Although the clinical benefit of these drugs has not yet been established, the intent of offering this treatment is to provide a possible therapeutic benefit, and thus the patient will be carefully monitored for tumor response and symptom relief in addition to safety and tolerability.

1.4 Phase IIb Secondary Objectives

1.4.1 To determine overall survival (OS), objective response rate (complete and partial responses), clinical benefit rate (response + stable disease), CA-125 reduction (percentage of patients with CA-125 reduction by ≥ 50%), and duration of response.

1.4.2 To assess the impact of the combination of atezolizumab and SGI-110 on anti-tumor immune responses.

1.4.3 To assess the impact of SGI-110 on NY-ESO-1 expression in tumor tissue

1.4.4 To assess toxicities associated with this combination (Cohort 2), as there is little human experience with this combination.

1.5 Exploratory/Translational Objectives

1.5.1 To determine the effectiveness of SGI-110 in combination with atezolizumab by assessing NY ESO 1 specific cellular and humoral immunity.

- Peripheral blood NY-ESO-1 specific CD8+ and CD4+ T cells.
- Peripheral blood NY-ESO-1 specific antibodies.
- Peripheral blood unrelated CTA specific antibodies (antigen spreading)
- Peripheral blood frequency of CD4+CD25+FOXP3+ regulatory T cells.
- Examine potential differential effect of NY-ESO-1 expression on PFS

1.5.2 To assess the impact on PDL1expression in tumor tissue.
1.5.3 Evaluation of therapeutic efficacy on immune cell phenotype
1.5.4 DNA methylation and DNA methylome: in pre- and on-treatment peripheral blood, serum (circulating DNA), and tumor biopsies.
- Global methylation: LINE-1, AluPromoter, Illumina 850K platform.
- Specific methylation of CTA genes: NY-ESO-1, MAGE-A3, CT45, PRAME, BORIS.

1.5.5 Pre-and on-treatment Density and location of tumor infiltrating CD3+ and CD8+ T cells
1.5.6 Evaluate the pre- and post-treatment mutational and neo-antigen load and therapeutic efficacy.
1.5.7 Pre- and post-treatment T cell receptor (TCR) repertoire to study the effect of TCR Vβ diversity due to combination of PDL1 blockade, epigenetic modification, and vaccination on therapeutic efficacy.
1.5.8 Gut microbiota at baseline and one on-treatment sample at C4D1 (cycle 4, Day 1) or at progression, whichever is earlier to evaluate the role of microbiota on the therapeutic efficacy of the proposed combination therapy.

Prioritization of Exploratory Endpoints:

Prioritization order	Exploratory Tests	Sample
1	NY-ESO-1-specific immune responses (ELISPOT and FACS of NY-ESO-1-specific T cells)	Tumor and PBMC
2	PDL1 expression in tumor tissue (IHC)	Tumor
3	AIM gene signatures (Nanostring Immune Profiling Panel)	Tumor and PBMC
4	Immune cell phenotype (FACS for frequency of immune cell subsets e.g. MDSC, Tregs, CTL, DC, macrophage and expression of co-stimulatory or co-inhibitory receptors (e.g. 4-1BB, OX40, GITR, TIM-3, LAG3, etc)	Tumor and PBMC
5	Neo-antigen and mutational antigen (Whole exome sequencing and RNAseq)	Tumor and PBMC
6	T cell receptor (TCR) repertoire (TCRseq for Vβ)	Tumor and PBMC
7	Microbiome (Microbiome sequencing and bacterial 16s RNA expression)	Stool

2. BACKGROUND

2.1 Study Disease: Epithelial Ovarian, Fallopian Tube, or Peritoneal Cancer

Ovarian cancer is the leading cause of death from gynecologic malignancies (1). More than 23,000 cases occur annually in the United States, and 14,000 women can be expected to succumb to the disease (2). Despite advances in surgery, and chemotherapy over the past 20 years, only modest progress has been made in improving the time to disease progression (3, 4), and the overall survival for patients with ovarian cancer has not changed significantly. Although the majority of women with advanced stage ovarian cancer respond to first-line chemotherapy, most of these responses are not durable and more than 70% of patients die of recurrent disease within 5 years of diagnosis. By contrast, patients with early stage ovarian cancer have excellent prognosis with currently available treatment modalities. The poor survival in advanced disease is due both to late diagnosis and lack of effective second-line cytotoxic therapy for the majority of patients who relapse following initial clinical complete response to platinum-based chemotherapy. The inability to "cure" ovarian cancer is attributed to intrinsic or acquired resistance to drug therapy which is combated with high-dose chemotherapeutic regimens complicated by significant drug related morbidity and mortality. Moreover, effective strategy for the prevention of recurrence in complete responders is lacking. Therefore, new treatment modalities and paradigms are needed before a significant impact can be expected in the prognosis of women diagnosed with epithelial ovarian cancer. Fallopian tube and primary peritoneal cancers are clinically and biologically similar to ovarian cancer, and are treated in the same fashion.

2.2 NY-ESO-1 as a Target for Immunotherapy

There is increasing evidence that the immune system has the ability to recognize tumor-associated antigens expressed in human malignancies and to induce antigen-specific humoral and cellular immune responses to these targets. Importantly, evidence from correlative human studies indicate that the presence of tumor-infiltrating lymphocytes may be associated with improved clinical outcome in several human cancers, including ovarian, melanoma, colon, gastric, liver, brain, lungs, kidney and prostate cancers. In addition, encouraging results from large-scale clinical trials of immune system-provoking therapies (5-8) have re-kindled the promise of harnessing the power of the immune system to attack cancers. Therefore, the development of strategies to enhance the potential of tumor antigen-specific CD8+ T cells and CD4+ T cells is urgently needed for extending remission rates in several cancer types. In this regard, cancer-testis antigens, a unique class of antigens that demonstrate high levels of expression in adult male germ cells but generally not in other normal adult tissues and aberrant expression in a variable proportion of a wide range of different cancer types, are promising candidates for immunotherapy. Among cancer-testis antigens, NY-ESO-1 is one of the most spontaneously immunogenic tumor antigens described so far (9, 10). The NCI antigen prioritization panel has recently ranked NY-ESO-1 in the top 10 antigens for further development of human cancer vaccines (11).

The NY-ESO-1 antigen was initially identified by serological analysis of recombinant cDNA expression libraries (SEREX), using tumor mRNA and autologous serum from a patient with squamous cell carcinoma of the esophagus. The full-length NY-ESO-1 cDNA was cloned, encoding a protein of 180 amino acids (9). The function of this protein is unknown. The gene encoding NY-ESO-1 has been mapped to chromosome Xq28 (12). A gene LAGE-1, encoding an antigen closely related to NY-ESO-1, has also been mapped to chromosome Xq28 (13).
NY-ESO-1 showed restricted mRNA expression in normal tissues, with high-level mRNA expression found only in the germ line cells in testis (9, 13). Reverse transcription-polymerase chain reaction (RT-PCR) analysis showed NY-ESO-1 mRNA expression in a variable proportion of a wide array of human cancers, including melanoma, breast cancer, prostate cancer, lung cancer, hepatocellular carcinoma, ovarian cancer, thyroid cancer and bladder cancer(14). Expression of

LAGE-1 parallels the expression of NY-ESO-1 in most cases (12). The pattern of NY-ESO-1 and LAGE-1 expression indicates that NY-ESO-1 family belongs to a growing family of immunogenic testicular antigens that are aberrantly expressed in a wide range of human cancers. These antigens, initially detected by either cytotoxic T cells (MAGE, BAGE, GAGE-1) or antibodies [HOM-MEL-40 (SSX2), NY-ESO-1], represent a pool of antigenic targets for cancer vaccination (15). The expression of these genes in many different types of cancer suggests that immunization against the antigens encoded by these genes could be applied to a large proportion of cancer patients.

NY-ESO-1 has 2 characteristics that make it a viable candidate as a tumor vaccine; (a) testis-restricted expression in normal tissues and; (b) its immunogenicity. The first characteristic also applies to MAGE and other genes with a similar expression pattern (see above). A serological survey has shown that patients with melanoma, ovarian cancer, lung cancer, or breast cancer produced antibodies against NY-ESO-1 more often (19 of 209) than antibodies against MAGE-1 (3 of 209) or MAGE-3 (2 of 209). Antibodies were not detected in 70 healthy blood donors (16). This is evidence for NY-ESO-1 immunogenicity in patients with cancer, supporting the expectation that immunity can be enhanced or induced by vaccination.

2.3 NY-ESO-1 and Ovarian Cancer

Data from 8 ovarian tumors by Chen *et al.* (9) showed aberrant expression of NY-ESO-1 in 25% (2 of 8) of epithelial ovarian cancers. Using a combination of RT-PCR and immunohistochemistry, data indicates aberrant expression of NY-ESO-1 in up to 43% (82 of 190) of patients with epithelial ovarian cancer (15).

2.4 Epigenetic reprogramming of NY-ESO-1 and the tumor microenvironment as a strategy to potentiate immunotherapy

The ability of the immune system to recognize and reject tumors is dependent on several factors that include the expression of immunogenic target antigens on tumor cells, generation of high frequencies of tumor antigen (TA) specific T cells with potent effector function, and capacity to overcome several mechanisms by which tumors escape immune attack. Although several *in vitro* and *in vivo* studies have demonstrated that the amount and duration of antigen stimulation can influence antigen-specific T cell responses (17, 18), the majority of cancer immunotherapy studies have focused on the generation of TA-specific T cells without concomitant manipulation of antigen expression. This is a critical consideration because (i) TAs are generally not expressed at 100% frequency in cancers; (ii) even when expressed, the antigen density may be low; and (iii) expression is often heterogeneous with areas of tumor that are distinctly positive or negative for the TA. Moreover, in the latter scenario, the generation of tumor specific immunity may result in tumor “immunosculpting” that could influence the ability to evade immune eradication. In this regard, a number of human immunotherapy clinical trials, have demonstrated evidence of immunoediting with either loss of antigen expression (e.g. MART-1) or MHC class I expression (19-21).

NY-ESO-1 expression is regulated by DNA methylation in EOC cells and tumor tissues, and decitabine treatment induces NY-ESO-1 expression in EOC cell lines (22). Additionally, heterogeneous intratumor expression of NY-ESO-1, which frequently occurs in EOC and other tumor types, is associated with the DNA methylation status of the *NY-ESO-1* promoter and the global methylation status of the tumor cells (22). These observations and other work suggest that antigen expression is an important limitation for CT antigen vaccine therapy, and epigenetic therapy is a means to overcome this limitation. In agreement, a link between demethylating agents and the immune system was made over a decade ago in experiments that showed that azacytidine (AZA) could demethylate and activate genes encoding MHC class I genes and tumor antigens.(23, 24) Interferon pathway genes were also upregulated by AZA, and this was correlated with increased expression of endogenous retroviral transcripts rather than de-repression of interferon pathway transcription factors. More recently it was shown that the most common set of genes induced by AZA in solid tumor cell lines were those involved in antigen presentation and interferon response (AZA-induced immune signature or AIM) (25). This finding was corroborated by the observation that patients who had previously received AZA for lung cancer subsequently had a response to immune checkpoint inhibitors (26).

In a recent study by Roulois *et al* (27) gene expression profiling of colorectal and ovarian carcinoma cells treated with low-dose AZA identified genes upregulated with delayed kinetics whose expression was not correlated with changes in DNA methylation, many of which were targets of the IRF7 protein, a known antiviral mediator. Probing upstream, these investigators found that AZA induced nuclear localization of IRF7 by stimulation of the MDA5 and RIG-I proteins that recognize double stranded viral RNA. The anti-proliferative response to AZA depended on the RIG-I/MDA5 RNA response pathway, and moreover, double-stranded RNA (dsRNA) itself mimicked the effect of AZA in inhibiting self-renewal of colorectal cells. In another report by Chiappinelli *et al* (28), colon cancer cells treated with AZA actually began to secrete interferon, but this was not linked to demethylation of interferon pathway genes. Again, the evidence pointed to activation of endogenous retroviral sequences, as transfection of a fresh culture of cells with dsRNA derived from AZA-treated cells, but not control cells, induced an antiviral response in recipient cells. These studies highlight another mode of action of demethylating agents in cancer and suggest new rational approaches to the use of such agents in immunotherapy. Indeed, Chiappinelli *et al* provide evidence that inhibition of DNA methylation could further sensitize a murine model of melanoma to anti-CTLA4 immune checkpoint therapy. Consequently, the current study will leverage the multi-faceted effects of a demethylating agent to promote the impact of checkpoint blockage in ovarian cancer patients.

2.4.1 Pre-clinical and clinical studies support the use of second generation DNMTi SGI-110 in combination with NY-ESO-1 vaccination for EOC

In a recent pre-clinical study of the effect of SGI-110 on CT antigens in ovarian cancer, (29) SGI-110 treatment at clinically relevant concentrations caused *LINE1* (global) and *NY-ESO-1* promoter hypomethylation, induced NY-ESO-1 mRNA and protein expression, and promoted NY-ESO-1 specific CTL response against EOC cell lines (30). Importantly, these effects were observed both in EOC cell lines treated *in vitro* as well as in EOC tumor xenografts in immuno-compromised

(SCID) mice. Three different SGI-110 treatment regimens were tested in EOC xenograft-bearing mice: (i) daily x 5 day; (ii) weekly x 3; (iii) twice weekly x 3, using a range of different SGI-110 doses delivered subcutaneously (30). All 3 treatment regimens led to significant DNA hypomethylation and NY-ESO-1 induction, with the first and the third regimen showing the greatest effects. The effects of SGI-110 were also compared to those of 5-azacytidine and decitabine *in vitro*, and to decitabine *in vivo*. There was a roughly equivalent effect of SGI-110 and decitabine *in vitro*, with lower activity of 5-azacytidine. In the xenograft studies, SGI-110 performed superior to decitabine for both DNA hypomethylation and NY-ESO-1 induction. Notably, SGI-110 treatment induced the cell surface expression of MHC I and the co-stimulatory molecule ICAM in EOC cells, and promoted NY-ESO-1 specific recognition of EOC cells by cytotoxic T lymphocytes. Taken together, the pre-clinical data provide strong support for the use of SGI-110 as an NY-ESO-1-specific immune modulator in EOC patients. Moreover, the ability of SGI-110 to induce NY-ESO-1 expression in-vivo in ovarian cancer patients has now been addressed in an on-going Phase1/2 clinical trial of SGI-110 + carboplatin for EOC (NCT01696032) (see below).

2.4.2 Phase I clinical trial of decitabine + NY-ESO-1 vaccine + liposomal doxorubicin for recurrent epithelial ovarian cancer (EOC)

NY-ESO-1 is a cancer testis antigen (CTA) and vaccine target in melanoma, Epithelial Ovarian Cancer (EOC), lung cancer, and other malignancies (31). As NY-ESO-1 expression is principally regulated by promoter DNA methylation (32), the possibility has been raised that DNA methyltransferase inhibitor (DNMTi) therapy may promote improved immunological and clinical responses to NY-ESO-1 vaccines (33). We recently tested this idea in a Phase I clinical trial of EOC (34). The treatment regimen consisted of decitabine (5-aza-2′-deoxycytidine), NY-ESO-1 vaccine, and liposomal doxorubicin (doxil) (34). Patients received decitabine on day 1, doxil on day 8, and NY-ESO-1 vaccine on d15, and the regimen was repeated monthly for a total of four cycles. From this study we reported a number of key findings: i) DNA hypomethylation in tumor tissues was associated with spontaneous production of circulating NY-ESO-1 antibodies in EOC patients, ii) the treatment regimen led to significant hypomethylation of *LINE1* (global methylation marker) and the *NY-ESO-1* promoter in peripheral blood leukocytes and circulating DNAs, the latter of which is likely to reflect tumor cell pharmacodynamics, iii) EOC patients demonstrated serological conversion to NY-ESO-1 antibody positive, while patients who had NY-ESO-1 antibodies prior to therapy maintained NY-ESO-1 antibody levels during and post-therapy, iv) EOC patients demonstrated antigen spreading to other CTAs, suggesting that DNMTi initiates serological responses to newly reactivated CTAs, v) NY-ESO-1 specific CD4 and CD8 positive T-cell responses occurred, and vi) a clinical benefit rate (i.e. stable disease + disease response) of 60%, which is greater than seen in historical data of doxil monotherapy for recurrent EOC (~20% clinical benefit rate) (34). Taken together, these striking findings provide strong support for a novel chemo-immuno therapeutic approach that integrates DNMTi treatment with vaccination against reactivated tumor antigens, including NY-ESO-1. However, it is unknown whether the regimen is clinically beneficial compared to standard second-line chemotherapy in EOC (doxil monotherapy), as the Phase I trial was not adequately powered to determine clinical benefit. Consequently, here we propose a Phase II evaluation of this approach.

2.5 Ovarian Cancer and the PD1/PDL1 pathway

Tumors employ the programmed cell death 1 (PD1) and its ligand (PDL1) inhibitory pathway to paralyze antitumor immune response. PDL1 (also known as B7-H1 or CD274) is a coregulatory molecule that is expressed on the surface of various types of cells, including immune cells and epithelial cells. By binding to its receptor PD1 on lymphocytes, it generates an inhibitory signal toward the T-cell receptor (TCR)-mediated activation of lymphocytes. Hamanishi et al reported high PD-L1 expression in ovarian cancer by immunohistochemistry (35). They looked at 70 tissue samples from ovarian cancer patients and found 88.5% to be positive for PD-L1 with high expression (score of 2 or 3) in 68.5%. Similarly, Akibo et al reported 44 out of 65 (67.69%) ovarian cancer patient samples to be positive for PD-L1(36). PDL1 expression in tumor cells has been shown to be an independent unfavorable prognostic factor in human ovarian cancer (35). Importantly, PDL1 expression showed the closest relation to unfavorable prognosis among other immunosuppressive molecules that have been tested in ovarian cancer (37). Moreover, direct *ex vivo* analysis of NY-ESO-1-specific $CD8^+$ T cells from TILs of ovarian cancer patients demonstrated impaired effector function, that was could be restored by blocking the PD1-PDL1 pathway (38). In preclinical models of ovarian cancer, PD1-PDL1 pathway blockade also enhanced the amplitude of tumor immunity by reprogramming suppressive and stimulatory signals that yielded more powerful cancer control (39, 40). These data suggest that PDL1 has a role in the clinical course of ovarian cancer by affecting the local immune microenvironment and that interruption of the PD1-PDL1 pathway could restore effective antitumor immunity in ovarian cancer patients. There are two published reports of the impact of PD1-PDL1 pathway blockade in ovarian cancer (41, 42), and two recent ASCO abstracts (43, 44). In the study by Brahmer *et al* (41) using the anti-PDL1 antibody MDX-1105, 1/17 (6%) patients had a partial response; 3/17 (18%) had stable disease lasting ≥ 24 weeks at the 10 mg/kg dose. The rate of PFS at 24 weeks was 22% (2-43%). More recently, Hamanishi *et al* (42) treated twenty patients with platinum-resistant ovarian cancer with an intravenous infusion of nivolumab (anti-PD1) every 2 weeks at a dose of 1 or 3 mg/kg (constituting two 10-patient cohorts). Patients received up to six cycles (four doses per cycle) of nivolumab treatment or received doses until disease progression occurred. In the 20 patients in whom responses could be evaluated, the best overall response was 15%, which included two patients who had a durable complete response (in the 3 mg/kg cohort). The disease control rate in all 20 patients was 45%. The median PFS time was 3.5 months (95% CI, 1.7 to 3.9 months), and the median overall survival time was 20.0 months (95% CI, 7.0 months to not reached) at study termination. The largest study of anti-PDL1 to date was recently presented by Disis *et al* (43), avelumab (MSB0010718C) was tested in patients with previously treated, recurrent or refractory ovarian cancer in a phase Ib, open-label expansion trial (n=75). The patients were heavily pretreated (81% ≥2 lines of chemotherapy). The ORR was 10.7%, based on 8 PRs by RECIST (n=2 additional PRs by irRC), SD was 44.0% and DCR was 54.7%. In another study by Varga *et al* (44), the antitumor activity and safety of pembrolizumab (anti-PD1) was evaluated in patients with PDL1 positive advanced ovarian cancer. The patients were also heavily pretreated (≈90% >3 lines of chemo). The ORR was 11.5% (n=1 CR, n=2 PR); Responses were durable and still ongoing, stable disease 23.1% (n=6) and DCR 34.6% (n=9).

2.6 CTEP IND Agent(s)

2.6.1 **<u>Atezolizumab (MPDL3280A)</u>**

Atezolizumab (MPDL3280A) is a human immunoglobulin (Ig) G1 monoclonal antibody consisting of two heavy chains (448 amino acids) and two light chains (214 amino acids) and is produced in Chinese hamster ovary cells (Investigator's Brochure, 2016). Atezolizumab was engineered to eliminate Fc-effector function via a single amino acid substitution (asparagine to alanine) at position 298 on the heavy chain, which results in a non-glycosylated antibody that has minimal binding to Fc receptors, thus eliminating detectable Fc-effector function and associated antibody-mediated clearance of activated effector T cells. Atezolizumab targets human programmed death-ligand 1 (PD-L1) and inhibits the interaction with its PD-L1 and its receptors, PD-1 and B7-1 (also known as CD80), both of which function as inhibitory receptors expressed on T cells.

Atezolizumab shows anti-tumor activity in both nonclinical models and cancer patients and is being investigated as a potential therapy in a wide variety of malignancies. Atezolizumab is being studies as a single agent in the advanced cancer and adjuvant therapy settings, as well as in combination with chemotherapy, targeted therapy, and cancer immunotherapy.

Atezolizumab is approved in the United States for the treatment of locally advanced or metastatic urothelial carcinoma and as well as metastatic non-small cell lung carcinoma.

2.6.1.1 <u>Mechanism of Action</u>

PD-L1 expression is prevalent in many human tumors (*e.g.*, lung, bladder, ovarian, melanoma, colon carcinoma), and its overexpression has been associated with poor prognosis in patients with several cancers (35, 45-47). PD-L1 binds to two known inhibitory receptors expressed on activated T cells (PD-1 and B7.1), and receptor expression is sustained in states of chronic stimulation such as chronic infection or cancer (48, 49). Ligation of PD-L1 with PD-1 or B7.1 inhibits T-cell proliferation, cytokine production, and cytolytic activity, leading to the functional inactivation or inhibition of T cells. Aberrant expression of PD-L1 on tumor cells has been reported to impede anti-tumor immunity, resulting in immune evasion (50). Therefore, interruption of the PD-L1/PD-1 and PD-L1/B7.1 pathway represents an attractive strategy to reinvigorate tumor-specific T-cell immunity.

Blockade of PD-L1 or PD-1 with monoclonal antibodies has been reported to result in strong and often rapid antitumor effects in several mouse tumor models (51). These data suggest that tumor-specific T cells may be present in the tumor microenvironment in an inactive or inhibited state, and blockade of the PD-L1/PD-1 pathway can reinvigorate tumor-specific T-cell responses.

Collectively, these data establish the PD-L1/PD-1 pathway as a promising new therapeutic target in patients with advanced tumors. Immune-related adverse events (AEs) reported from the two recent studies were consistent with the role of the PD-L1/PD-1 pathway in regulating peripheral tolerance.

2.6.1.2 Summary of Nonclinical Experience

The safety, pharmacokinetics (PK), and toxicokinetics of atezolizumab were investigated in mice and cynomolgus monkeys to support intravenous (IV) administration and to aid in projecting the appropriate starting dose in humans. Given the similar binding of atezolizumab for cynomolgus monkey and human PD-L1, the cynomolgus monkey was selected as the primary and relevant nonclinical model for understanding the safety, PK, and toxicokinetics of atezolizumab.

Overall, the nonclinical PK and toxicokinetics observed for atezolizumab supported entry into clinical studies, including providing adequate safety factors for the proposed phase 1 starting doses. The results of the toxicology program were consistent with the anticipated pharmacologic activity of down-modulating the PD-L1/PD-1 pathway and supported entry into clinical trials in patients.

Refer to the Atezolizumab Investigator's Brochure for details on the nonclinical studies.

2.6.1.3 Summary of Clinical Experience

A summary of clinical data from company-sponsored atezolizumab trials is presented below. Details of all ongoing studies can be found in the Atezolizumab Investigator's Brochure.

2.6.1.3.1. Clinical PK and Immunogenicity

On the basis of available preliminary PK data (0.03–20 mg/kg), atezolizumab shows linear PK at doses ≥1 mg/kg (Investigator's Brochure, 2016). Based on an analysis of exposure, safety, and efficacy data, the following factors had no clinically relevant effect: age (21−89 years), body weight, gender, positive ATA status, albumin levels, tumor burden, region or ethnicity, renal impairment, mild hepatic impairment, level of PD-L1 expression, or ECOG status. No formal PK drug-drug interaction studies have been conducted with atezolizumab, and the interaction potential is unknown. Further details can be found in the current Investigator's Brochure.

The development of anti-therapeutic antibodies (ATAs) has been observed in patients in all dose cohorts and was associated with changes in PK for some patients in the lower dose cohorts (0.3, 1, and 3 mg/kg) (Investigator's Brochure, 2016). Patients dosed at ≥10 mg/kg maintained C_{min} values well above the target serum concentration of 6 mcg/mL despite the detection of ATAs. Accordingly, the development of detectable ATAs does not appear to have a clinically significant impact on PK for doses above 10 mg/kg. To date, no relationship between the development of measurable ATAs and safety or efficacy has been observed.

2.6.1.3.2. Clinical Safety Summary

As of May 10, 2016, atezolizumab has been administered (alone or in combination with other agents) to approximately 6053 patients with solid tumors and hematologic malignancies (Investigator's Brochure, 2016). The first-in-human monotherapy study PCD4989g (in patients with locally advanced or metastatic solid tumors or hematologic malignancies) provides the majority of monotherapy safety data, with 629 safety-evaluable patients as of the data extraction date. Currently, no maximum tolerated dose (MTD), no dose-limiting toxicities (DLTs), and no clear dose-related trends in the incidence of AEs have been determined.

Fatigue, decreased appetite, nausea, diarrhea, constipation, and cough were commonly reported AEs in single and combination therapy (Investigator's Brochure, 2016). AE profiles are similar across tumor types studied, including non-small cell lung cancer (NSCLC), renal cell carcinoma (RCC), triple-negative breast cancer (TNBC), and urothelial carcinoma (UC), and are consistent with the mechanism of action of atezolizumab. The overall immune-mediated AEs reported were considered moderate in severity, and the majority of patients were able to continue on atezolizumab therapy.

As of the data extraction date of December 15, 2015, there were 629 safety-evaluable patients from the first-in-human phase 1a study PCD4989g (Investigator's Brochure, 2016). The median age was 61 years. Of the 629 patients, 619 patients (98.4%) reported at least one AE of any grade or attribution to atezolizumab, and 316 patients (50.2%) experienced at least one grade 3 or 4 AE of any attribution. A total of 444 patients (70.6%) reported at least one treatment-related AE, and 86 patients (13.7%) experienced at least one treatment-related grade 3 or 4 AE. The most frequently observed AEs of any grade and attribution (occurring in ≥10% of treated patients) include fatigue, decreased appetite, nausea, pyrexia, constipation, cough, dyspnea, diarrhea, anemia, vomiting, asthenia, back pain, headache, arthralgia, pruritus, rash, abdominal pain, insomnia, peripheral edema, and dizziness.

Serious AEs (SAEs) have been reported in 261 patients (41.5%) in study PCD4989g (Investigator's Brochure, 2016). Reported SAEs were consistent with the underlying disease. Treatment-related SAEs (57 patients [9.1%]) included pyrexia, dyspnea, pneumonitis, malaise, fatigue, hypoxia, colitis, and bone pain. Pooled single-agent safety data from 1978 patients with UC, NSCLC, and other indications (including trial PCD4989g) indicate that the most frequent (>1% of patients) serious adverse drug reactions (regardless of grade) include dyspnea (3.0%), back pain (1.2%), and abdominal pain (1.1%). A list of AEs considered "expected" for atezolizumab is presented in Section 7.1.1.1.

2.6.1.3.3. Immune-Related Adverse Events

Given the mechanism of action of atezolizumab, events associated with inflammation and/or immune-mediated AEs have been closely monitored during the atezolizumab clinical program (Investigator's Brochure, 2016). To date, immune-related adverse events associated with atezolizumab include hepatitis, pneumonitis, colitis, pancreatitis, diabetes mellitus, hypothyroidism, hyperthyroidism, adrenal insufficiency, Guillain-Barré syndrome, myasthenic syndrome/myasthenia gravis, and meningoencephalitis.

For further details, see the most recent Atezolizumab Investigator's Brochure.

2.6.1.3.4. Clinical Efficacy Summary

Patients with multiple tumor types were included in study PCD4989g, with the largest cohorts consisting of patients with non-small cell lung cancer (NSCLC), renal cell carcinoma (RCC), and urothelial carcinoma (UC) cancer (Investigator's Brochure, 2016). Objective responses with atezolizumab monotherapy were observed in a broad range of malignancies, including NSCLC, RCC, melanoma, UC, colorectal cancer, head and neck cancer, gastric cancer, breast cancer, and sarcoma. Both the preliminary and more mature efficacy data available suggest that treatment with atezolizumab as a single agent or in combination with other therapeutic agents results in anti-tumor activity across a range of tumor types and hematologic malignancies (UC, NSCLC, RCC, TNBC, melanoma, CRC, and NHL) and across lines of therapy. Clinical benefit was observed in terms of objective responses, durability of responses, and overall survival (OS). Improved efficacy of atezolizumab was observed in the unselected patient population, as well as in patients with higher PD-L1 expression on TCs or ICs (*e.g.*, NSCLC) or on ICs only (*e.g.*, mUC, RCC).

2.6.2 **SGI-110**

SGI-110 is a novel DNA methylation inhibitor whose active metabolite is the FDA approved drug 5-Aza-2'-deoxyazacitidine (Decitabine, Dac, Dacogen, Eisai, Inc.). This novel drug was synthesized by Astex Pharmaceuticals, Inc. and was rationally designed to be more stable than Dac in order to provide enhanced pharmacokinetic and pharmacodynamics properties. The decitabine moiety is linked through a phosphodiester bond to deoxyguanosine and this linkage results in reduced susceptibility to immediate inactivation by cytidine deaminases and a prolonged plasma half-life compared to the parent drug. Following the cleavage of the phosphodiester bond, free decitabine undergoes intracellular activation by nucleotide kinases and incorporation into cellular DNAin the standard fashion. Due to its chemical structure, the molar equivalent of a 1 mg dose of decitabine is approximately 2.5 mg of SGI-110. Pre-clinical studies both *in vitro* and in pre-clinical animal models with SGI-110 demonstrate that exposure to this drug induces similar effects on both gene specific and global methylation when compared with Aza and Dac. Furthermore early pharmacokinetic results suggest an improved exposure window with similar AUC exposure profiles between i.v. Dac and s.q. SGI-110, albeit with lower maximal concentrations (Cmax) based on molar equivalence in dose (52) Presently, SGI-110 is being tested against both myeloid malignancies as well as solid tumors in Phase I and II clinical trials.

There is significant interest in the possibility that re-expression of epigenetically silent TSGs could re-sensitize resistant cancer cells to chemotherapy induced death. This concept has been demonstrated previously with Dac in combination with carboplatin in ovarian cancer, and given the potentially superior pharmacokinetic/pharmacodynamic (longer half-life/demethylation effect) and administration profile (s.q.) of SGI-110, pre-clinical demonstration of a similar profile was necessary in order to facilitate clinical development of this concept (53-55) A recently published abstract indicated that the platinum resistant A2780 ovarian cancer cell line could be re-sensitized to carboplatin following a 48 h pre-incubation with 5 μM SGI-110 in vitro (56). These authors also demonstrated that the combination of SGI-110 (2-5 mg/kg/d) and carboplatin (2mg/kg/d) was well tolerated in tumor-bearing mouse xenografts and that the combination resulted in both global and gene-specific hypomethylation. These data as well as the previously published have been sufficient to prompt a Phase I/II study of SGI-110 in combination with conventional therapies in relapsed/refractory ovarian cancer patients.

The pharmacokinetics, bioavailability and safety of SGI-110 have been studied in a variety of different animal model systems. Tolerability has been demonstrated in mouse, rat and rabbit models using multiple dose routes and schedules (57) These studies demonstrate that a majority of the SGI 110 dose is converted to Dac and the relative bioavailability in mice and rats was near 100% when the drug was dosed s.q. compared with i.v.

In addition to studies in rodents, two studies of SGI-110 have demonstrated tolerability and bioavailability in non-human primates. The first study was done to assess the ability of SGI-110 to induce the fetal hemogolobin (HgF) gene. The authors first bled two baboons for eight days to a hematocrit of 20%. Animals were given either Dac 0.5 mg/kg or SGI-110 1 mg/kg for 9 days. Animals were treated with each drug sequentially on the same protocol to allow for intra-animal comparison of drug pharmacokinetics. In both animals, HgF was induced with a latency of about 15 days following drug exposure. Induction of HgF in treated baboons was associated with hypomethylation of the gene promoter to 28.9% following SGI-110 and 36.5% following Dac. Another study of the drug in comparison to Dac was undertaken in Cynomolgus monkeys in which male monkeys were injected for 5 days with either 1.7 mg/kg of i.v. Dac, 1.7 mg/kg of s.q. SGI-110 or 3.0 mg/kg of s.q. SGI-110 (58) This study was done specifically to demonstrate safety, pharmacokinetics and pharmacodynamics in comparison with Dac in order to facilitate the Phase I first in human study of SGI-110. The dose of Dac was chosen to mimic the current clinically utilized Dac schedule of 20 mg/m2/d x 5 days while the SGI-110 schedules reflect 42 and 75% of the molar equivalent Dac doses. Plasma levels of Dac and SGI-110 were evaluated every 30 minutes following drug treatment. In the monkeys, SGI-110 administered s.q. was rapidly converted to Dac producing a similar exposure window when compared with i.v. Dac, for the groups treated with s.q. SGI-110, the Cmax ranged from 157-200 (1.7 mg/kg) or 190-378 ng/mL (3 mg/kg) for SGI-110; and 52-82 (1.7 mg/kg) or 60-163 ng/mL (3.0 mg/kg) for Dac. For i.v. Dac, the Cmax was 215-525 ng/mL. The AUCs for the SGI-110 groups were ranged 90-101 (1.7 mg/kg) or 123-324 (3 mg/kg) ng*hr/mL; while decitabine AUCs were 37-68 (1.7 mg/kg) or 69-155 (3 mg/kg). AUC for i.v. DAC was 99-221. Treated monkeys developed reversible decreases in hemoglobin, neutrophils and total white blood cell counts which nadired between days 8 and 14 and recovered between days 21 and 28; Dac treated monkeys had lower nadirs when compared with either of the two SGI-110 treated monkey groups. The doses of SGI-110 tested produced superimposable changes in LINE-1 and gene specific methylation (miR29b) when compared with Dac. A fourth cohort of monkeys was treated with a schedule of 3.0 mg/kg of s.q. SGI-110 once per week for 3 weeks. This schedule produced significantly less impact on the hemogram of the monkeys compared with Dac and a less robust reduction in LINE-1 and miR29b methylation. The treated monkeys tolerated both drugs well.

Two recent publications in the same issue of cell (32, (28) show that DNA methyltransferase inhibitors (DNMTis) upregulate immune signaling in cancer through the viral defense pathway. In ovarian cancer (OC), DNMTis trigger cytosolic sensing of double-stranded RNA (dsRNA) causing a type I interferon response and apoptosis. These studies highlight another mode of action of demethylating agents in cancer and suggest new rational approaches to the use of such agents in immunotherapy. Most EOC are p53 mutant (59) and selected drugs that inhibit DNA methylation can preferentially kill p53 deficient cells (60).

SGI-110 Clinical Data Summary

Study SGI-110-01 was a first-in-human (FIH) study in subjects with MDS or AML. It was conducted in 2 phases and enrolled approximately 400 subjects:
• Phase 1 Dose Escalation in 93 subjects established the maximal tolerated dose (MTD) and biologically effective dose (BED) based on long interspersed nucleotide element-1 (LINE-1) demethylation. LINE-1 demethylation relative to baseline in peripheral blood mononuclear cells (PBMC) provides a reliable surrogate of global hypomethylation by hypomethylating agents (HMAs). Phase 1 tested 28-day regimens of SGI-110 of (1) once daily for the first 5 days (Daily×5) or (2) once weekly for 3 weeks. LINE-1 demethylation was greater for the Daily×5 regimen. Subsequently, a third 28-day regimen was studied: twice weekly SGI-110 for 3 weeks.
• Phase 2 Dose Expansion in over 300 subjects established response rate and safety in different subject populations of MDS and AML using 3 different doses/regimens. Five-day regimens of 60 and 90 mg/m2/day were tested in both AML and MDS subjects. After safety and activity were established with the 5-day regimen, a single-arm 10-day regimen with 60 mg/m2/day was studied in AML subjects.

2.6.2.1.1. Phase I (SGI-110-01 Dose Escalation)

Dose escalation was in subjects with refractory or relapsed MDS or AML who had ECOG PS 0-2 and acceptable liver and kidney function (52). Co-primary endpoints were MTD and BED; results are described below.

- MTD
- Daily×5 MTD was reached for MDS subjects at 90 mg/m2/day but was not reached for AML subjects up to 125 mg/m2. Of 12 subjects enrolled at the 125 mg/m2 dose level, 3 had MDS, and 9 had AML. Although none of the subjects with AML experienced DLT, 2 of 3 subjects with MDS treated at 125 mg/m2 Daily×5 experienced DLTs (febrile neutropenia + bacteremia in 1 subject, and febrile neutropenia + thrombocytopenia + fatal sepsis in another subject).
- The MTD was not reached for the once weekly (up to 125 mg/m2/dose) or the twice weekly (up to 90 mg/m2/dose) regimens.
- BED (defined as the minimum dose that achieves maximal demethylation of LINE-1 from 3 successive cohorts)
- The Daily×5 regimen had the greatest effect on LINE-1 demethylation, and the BED was 60 mg/m2/day. Similar LINE-1 demethylation occurred at 60, 90, and 125 mg/m2.

- Less demethylation was observed with the Once Weekly regimen; the Twice Weekly regimen did not lead to better LINE-1 demethylation than the Daily×5 regimen.

2.6.2.1.2. Phase 2 (SGI-110-01 Dose Expansion)

Phase 2 dose expansion was opened as a multicenter, open-label, randomized dose-response comparison of 60 vs 90 mg/m2 SGI-110 SC Daily×5. Subjects were stratified by disease type (treatment-naïve elderly AML, relapsed/refractory AML, treatment-naïve MDS, and relapsed/refractory MDS). Approximately 50 subjects were enrolled for each disease cohort. The 60 mg/m2 Daily×5 dose was chosen because it represented the BED from Phase 1, while 90 mg/m2 was chosen to explore benefit from a higher dose that was still well tolerated in both MDS and AML subjects.

2.6.2.2 Ongoing Phase 2 Study SGI-110-02 in Ovarian Cancer (Combination Therapy)

In protocol SGI-110-02, ovarian cancer patients were dosed with guadecitabine SC (Cohort 1: 45 mg/m2, Cohort 2: 30 mg/m2) on Days 1-5 and with carboplatin IV (AUC 5 or AUC 4) on Day 8 (61). Blood samples were collected from a total of 19 subjects (n=5 in Cohort-1 and n=14 in Cohort-2). The pharmacokinetic profile of guadecitabine and decitabine appeared similar to what was seen in study SGI-110-01 where a higher dose of guadecitabine was used. Guadecitabine was still present in circulation at 8 hours in most subjects and guadecitabine slowly released decitabine, which had an exposure window of over 8 hours. Gene-specific methylation and expression was evaluated in DNA and RNA extracted from matched tumor biopsies or cell pellets from ascitic fluid (when available); biopsies or ascites were obtained at Screening and at Cycle 2, Day 8. Quantitative reverse-transcription polymerase chain reaction was used to evaluate the expression of the genes pre- and post-treatment. Overall, post-treatment demethylation and an increased expression of the demethylated genes were observed in all genes tested including CTAs (NY-ESO-1, MAGE A3, MAGE A11, and MAGE A2) when compared to baseline expression for the same transcripts. This shows that a similar extent of demethylation could be achieved with lower doses of 45 and 30 mg/m2 in ovarian cancer as compared to the higher doses of 60 to 125 mg/m2 in myeloid malignancies.

A detailed discussion of the preclinical and clinical pharmacology, pharmacokinetics, and toxicology of SGI-110 can be found in the Investigator's Brochure.

2.7 Rationale

Rationale for combining Atezolizumab and SGI-110:

Treatment options are often still limited for patients with recurrent ovarian cancer and the prognosis for patients with relapsed ovarian cancer is extremely poor. Since immunotherapeutic strategies have demonstrated few side effects in the studies undertaken to date, and since there is now compelling evidence that the immune system plays an important role in delaying disease progression (62), it would seem reasonable to offer such a therapy to patients with few options. In this trial, we seek to address the question:

> Does epigenetic therapy (with associated induction of AIM genes/↑ Ag density & minimize Ag loss) sensitize patients for PD1-PDL1 blockade?

To address this question, we will test in a randomized Phase I/IIb clinical trial a multimodal strategy that will include two patient cohorts: (i) the use of checkpoint blockade alone, Atezolizumab (MPDL3280A) to block PDL1 signaling thereby allowing efficient antitumor effector function of TILs and (ii) the use of Atezolizumab in combination with SGI-110 to enhance tumor antigen expression, effector T cell activation, immune gene upregulation for CTA presentation on MHC. In preliminary studies, we and others have shown that (i) treatment of ovarian cancer cell lines in vitro with low micromolar concentrations of DNMTi, including SGI-110, leads to robust expression tumor antigens, including NY-ESO-1 (23, 32, 63-65), (ii) SGI-110 treatment effectively hypomethylates the promoter of the tumor antigen genes in vivo, (iii) DNMTi may directly enhance effector differentiation of antigen-specific T cells, and (iv) upregulation of immune-signature pathways such that epigenetic therapy may further sensitize the tumor microenvironment to immunotherapy. Therefore, we hypothesize that a multimodal strategy consisting of SGI-110 and Atezolizumab would (i) enhance antigenic dose to antigen presenting cells (APCs) and promote clonal expansion of antigen specific cytotoxic CD8+ T cells, (ii) prevent loss of effector function by TILs via engagement of PD1-PDL1, and (iii) minimize immune escape via tumor antigen loss. We predict that the combination of SGI-110 and Atezolizumab will be safe, and lead to enhanced induction of anti-tumor CD8+ and CD4+ T cells. The demonstration of little toxicity combined with the development of a strong cellular and humoral immune response may contribute important information on the role of DNMTi induced immune gene upregulation/CTA presentation on MHC for augmenting checkpoint blockade therapy for recurrent ovarian cancer.

Hypotheses

The combination of PDL1 blockade (MPDL3280A) and epigenetic modification (SGI-110) will be safe and improve progression free survival (PFS) and/or overall survival (OS) in women who have recurrent/persistent platinum-resistant ovarian, primary peritoneal, or fallopian tube cancers.

Epigenetic therapy (SGI-110) will increase CTA expression and induce an immune signature that would further sensitize patients for PDL1 blockade (MPDL3280A).

2.8 Correlative Studies Background

2.8.1 NY-ESO-1 expression

IHC and qPCR expression of NY-ESO-1 will be assessed for pre-treatment and 4 weeks on treatment (and at progression when available) tumor samples from all cohorts to check the impact of SGI-110 (epigenetic modification by DNMTi) on CTA expression in tumor tissue and comparison within and across different cohorts. In order to identify NY-ESO-1 antigen expression level in the pre- and on treatment samples, paraffin-embedded tissue samples from patients are used to test for NY-ESO-1 expression by immunohistochemistry (IHC) and detection of the NY-ESO-1 transcript by qRT-PCR. Neither test has 100% sensitivity and a dual approach by both IHC and qRT-PCR is the optimal solution to identify the most accurate and precise NY-ESO-1 expression. This test will be performed at the Center for Personalized Medicine (CPM) at the Roswell Park Cancer Institute (RPCI).

2.8.2 AIM gene signatures

The NanoString nCounter® PanCancer Immune Profiling will be utilized for this purpose to evaluate the therapeutic efficacy on immune cell signature due to combination of PDL1 blockade, and epigenetic modification. This panel is a novel new gene expression panel that enables developing profiles of the human immune response in all cancer types. The 770 gene panel combines markers for 24 different immune cell types and populations, 30 common cancer antigens and genes that represent all categories of immune response including key checkpoint blockade genes. These include the set of genes commonly referred to as AZA-induced immune signature or AIM, which have been shown to be upregulated by treatment with demethylating agents such as azacytidine. The nCounter Analysis System utilizes a novel digital color-coded barcode technology that is based on direct multiplexed measurement of gene expression and offers high levels of precision and sensitivity (<1 copy per cell). The technology uses molecular "barcodes" and single molecule imaging to detect and count hundreds of unique transcripts in a single reaction. Each color- coded barcode is attached to a single target-specific probe corresponding to a gene of interest. Mixed together with controls, they form a multiplexed CodeSet. NanoString's technology employs two ~50 base probes per mRNA that hybridize in solution. The Reporter Probe carries the signal; the Capture Probe allows the complex to be immobilized for data collection. For each positive control, *in vitro* transcribed RNA targets are pre-mixed with the Reporter CodeSet during manufacturing. After hybridization, the excess probes are removed and the probe/target complexes aligned and immobilized in the nCounter Cartridge. Sample Cartridges are placed in the Digital Analyzer for data collection. Color codes on the surface of the cartridge are counted and tabulated for each target molecule. All the required reagents, positive negative controls and chips are included in the kit and supplied by NanoString. Pre-and 4 weeks on-treatment (and at progession when available) tumor and PBMC samples from all cohorts will be subjected to this assay. This assay will be performed by the Genomics Shared Resource (GSR) at the Roswell Park Cancer Institute (RPCI).

2.8.3 NY-ESO-1-specific immune responses

ELISPOT, ELISA, and FACS of NY-ESO-1 specific T cells will be done on pre-treatment and on treatment biopsy (baseline, 4 weeks) and PBMC samples (every 4 weeks) of patients from all cohorts for comparison to evaluate therapeutic efficacy on T cell function (e.g. IFN-γ production by NY-ESO-1 tetramer^{+} or peptide-specific) and phenotype (e.g. effector T cell markers and presence of PD1) due to combination of PDL1 blockade and epigenetic modification. These assays will be done at the Immune Analysis Facility (IAF) at the Center for Immunotherapy, Roswell Park Cancer Institute.

2.8.4 PD-L1 expression

IHC for PD-L1 analysis will be done at a Genentech-designated external laboratory for pre-treatment samples for all cohorts. Clone SP142 will be used for assessment of PD-L1 expression on tumor cells and immune cells using a predefined algorithm. (66)

2.8.5 Gut Microbiota

Microbiome sequencing and bacterial 16s RNA expression will be studied for all cohorts on pre- and cycle 4 day 1 on-treatment stool samples to explore the therapeutic efficacy on changes in gut microbiota due to combination of PDL1 blockade, epigenetic modification, and vaccination. Samples will be stored at the DBBR laboratory at Roswell Park cancer Institute and microbiome sequencing will be done at the Genomics Shared Resource (GSR), Roswell Park cancer Institute.

The human microbiome plays an essential role in digestion and nutrient absorption, epithelial homeostasis, angiogenesis, and in the proper function of nervous and immune systems (67, 68). Disruption of the microbiome could lead to chronic inflammation, altered immune responses and carcinogenesis (69, 70). Most recent research shows that there is a complex reciprocal modulation between gut microbiota and the human brain, which involves interaction of the central nervous system, sympathetic and parasympathetic branches of the autonomic nervous system, the enteric nervous system, and elements of the neuroendocrine and neuroimmune systems. For example, some microbiota synthesize neurotransmitters directly (e.g., gamma-amino butyric acid) while many modulate the synthesis of neurotransmitters, such as serotonin, dopamine and norepinephrine, and brain-derived neurotropic factor (71). Microbiotic abnormalities have been linked to reductions in neurotransmitter levels, cognitive deficits, anxiety, depression, chronic pain and fatigue (72, 73).

The composition of the microbiota also regulates the levels of tryptophan in the systemic circulation as well levels and nature of tryptophan catabolites (TRYCATs). These catabolites influence epithelial barrier integrity and the presence of an inflammatory or tolerogenic environment in the intestine and beyond (74). The composition of the microbiota also determines the levels and ratios of short chain fatty acids, such as butyrate and propionate, which are key energy source for colonocytes (75). Dysbiosis (altered microbiome) can lead to reduced levels of butyrate, and therefore may have adverse effects on epithelial barrier integrity, increase bacterial translocation into the systemic circulation, effects energy homeostasis, and the T helper 17/regulatory/T cell balance. (71, 75).

Increasing evidence suggests that different gut, skin, and vaginal microbiota profiles may be related to the etiology of certain gynecological cancers, such as cervical cancer, uterine cancer, and ovarian cancer (76). Patients' severity of side effects and response to certain cancer treatments can also be impacted by gut microbiome composition (69). In a recent study by Vetizou *et al.* it was shown that anticancer immunotherapy by CTLA-4 blockade relies on the gut microbiota and the authors showed the key role for *Bacteroidales* in the immunostimulatory effects of CTLA-4 blockade both in mice and patients (77). In another recent landmark paper by Sivan *et al.* (78) was shown that *Bifidobacterium* as associated with the antitumor effects and oral administration of

Bifidobacterium alone improved tumor control to the same degree as PD-L1-specific antibody therapy, and combination treatment nearly abolished tumor outgrowth. Thus, the authors suggested that manipulating the microbiota may modulate cancer immunotherapy (78). Since data is more mature for gut microbiome and much less is known on pelvic/peritoneal or vaginal microbiome, only samples for gut microbiota will be collected in his study.

We will also ask our patients to complete basic surveys (refer to APPENDICES B and C) on their nutrition and lifestyle in order to gather a more comprehensive data on the interaction of microbiome and immune system.

2.8.6 Neo-antigen and mutational antigen

For neo-antigen load analysis, additional exploratory analysis may be done using whole exome sequencing (WES) and RNAseq on pre- and on treatment tumor (4 weeks and at progression when available)) and PBMC (every 4 weeks) samples for all cohorts. This will be done at the Genomics Shared Resource (GSR), Roswell Park cancer Institute.

2.8.7 T cell receptor (TCR) repertoire

For evaluation of therapeutic efficacy on TCR Vβ diversity due to combination of PDL1 blockade and epigenetic modification,, TCRseq for Vβ will be carried out at the Center for Immunotherapy at Roswell Park cancer Institute on pre- and on treatment tumor (4 weeks and at progression when available)) and PBMC (every 4 weeks) samples for all cohorts.

3. PATIENT SELECTION

3.1 Eligibility Criteria

3.1.1 Women with epithelial ovarian, fallopian tube, or primary peritoneal carcinoma with platinum-resistant disease (defined as having relapsed within 6 months of last platinum-containing regimen because we would like to include both primary and secondary resistance). Patients are allowed to have had more than 2 prior cytotoxic treatment regimens. All patients should have received standard of care agents, which confer clinical benefit.

3.1.2 Presence of biopsiable disease and patient able to undergo pre-treatment and on-treatment biopsy.

3.1.3 Tissue available from primary and/or recurrent disease to evaluate tumor expression of NY-ESO-1 or PDL1 by IHC and/or RT-PCR, and for measurement of DNA methylation.

3.1.4 No requirement for tumor expression of NY-ESO-1.

3.1.5 Life expectancy > 6 months as assessed by study physician.

3.1.6 Age ≥18 years. Because no dosing or adverse event data are currently available on the use of Atezolizumab in combination with SGI-110 in patients < 18 years of age, children are excluded from this study, but may be eligible for future pediatric trials.

3.1.7 ECOG performance status ≤ 2. Refer to Appendix A.

3.1.8 Have been informed of other treatment options.

3.1.9 Have measurable disease outside of biopsy site present per irRECIST criteria. (Rationale: Biopsy may also induce an inflammatory response and bias outcome measurements.)

3.1.10 Patients may have received previous NY ESO 1 vaccine therapy. Patients who received bevacizumab or other experimental therapies are eligible for enrollment provided they have discontinued therapy (at least 4 weeks) prior to treatment start or randomization and recovered from toxicities to less than Grade 2.

3.1.11 Patients must have normal organ and marrow function as defined below:

– leukocytes	≥2,500/mcL
– absolute neutrophil count	≥1,500/mcL
– platelets	≥100,000/mcL
– hemoglobin	≥10 g/dL
– total bilirubin	≤1.5 × institutional upper limit of normal (ULN) (however, patients with known Gilbert disease who have serum bilirubin level ≤3 × ULN may be enrolled)
– AST(SGOT)/ALT(SGPT)	≤1.5 × ULN (AST and/or ALT ≤3 × ULN for patients with liver involvement)
– alkaline phosphatase	≤2 × ULN (≤5 × ULN for patients with documented liver involvement or bone metastases)
– creatinine clearance	≥30 mL/min/1.73 m^2 by Cockcroft-Gault: $\frac{(140 - \text{age}) \times (\text{weight in kg}) \times (0.85 \text{ if female})}{72 \times (\text{serum creatinine in mg/dL})}$ (Adjusted for BSA by 1.73 m^2/BSA)
– PT/INR and aPTT	≤1.5 × ULN

3.1.12 Administration of the drugs used in this study may have an adverse effect on pregnancy and poses a risk to the human fetus, including embryo-lethality. Women of child-bearing potential must agree to use adequate contraception (hormonal or barrier method of birth control; abstinence) prior to study entry, for the duration of study participation, and for 5 months (150 days) after the last dose of study agent. Should a woman become pregnant or suspect she is pregnant while she is participating in this study, she should inform her treating physician immediately.

3.1.13 Participant or legal representative must understand the investigational nature of this study and sign an Independent Ethics Committee/Institutional Review Board approved written informed consent form prior to receiving any study related procedure

3.2 Exclusion Criteria

3.2.1 Patients with prior allogeneic bone marrow transplantation or prior solid organ transplantation.

3.2.2 Patients who have had chemotherapy or radiotherapy including complementary and alternative medicine treatments (CAMs) within 4 weeks prior to entering the study or those who have not recovered from adverse events (other than alopecia) due to agents administered more than 4 weeks earlier.

3.2.3 Prior treatment with anti-PD-1, or anti-PD-L1 therapeutic antibody or pathway-targeting agents.

- Patients who have received prior treatment with anti-CTLA-4 may be enrolled, provided the following requirements are met:
  - Minimum of 12 weeks from the first dose of anti-CTLA-4 and >6 weeks from the last dose
  - No history of severe immune-related adverse effects from anti-CTLA-4 (NCI CTCAE Grade 3 and 4)

3.2.4 Treatment with any other investigational agent within 4 weeks prior to Cycle 1, Day 1.

3.2.5 Treatment with systemic immunostimulatory agents (including, but not limited to, interferon [IFN]-α or interleukin [IL]-2) within 6 weeks prior to Cycle 1, Day 1.

3.2.6 Treatment with systemic immunosuppressive medications (including, but not limited to, prednisone, cyclophosphamide, azathioprine, methotrexate, thalidomide, and anti-tumor necrosis factor [anti-TNF] agents) within 2 weeks prior to Cycle 1, Day 1.

- Patients who have received acute, low dose, systemic immunosuppressant medications (*e.g.*, a one-time dose of dexamethasone for nausea) may be enrolled.
- The use of inhaled corticosteroids and mineralocorticoids (*e.g.*, fludrocortisone) for patients with orthostatic hypotension or adrenocortical insufficiency is allowed.

3.2.7 Patients taking bisphosphonate therapy for symptomatic hypercalcemia. Use of bisphosphonate therapy for other reasons (*e.g.*, bone metastasis or osteoporosis) is allowed.

3.2.8 Concomitant systemic treatment with chronic use of anti-histamine or non-steroidal anti-inflammatory drugs and other platelet inhibitory agents and patients on oral anticoagulant (e.g. warfarin). Exception: Patients on therapeutic anticoagulation therapy such as low-molecular-weight heparin or warfarin at a stable dose level are allowed on study. See section 5.3.1.

3.2.9 Patients with known primary central nervous system (CNS) malignancy or symptomatic CNS metastases are excluded.

3.2.10 Known hypersensitivity to Chinese hamster ovary cell products or other recombinant human antibodies.

3.2.11 History of severe allergic, anaphylactic, or other hypersensitivity reactions to chimeric or humanized antibodies or fusion proteins

3.2.12 History of allergic reactions attributed to compounds of similar chemical or biological composition to other agents used in this study.

3.2.13 Subjects who have received prior therapy with hypomethylating agents (5-azacytidine, decitabine, SGI-110).

3.2.14 Mental impairment that may compromise the ability to give informed consent and comply with the requirements of the study.

3.2.15 Lack of ability of a patient for immunological and clinical follow-up assessment.

3.2.16 Evidence of current drug or alcohol abuse or psychiatric impairment, which in the Investigator's opinion will prevent completion of protocol therapy or follow-up.

3.2.17 Due to unknown effects on the developing fetus or newborn, pregnant or nursing female patients are excluded from this study.

3.2.18 Unwilling or unable to follow protocol requirements.

3.2.19 Any condition which in the Investigator's opinion deems the participant an unsuitable candidate to receive study drug. (i.e., any significant medical illness or abnormal laboratory finding that would increase the patient's risk by participating in this study).

3.2.20 Known clinically significant liver disease, including active viral, alcoholic, or other hepatitis; cirrhosis; fatty liver; and inherited liver disease.

- Patients with past or resolved hepatitis B infection (defined as having a negative hepatitis B surface antigen [HBsAg] test and a positive anti-HBc [antibody to hepatitis B core antigen] antibody test) are eligible.

- Patients positive for hepatitis C virus (HCV) antibody are eligible only if polymerase chain reaction (PCR) is negative for HCV RNA.

3.2.21 History or risk of autoimmune disease, including, but not limited to, systemic lupus erythematosus, rheumatoid arthritis, inflammatory bowel disease, vascular thrombosis associated with antiphospholipid syndrome, Wegener's granulomatosis, Sjögren's syndrome, Bell's palsy, Guillain-Barré syndrome, multiple sclerosis, autoimmune thyroid disease, vasculitis, or glomerulonephritis.

- Patients with a history of autoimmune hypothyroidism on a stable dose of thyroid replacement hormone may be eligible.
- Patients with controlled Type 1 diabetes mellitus on a stable insulin regimen may be eligible.
- Patients with eczema, psoriasis, lichen simplex chronicus of vitiligo with dermatologic manifestations only (*e.g.*, patients with psoriatic arthritis would be excluded) are permitted provided that they meet the following conditions:
    - Patients with psoriasis must have a baseline ophthalmologic exam to rule out ocular manifestations
    - Rash must cover less than 10% of body surface area (BSA)
    - Disease is well controlled at baseline and only requiring low potency topical steroids (*e.g.*, hydrocortisone 2.5%, hydrocortisone butyrate 0.1%, flucinolone 0.01%, desonide 0.05%, aclometasone dipropionate 0.05%)
    - No acute exacerbations of underlying condition within the last 12 months (not requiring psoralen plus ultraviolet A radiation [PUVA], methotrexate, retinoids, biologic agents, oral calcineurin inhibitors; high potency or oral steroids)

3.2.22 History of idiopathic pulmonary fibrosis, pneumonitis (including drug induced), organizing pneumonia (*i.e.*, bronchiolitis obliterans, cryptogenic organizing pneumonia, etc.), or evidence of active pneumonitis on screening chest computed tomography (CT) scan. History of radiation pneumonitis in the radiation field (fibrosis) is permitted.

3.2.23 Patients with active tuberculosis (TB) are excluded.

3.2.24 Severe infections within 4 weeks prior to Cycle 1, Day 1, including, but not limited to, hospitalization for complications of infection, bacteremia, or severe pneumonia.

3.2.25 Signs or symptoms of infection within 2 weeks prior to Cycle 1, Day 1.

3.2.26 Received oral or intravenous (IV) antibiotics within 2 weeks prior to Cycle 1, Day 1. Patients receiving prophylactic antibiotics (*e.g.*, for prevention of a urinary tract infection or chronic obstructive pulmonary disease) are eligible.

3.2.27 Major surgical procedure within 28 days prior to Cycle 1, Day 1 or anticipation of need for a major surgical procedure during the course of the study.

3.2.28 Administration of a live, attenuated vaccine within 4 weeks before Cycle 1, Day 1 or anticipation that such a live, attenuated vaccine will be required during the study and up to 5 months after the last dose of atezolizumab.

- Influenza vaccination should be given during influenza season only (approximately October to March). Patients must not receive live, attenuated influenza vaccine within 4 weeks prior to Cycle 1, Day 1 or at any time during the study and until 5 months after the last dose of atezolizumab.

3.2.29 Uncontrolled intercurrent illness including, but not limited to, ongoing or active infection, symptomatic congestive heart failure, unstable angina pectoris, cardiac arrhythmia, or psychiatric illness/social situations that would limit compliance with study requirements.

3.2.30 Patients positive for human immunodeficiency virus (HIV) are <u>NOT</u> excluded from this study, but HIV-positive patients must have:

a. A stable regimen of highly active anti-retroviral therapy (HAART)
b. No requirement for concurrent antibiotics or antifungal agents for the prevention of opportunistic infections
c. A CD4 count above 250 cells/mcL and an undetectable HIV viral load on standard PCR-based tests

3.2.31 Patients requiring treatment with a RANKL inhibitor (*e.g.* denosumab) who cannot discontinue it before treatment with atezolizumab.

3.3 Inclusion of Women and Minorities

NIH policy requires that women and members of minority groups and their subpopulations be included in all NIH-supported biomedical and behavioral research projects involving NIH-defined clinical research unless a clear and compelling rationale and justification establishes to the satisfaction of the funding Institute & Center (IC) Director that inclusion is inappropriate with respect to the health of the subjects or the purpose of the research. Exclusion under other circumstances must be designated by the Director, NIH, upon the recommendation of an IC Director based on a compelling rationale and justification. Cost is not an acceptable reason for exclusion except when the study would duplicate data from other sources. Women of childbearing potential should not be routinely excluded from participation in clinical research. Please see http://grants.nih.gov/grants/funding/phs398/phs398.pdf.

4. REGISTRATION PROCEDURES

4.1 Investigator and Research Associate Registration with CTEP

Food and Drug Administration (FDA) regulations and National Cancer Institute (NCI) policy require all individuals contributing to NCI-sponsored trials to register and to renew their registration annually. To register, all individuals must obtain a Cancer Therapy Evaluation Program (CTEP) Identity and Access Management (IAM) account (https://ctepcore.nci.nih.gov/iam). In addition, persons with a registration type of Investigator (IVR), Non-Physician Investigator (NPIVR), or Associate Plus (AP) (i.e., clinical site staff requiring write access to OPEN or RAVE or acting as a primary site contact) must complete their annual registration using CTEP's web-based Registration and Credential Repository (RCR) (https://ctepcore.nci.nih.gov/rcr). Documentation requirements per registration type are outlined in the table below.

Documentation Required	**IVR**	**NPIVR**	**AP**	**A**
FDA Form 1572	✔	✔		
Financial Disclosure Form	✔	✔	✔	
NCI Biosketch (education, training, employment, license, and certification)	✔	✔	✔	
HSP/GCP training	✔	✔	✔	
Agent Shipment Form (if applicable)	✔			
CV (optional)	✔	✔	✔	

An active CTEP-IAM user account and appropriate RCR registration is required to access all CTEP and CTSU (Cancer Trials Support Unit) websites and applications. In addition, IVRs and NPIVRs must list all clinical practice sites and IRBs covering their practice sites on the FDA Form 1572 in RCR to allow the following:

- Added to a site roster
- Assigned the treating, credit, consenting, or drug shipment (IVR only) tasks in OPEN
- Act as the site-protocol PI on the IRB approval

Additional information can be found on the CTEP website at < https://ctep.cancer.gov/investigatorResources/default.htm.>. For questions, please contact the RCR ***Help Desk*** by email at < RCRHelpDesk@nih.gov >.

4.1.1 Investigator Brochure Access for CTEP IND agents

The current version of the Investigator Brochure (IB) will be accessible to site investigators and research staff through the PMB Online Agent Order Processing (OAOP) application. Access to OAOP requires the establishment of a CTEP Identity and Access

Management (IAM) account and the maintenance of an "active" account status and a "current" password. Questions about IB access may be directed via email to IBcoordinator@mail.nih.gov or by phone (240) 276-6575 Monday through Friday between 8:30 am and 4:30 pm (ET).

4.2 Site Registration

This study is supported by the NCI Cancer Trials Support Unit (CTSU).

Each investigator or group of investigators at a clinical site must obtain IRB approval for this protocol and submit IRB approval and supporting documentation to the CTSU Regulatory Office before they can be approved to enroll patients. Assignment of site registration status in the CTSU Regulatory Support System (RSS) uses extensive data to make a determination of whether a site has fulfilled all regulatory criteria including but not limited to the following:

- An active Federal Wide Assurance (FWA) number
- An active roster affiliation with the Lead Network or a participating organization
- A valid IRB approval
- Compliance with all protocol specific requirements.

In addition, the site-protocol Principal Investigator (PI) must meet the following criteria:

- Active registration status
- The IRB number of the site IRB of record listed on their Form FDA 1572
- An active status on a participating roster at the registering site

Sites participating on the NCI CIRB initiative that are approved by the CIRB for this study are not required to submit IRB approval documentation to the CTSU Regulatory Office. For sites using the CIRB, IRB approval information is received from the CIRB and applied to the RSS in an automated process. Signatory Institutions must submit a Study Specific Worksheet for Local Context (SSW) to the CIRB via IRBManager to indicate their intent to open the study locally. The CIRB's approval of the SSW is then communicated to the CTSU Regulatory Office. In order for the SSW approval to be processed, the Signatory Institution must inform the CTSU which CIRB-approved institutions aligned with the Signatory Institution are participating in the study.

4.2.1 Downloading Regulatory Documents

Site registration forms may be downloaded from the #10017 protocol page located on the CTSU Web site. Permission to view and download this protocol is restricted and is based on person and site roster data housed in the CTSU RSS. To participate, Investigators and Associates must be associated with the Corresponding or Participating protocol organization in the RSS.

- Go to https://www.ctsu.org and log in using your CTEP-IAM username and password.

- Click on the Protocols tab in the upper left of your screen.
- Either enter the protocol # in the search field at the top of the protocol tree, or
- Click on the By Lead Organization folder to expand, then select EDDO-NY158, and protocol #10017.
- Click on LPO Documents, select the Site Registration documents link, and download and complete the forms provided. (Note: For sites under the CIRB initiative, IRB data will load to RSS as described above.)

4.2.2 Requirements For #10017 Site Registration:

- IRB approval (For sites not participating via the NCI CIRB; local IRB documentation, an IRB-signed CTSU IRB Certification Form, Protocol of Human Subjects Assurance Identification/IRB Certification/Declaration of Exemption Form, or combination is accepted)

4.2.3 Submitting Regulatory Documents

Submit required forms and documents to the CTSU Regulatory Office, where they will be entered and tracked in the CTSU RSS.

Regulatory Submission Portal: www.ctsu.org (members' area) → Regulatory Tab →Regulatory Submission

When applicable, original documents should be mailed to:

CTSU Regulatory Office
1818 Market Street, Suite 3000
Philadelphia, PA 19103

Institutions with patients waiting that are unable to use the Portal should alert the CTSU Regulatory Office immediately at 1-866-651-2878 in order to receive further instruction and support.

4.2.4 Checking **Site** Registration Status

You can verify your site registration status on the members' section of the CTSU website.

- Go to https://www.ctsu.org and log in to the members' area using your CTEP-IAM username and password
- Click on the Regulatory tab at the top of your screen
- Click on the Site Registration tab
- Enter your 5-character CTEP Institution Code and click on Go

Note: The status given only reflects compliance with IRB documentation and institutional compliance with protocol-specific requirements as outlined by the Lead Network. It does not reflect compliance with protocol requirements for individuals participating on the protocol or the enrolling investigator's status with the NCI or their affiliated networks.

4.3 Patient Registration

4.3.1 OPEN / IWRS

Patient enrollment will be facilitated using the Oncology Patient Enrollment Network (OPEN). OPEN is a web-based registration system available to users on a 24/7 basis. It is integrated with the CTSU Enterprise System for regulatory and roster data interchange and with the Theradex Interactive Web Response System (IWRS) for retrieval of patient registration/randomization assignment. Patient enrollment data entered by Registrars in OPEN / IWRS will automatically transfer to the NCI's clinical data management system, Medidata Rave.

For trials with slot reservation requirements, OPEN will connect to IWRS at enrollment initiation to check slot availability. Registration staff should ensure that a slot is available and secured for the patient before completing an enrollment. Slot Reservation is **required** for the phase I portion of this study. Reservations will expire 14 days after the slot has been reserved.

The OPEN system will provide the site with a printable confirmation of registration and treatment information. Please print this confirmation for your records.

4.3.2 OPEN/IWRS User Requirements

OPEN/IWRS users must meet the following requirements:

- Have a valid CTEP-IAM account (*i.e.*, CTEP username and password).
- To enroll patients: Be on an ETCTN Corresponding or Participating Organization roster with the role of Registrar. Registrars must hold a minimum of an AP registration type.
- Have regulatory approval for the conduct of the study at their site.

Prior to accessing OPEN/IWRS, site staff should verify the following:

- All eligibility criteria have been met within the protocol stated timeframes.
- If applicable, all patients have signed an appropriate consent form and HIPAA authorization form.

4.3.3 OPEN/IWRS Questions?

Further instructional information on OPEN is provided on the OPEN tab of the CTSU website at https://www.ctsu.org or at https://open.ctsu.org. For any additional questions contact the CTSU Help Desk at 1-888-823-5923 or ctsucontact@westat.com.

Theradex has developed a Slot Reservations and Cohort Management User Guide, which is available on the Theradex website: http://www.theradex.com/clinicalTechnologies/?National-Cancer-Institute-NCI-11. This link to the Theradex website is also on the CTSU website OPEN tab. For questions about the use of IWRS for slot reservations, contact the Theradex Helpdesk at 609-619-7862 or Theradex main number 609-799-7580; CTMSSupport@theradex.com.

4.4 General Guidelines

Following registration, patients should begin protocol treatment within 28 days, keeping in consideration that Baseline assessments should be within 28 days prior to treatment start. Issues that would cause treatment delays should be discussed with the Principal Investigator. If a patient does not receive protocol therapy following registration, the patient's registration on the study may be canceled. The Study Coordinator should be notified of cancellations as soon as possible.

5. TREATMENT PLAN

5.1 Agent Administration

Reported adverse events and potential risks are described in Section 7. Appropriate dose modifications are described in Section 6. No investigational or commercial agents or therapies other than those described below may be administered with the intent to treat the patient's malignancy.

For the Phase I safety lead-in dose escalation component of the study, 3 eligible patients will receive SGI-110 (guadecitabine), on days 1-5 at an initial dose of 30mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles. The following decision rules will be followed: (i) If 0/3 dose limiting toxicities (DLTs) are observed, proceed to the next dose level of SGI-110 in combination with Atezolizumab (i.e. SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). (ii) If 1/3 DLTs are observed enroll 3 more patients at the same dose of SGI-110 (i.e. 30mg/m^2). If 1/6 DLTs are observed, proceed to the next dose level of SGI-110 in combination with atezolizumab (i.e. SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). At the next dose level of 45mg/m^2, if 0 or 1 out of 3 patients experiences a DLT, 3 more patients will be added. If 0 or 1 out of the 6 patients experiences a DLT, this will be declared as a safe dose for the Phase II. If $\geq$ 2 DLTs are experienced, the treatment will be considered too toxic and the lower dose-level will be considered the MTD. Given that the patients have received 2 prior lines of chemotherapy, they will be sensitive to the myelosuppression of SGI-110 and therefore, all patients receiving SGI-110 will receive G-CSF prophylactically, filgrastim (neupogen) or pegfilgrastim (neulasta) or biosimilar. The recommended dose of neupogen is 5 mcg/kg subcutaneously per day, rounding off the dose to the nearest vial size, until post-nadir ANC recovery to normal or near normal levels ($\geq$ 1,500/mcL).

Start filgrastim the next day, no later than 3 days after completion of SGI110, and treat through post-nadir recovery. The recommended dose of pegfilgrastim is 6 mg subcutaneously once per cycle, to be given within 24 hours after completing SGI110 or as per institutional guidelines.

A total of 9-12 subjects will be enrolled into the Phase I component of the study. The DLT monitoring period is defined as 1 cycle of treatment.

Considering the number of subjects who experienced neutropenia on the study at the SGI-110 30mg/m^2 dose level, the FDA advised to de-escalate to SGI-110 20mg/m^2 and study 3 to 6 subjects as an expansion of the Phase 1 portion of the study (Dose Level -1 in the table below). If 1 out of 3 patients experience a DLT, 3 more patients will be added. If 0 out of 3 or 1 out of the 6 patients experience a DLT, this will be declared as a safe dose for the Phase II.
Phase I part is now complete and the SGI-110 20mg/m^2 has been declared as the safe dose for Phase II, in combination with Atezolizumab 840 mg.

Phase I Dosing

Dose Level	**Dose**	
Dose Level 1	SGI-110: 30 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles
Dose Level 2	SGI-110: 45 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles
Dose Level -1	SGI-110: 20 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles.

In patients receiving therapeutic benefit, dose reduction with continued treatment can be considered regardless of degree and type of toxicity at the discretion of the Principal Investigator.

In the randomized Phase IIb, an additional n=66 patients will be enrolled, allocated n=33 each for cohort 1 (Atezolizumab alone) and cohort 2 (Atezolizumab + SGI-110). Eligible patients will be randomized to one of the following three treatment groups:

- Cohort 1: Atezolizumab 840 mg IV every 2 weeks; on days 1 and 15 of each 28 day cycle.
- Cohort 2: SGI-110 subcutaneous on days 1-5 of each cycle for a total of 6 cycles plus Atezolizumab 840 mg IV every 2 weeks, on days 8 and 22, for a total of 24 cycles.

  (1 cycle = 28 days)

Accrual into the randomized Phase IIb study (Cohort 2) will utilize the dose of SGI-110 that has been determined to be safe in the Phase I trial, which is 20 mg/m^2. The phase IIb component of this trial is designed as a randomized partial crossover design. Subjects randomized to Cohort 1 will be crossed over at the six months time point, following the primary measure of efficacy, and will be eligible to receive SGI-110 in addition to Atezolizumab.

Randomized Phase IIb Dosing

Treatment Cohorts	# of Patients	SGI-110	Atezolizumab
Cohort 1	33	None	840 mg IV Day 1 and 15 of each 28 day cycle for a total of 24 cycles.
Cohort 2	33	20 mg/m^2 SC on days 1-5 of each 28 day cycle for a total of 6 cycles.	840 mg IV Day 8 and 22 of each 28 day cycle for a total of 24 cycles.

In patients receiving therapeutic benefit, dose reduction with continued treatment can be considered regardless of degree and type of toxicity at the discretion of the Principal Investigator.

5.1.1 Atezolizumab

The atezolizumab dose for this study is 840 mg IV every 2 weeks (start day depends on the cohort).

Administration of atezolizumab will be performed in a setting with emergency medical facilities and staff who are trained to monitor for and respond to medical emergencies.

The initial dose of atezolizumab will be delivered over 60 ($\pm$15) minutes. If the first infusion is tolerated without infusion-associated AEs, the second infusion may be delivered over 30 ($\pm$10) minutes. If the 30-minute infusion is well tolerated, all subsequent infusions may be delivered over 30 ($\pm$10) minutes. For the first infusion, the patient's vital signs (heart rate, respiratory rate, blood pressure, and temperature) should be determined within 60 minutes before, during (every 15 [$\pm$5] minutes), and 30 ($\pm$10) minutes after the infusion. For subsequent infusions, vital signs will be collected within 60 minutes before and within 30 minutes after the infusion. Vital signs should be collected during the infusion only if clinically indicated. Patients will be informed about the possibility of delayed post-infusion symptoms and instructed to contact their study physician if they develop such symptoms.

Premedication is not permitted for the first dose of atezolizumab. Premedication with antihistamines or antipyretics/analgesics (*e.g.*, acetaminophen) may be administered for subsequent infusions at the discretion of the treating physician. The management of Infusion Related Reactions will be according to severity as follows:

- In the event that a patient experiences a Grade 1 Infusion Related Reaction during Cycle 1, the infusion rate should be reduced to half the rate being given at the time of event onset. Once the event has resolved, the investigator should wait for 30 minutes while delivering the infusion at the reduced rate. If tolerated, the infusion rate may then be increased to the original rate.

- In the event that a patient experiences a Grade 2 Infusion Related Reaction, or flushing, fever, or throat pain, the infusion should be immediately interrupted and the patient should receive aggressive symptomatic treatment. The infusion should be restarted only after the symptoms have adequately resolved to baseline grade. The infusion rate at restart should be half of the infusion rate that was in progress at the time of the onset of the Infusion Related Reaction. For subsequent infusions, administer oral premedication with antihistamine and anti-pyretic and monitor closely for Infusion Related Reactions.

- For Grade 3 or 4 Infusion Related Reactions, the infusion should be stopped immediately, and aggressive resuscitation and supportive measures should be initiated (*e.g.*, oral or IV antihistamine, anti-pyretic, glucocorticoids, epinephrine, bronchodilators, oxygen). Atezolizumab should be permanently discontinued. Resumption of atezolizumab may be considered in patients who are deriving benefit and have fully recovered from the immune-related event; retreatment requires consultation with, and consent of, the trial Principal Investigator (PI).

For anaphylaxis precautions, use the following procedure:

Equipment Needed

- Tourniquet
- Oxygen
- Epinephrine for subcutaneous, intravenous, and/or endotracheal use in accordance with standard practice
- Antihistamines
- Corticosteroids
- Intravenous infusion solutions, tubing, catheters, and tape

Procedures

In the event of a suspected anaphylactic reaction during atezolizumab infusion, the following procedures should be performed:

1. Stop the study drug infusion.
2. Apply a tourniquet proximal to the injection site to slow systemic absorption of study drug. Do not obstruct arterial flow in the limb.

3. Maintain an adequate airway.
4. Administer antihistamines, epinephrine, or other medications as required by patient status and directed by the physician in charge.
5. Continue to observe the patient and document observation.

5.1.2 SGI-110

The SGI-110 regimen for this study is 30 mg/m^2 or 45 mg/m^2 or 20 mg/m^2 for Dose Level 1, 2 and -1 respectively in the Phase I part and the MTD determined in Phase I for the Phase II part of the study on days 1-5 of each 28 day cycle. Administer SGI-110 by SC injection, preferably in the abdominal area, upper thigh, or arm. Injection site should be rotated for the 5 injections in each cycle. Due to the viscosity of the SGI-110, a 23 gauge, 5/8 inch needle is recommended for administration. SGI-110 should be administered in the clinic by a health professional. The total amount (in mg) of SGI-110 to be administered is determined by body surface area (BSA). In calculating BSA, use actual heights and weights. Do not adjust to "ideal" body weight. Utilize the dose calculated for the baseline BSA throughout the protocol unless weight decreases by 10% or more, at which time it may be adjusted. The institutional standard for calculating BSA is acceptable.

5.1.2.1 Risks/Precautions Associated with SGI-110

Refer to the SGI-110 IB for the most current risks and precautions, as well as a complete list of AEs considered expected with SGI-110 therapy. Injection site reactions, such as pain, irritation, inflammation, erythema, and burning have been reported in the AML/MDS population and in subjects with solid tumors. Injection site reactions are related to SGI-110 SC administration. Care must be taken to avoid intradermal injection. Administer SGI-110 by slow SC injection. If injection site pain is reported upon injection, apply ice packs to the injection site both before and after injection. If injection site AEs are reported at subsequent injections despite slow injection and use of ice packs, pretreatment with topical or systemic analgesics can be considered.

5.2 Definition of Dose-Limiting Toxicity

Toxicity	DLT Criteria
Hematology	CTCAE grade 4 neutropenia lasting more than 7 consecutive days
	CTCAE grade 4 thrombocytopenia
	CTCAE grade 3 thrombocytopenia with bleeding
	CTCAE grade 3 or 4 febrile neutropenia
Cardiac	Cardiac toxicity ≥ CTCAE grade 3 Clinical signs of cardiac disease, such as unstable angina or myocardial infarction
Gastro-intestinal	≥ CTCAE grade 3 vomiting ≥48 hours despite optimal anti-emetic therapy ≥ CTCAE grade 3 diarrhea ≥48 hours despite optimal anti-diarrhea treatment

Hepato-biliary	≥ CTCAE grade 2 total bilirubin for more than 7 consecutive days ≥ CTCAE grade 3 total bilirubin ≥ CTCAE grade 2 ALT with a ≥grade 2 bilirubin elevation of any duration in the absence of liver metastases ≥ CTCAE grade 3 ALT for >4 consecutive days CTCAE grade 4 ALT or AST Grade 4 serum alkaline phosphatase >7 consecutive days
Renal	≥ CTCAE grade ≥3 serum creatinine
Infusion Related Reaction	≥ CTCAE grade 3
Non-hematologic events	≥ CTCAE grade 3, except for the exclusions noted below:
Any immune-related reaction	≥ CTCAE grade 2, except for grade 2 or higher endocrine reactions (thyroid, pituitary, and/or adrenal insufficiency) that is managed with or without systemic corticosteroid therapy and/or hormone replacement therapy and the subject is asymptomatic
Exceptions to DLT criteria	Grade 3 alopecia
	<5 days of CTCAE grade 3 fatigue
	Grade 3 fever or infection without neutropenia <5 days duration
	Any > Grade 3 laboratory abnormalities that are responsive to oral supplementation or IV supplement and resolves to < Grade 1 with 48 hours.
CTCAE version 5.0 should be used for grading. To be considered a DLT, the toxicity must be determined by the investigator to be definitely, probably or possibly **related to the study drugs SGI-110 and/or Atezolizumab**. Optimal therapy for vomiting and diarrhea should be based on institutional guidelines with consideration of the prohibited medications listed in these protocol guidelines.	

*Management and dose modifications associated with the above adverse events are outlined in Section 6.
** Adverse reactions will be reported to FDA according to 21 CFR 312.32.

5.2.1 Grade 3 or 4 Adverse Events that do not meet the DLT criteria or exceptions listed above, may require a treatment interruption at the discretion of the PI.

5.2.2 Persistent treatment related toxicity (as defined in the table above), which delays any cycle 1 atezolizumab dose or Cycle 2 day 1 dose of SGI-110 for > 14 days is a DLT.
Any subject experiencing DLT will be permanently removed from the study, and will not be replaced.

5.2.3 Patients who do not have a DLT and who do not complete one full (100% of doses) 28-day cycle of treatment will be replaced in the study. If a patient is replaced they may continue on treatment per the investigator's discretion. All patients will be included in the analysis results.

5.2.4 Dose escalation will proceed within each cohort according to the following scheme. Dose-limiting toxicity (DLT) is defined above.

Number of Patients with DLT at a Given Dose Level	**Escalation Decision Rule**
0 out of 3	Enter 3 patients at the next dose level.
≥2	Dose escalation will be stopped. This dose level will be declared the maximally administered dose (highest dose administered). Three (3) additional patients will be entered at the next lowest dose level if only 3 patients were treated previously at that dose.
1 out of 3	Enter at least 3 more patients at this dose level. • If 0 of these 3 patients experience DLT, proceed to the next dose level. • If 1 or more of this group suffer DLT, then dose escalation is stopped, and this dose is declared the maximally administered dose. Three (3) additional patients will be entered at the next lowest dose level if only 3 patients were treated previously at that dose.
≤1 out of 6 at highest dose level below the maximally administered dose	This is generally the recommended phase 2 dose. At least 6 patients must be entered at the recommended phase 2 dose.

5.2.5 Dose Expansion

Once the RP2D is reached, patients will continue to be monitored closely during the phase 2 portion of the study. If severe side effects are noted, the Principal Investigator will discuss with all study investigators and with CTEP whether further addition of patients is needed to re-assess the RP2D. Monitoring of all safety and toxicity data is done by the Principal Investigator and the Corresponding Organization in a timely manner as data are entered into Medidata Rave using the Web Reporting Module. All participating sites are expected to notify the Principal Investigator when a DLT has occurred.

5.2.6 Study Stopping Rules:

Any investigational drug related Grade 4 DLT event will trigger a study pause for data review prior to further enrollment. In addition, the following stopping rule for treatment-related SAEs will be used:

- Accrual to study will be suspended to allow protocol review with institutional IRB, the Phase I Committee, and the FDA review team if any of the following occurs among patients who receive experimental agents:

- Any treatment-related death, judged to be possibly, probably, or definitely related to atezolizumab or SGI-110.
- Grade 4 events considered to be possibly, probably, or definitely related to the experimental agents.

5.2.7 Phase II DLT Monitoring Rules

We will utilize a sequential probability ratio test to monitor for higher than expected grade 3, 4, or 5 drug related AEs in the cohort 2 safety data. The null rate of severe and drug related AEs based on past studies will be set at 15% and the alternative rate will be set at 30%. We are setting the alpha=Type I error rate to be 0.3 and the beta=Type II error rate to be 0.1. The stopping boundary for rejecting the null rate will be (1-beta)/alpha=3. If the likelihood ratio statistic (LR) of the alternative likelihood over the null likelihood, evaluated after each subject has been evaluated, crosses the stopping boundary at any given point in time enrollment will be suspended. If in fact the true severe AE rate is 30%, we would require an average of 15.4 cohort 2 subjects to suspend accrual with the probability of stopping after 6 subjects have been enrolled in cohort 2 to be 0.32. As alternative way of understanding this decision rule with a true severe AE rate of 30% consider the scenario where the sequence in the first 3 subjects was no AE, AE, AE. Then the decision rule would yield LR values of 0.8, 1.6, 3.3, after 3 subjects, which would cross the boundary of 3 alerting us that the rate is statistically higher than expected. If we had a scenario for the first 5 subjects of AE, no AE, no AE, AE, AE then the decision rule would yield LR values of 2.0, 1.6, 1.3, 2.7 and 5.4 after 5 subjects, which would cross the boundary of 3 alerting us that the rate is statistically higher than expected.

If the true severe AE rate was higher at 40%, we would require an average of 8.6 cohort 2 subjects to suspend accrual with the probability of stopping after 6 subjects have been enrolled in cohort 2 to be 0.51. If the true severe AE rate was more extreme at 60% we would require an average of 4.0 cohort 2 subjects to suspend accrual with the probability of stopping after 6 subjects have been enrolled in cohort to be 0.85. The trial enrollment will not be stopped in order to assess safety in a continuous fashion. Alternatively if the LR crossed the boundary beta/(1-alpha) there would be strong evidence that the AE rate in the combination therapy was in line with giving Atezolizumab alone.

Management and dose modifications associated with the above adverse events are outlined in Section 6.
Adverse reactions will be reported to FDA according to 21 CFR 312.32.

5.3 General Concomitant Medication and Supportive Care Guidelines

Because there is a potential for interaction of Atezolizumab and SGI-110 with other concomitantly administered drugs, the case report form must capture the concurrent use of all other drugs, over-the-counter medications, or alternative therapies. The Principal Investigator should be alerted if the patient is taking any agent known to affect or with the potential for drug interactions.

5.3.1 **Atezolizumab General Concomitant Medication Guidelines**

Concomitant therapy includes any prescription medications or over the counter preparations used by a patient between the 7 days preceding the screening evaluation and the treatment discontinuation visit.

Patients who experience infusion-associated symptoms may be treated symptomatically with acetaminophen, ibuprofen, diphenhydramine, and/or cimetidine or another H2 receptor antagonist, as per standard practice (for sites outside the United States, equivalent medications may be substituted per local practice). Serious infusion associated events manifested by dyspnea, hypotension, wheezing, bronchospasm, tachycardia, reduced oxygen saturation, or respiratory distress should be managed with supportive therapies as clinically indicated (*e.g.*, supplemental oxygen and β_2-adrenergic agonists; see **Section 5.1.1**).

Systemic corticosteroids and TNFα inhibitors may attenuate potential beneficial immunologic effects of treatment with atezolizumab but may be administered at the discretion of the treating physician. If feasible, alternatives to corticosteroids should be considered. Premedication may be administered for Cycles $\geq$2 at the discretion of the treating physician. The use of inhaled corticosteroids and mineralocorticoids (*e.g.*, fludrocortisone) for patients with orthostatic hypotension or adrenocortical insufficiency is allowed. Megestrol administered as appetite stimulant is acceptable while the patient is enrolled in the study.

Patients who use oral contraceptives, hormone-replacement therapy, prophylactic or therapeutic anticoagulation therapy (such as low-molecular-weight heparin or warfarin at a stable dose level), or other allowed maintenance therapy should continue their use. Females of reproductive potential should use highly effective means of contraception.

5.3.2 **Atezolizumab Excluded Therapies**

Any concomitant therapy intended for the treatment of cancer, whether health authority-approved or experimental, is prohibited. This includes but is not limited to the following:

- Chemotherapy, hormonal therapy, immunotherapy, radiotherapy, investigational agents, or herbal therapy (except for maintenance therapies outlined in **Section 5.3.1**).
- Treatment for progressive pain will be considered as evidence of disease progression and will necessitate removal from study.

It is strongly recommended that:

- Traditional herbal medicines not be administered because the ingredients of many herbal medicines are not fully studied and their use may result in unanticipated drug-drug interactions that may cause, or confound assessment of, toxicity.
- The use of a RANKL inhibitor (denosumab) be discontinued during the study; this agent could potentially alter the activity and the safety of atezolizumab.

Patients are not allowed to receive immunostimulatory agents, including, but not limited to, IFN-α, IFN-γ, or IL-2, during the entire study. These agents, in combination with atezolizumab, could potentially increase the risk for autoimmune conditions.

Patients should also not be receiving immunosuppressive medications, including, but not limited to, cyclophosphamide, azathioprine, methotrexate, and thalidomide. These agents could potentially alter the activity and the safety of atezolizumab.

In addition, all patients (including those who discontinue the study early) should not receive other immunostimulatory agents for 10 weeks after the last dose of atezolizumab.

5.3.3 SGI-110 General Concomitant Medication Guidelines

There have been to date no studies of drug interactions for SGI-110. Overall, the rapid clearance from plasma, lack of CYP-inhibition, and the metabolism of the drug both by hydrolysis and the active metabolite as well as by cytidine deaminase suggest that, like Dac and Aza, SGI-110 is unlikely to have significant drug interactions. Furthermore, guadecitabine nor the active metabolite decitabine inhibit major human drug transporters.

The investigator is permitted to prescribe supportive treatment(s) at his or her discretion. Appropriate hydration and supportive care, including blood and platelet transfusions, may be administered according to study-center standards. Antibiotics and/or anti-fungals may be used to manage febrile neutropenia based on institutional standard practice. Use is permitted if deemed to be medically necessary by the treating physician and should be guided by accepted practice or institutional guidelines.

Cytotoxic chemotherapy and investigational treatments other than the combination from this study are prohibited for as long as subjects remain on study treatment. Vaccination with live vaccines is prohibited while subjects remain on study treatment.

5.4 Duration of Therapy

In the absence of treatment delays due to adverse event(s), treatment may continue for up to two years (6 cycles of SGI-110 and 24 cycles of Atezolizumab) or until one of the following criteria applies:

- Disease progression (confirmation of PD is required by irRECIST)
- Intercurrent illness that prevents further administration of treatment,
- Unacceptable adverse event(s),
- Patient decides to withdraw from the study, or

- General or specific changes in the patient's condition render the patient unacceptable for further treatment in the judgment of the investigator.
- Patient Death

5.5 Duration of Follow Up

Patients will be followed at 30 days after the last dose of investigational drug and then long term follow-up every 2 months for up to 1 year, unless the consent is withdrawn. If a disease progression or change in therapy is documented, the patient will continue to be monitored for survival up to 1 year after the last dose of investigational drug.

5.6 Criteria for Removal from Study

Patients will be removed from study at one year after the last dose of treatment, if/when the patient withdraws consent or at the time of death, whichever comes first. The reason for study removal and the date the patient was removed must be documented in the Case Report Form. However, patients who have to discontinue treatment due to PD or AE may be followed up at defined schedule as specified for patient on study.

6. DOSING DELAYS/DOSE MODIFICATIONS

During the DLT period, no dose modifications are allowed. Refer to section 5.2 regarding dosing delays. The following modifications will apply after the DLT period (1st cycle) in phase I and for all cycles in phase 2. If one agent is stopped/delayed due to suspected toxicity, all agents will be stopped/delayed.

6.1 Atezolizumab (MPDL3280A)

6.1.1 General AE Management and Dose Modification Guidelines

There will be no dose reduction for atezolizumab in this study.

Patients may temporarily suspend study treatment for up to 84 days (12 weeks) beyond the scheduled date of delayed infusion if study drug-related toxicity requiring dose suspension is experienced. If atezolizumab is held because of AEs for >84 days beyond the scheduled date of infusion, the patient will be discontinued from atezolizumab and will be followed for safety and efficacy as specified in this protocol. If the AE resolves within 84 days and the patient is receiving corticosteroid therapy for the event, atezolizumab may be held for longer than 84 days (up to 4 weeks) in order to allow tapering of the steroid dose to ≤10 mg oral prednisone or equivalent.

Dose interruptions for reasons other than toxicity, such as surgical procedures, may be allowed. The acceptable length of interruption will be at the discretion of the study PI in consultation with CTEP.

Atezolizumab must be **permanently discontinued** if the patient experiences any of the following events, regardless of benefit:

- Grade 4 pneumonitis
- AST or ALT >5×ULN or total bilirubin >3×ULN
- Grade 4 diarrhea or colitis
- Grade 4 hypophysitis
- Any grade myasthenic syndrome/myasthenia gravis, Guillain-Barré or meningoencephalitis
- Grade 4 ocular inflammatory toxicity
- Grade 4 pancreatitis or any grade of recurrent pancreatitis
- Grade 4 rash
- Any grade myocarditis

Treatment may, under limited and compelling circumstances, be resumed in patients who have recovered from the following events, <u>but only after consultation with the trial Principal Investigator</u>:

- Grade 3 pneumonitis
- Grade 3 ocular inflammatory toxicity
- Grade 3 or 4 infusion-related reactions (atezolizumab should only be resumed if the infusion-reaction occurred in a subject who was not pre-medicated.)

Any toxicities associated or possibly associated with atezolizumab treatment should be managed according to standard medical practice. Additional tests, such as autoimmune serology or biopsies, may be used to determine a possible immunogenic etiology. Although most immune-related adverse events (irAEs) observed with immunomodulatory agents have been mild and self-limiting, such events should be recognized early and treated promptly to avoid potential major complications (Di Giacomo *et al.*, 2010). Discontinuation of atezolizumab may not have an immediate therapeutic effect, and there is no available antidote for atezolizumab. In severe cases, immune-related toxicities may be acutely managed with topical corticosteroids, systemic corticosteroids, or other immunosuppressive agents. The investigator should consider the benefit-risk balance prior to further administration of atezolizumab.

For detailed information regarding management of adverse events associated with atezolizumab, please refer to the most current version of the Atezolizumab Investigator's Brochure and the FDA product label.

The primary approach to grade 1 to 2 irAEs is supportive and symptomatic care with continued treatment with atezolizumab; for higher-grade irAEs, atezolizumab should be withheld and oral and/or parenteral steroids administered. Recurrent grade 2 irAEs may also mandate withholding atezolizumab or the use of steroids. Assessment of the benefit-risk balance should be made by the investigator, with consideration of the totality of information as it pertains to the nature of the toxicity and the degree of clinical benefit a given patient may be experiencing prior to further administration of atezolizumab. Atezolizumab should be permanently discontinued in patients with life-threatening irAEs.

Patients should be assessed clinically (including review of laboratory values) for toxicity prior to, during, and after each infusion. If unmanageable toxicity due to atezolizumab occurs at any time during the study, treatment with atezolizumab should be discontinued.

Systemic Immune Activation

Systemic immune activation is a rare condition characterized by an excessive immune response. Given the mechanism of action of atezolizumab, systemic immune activation is considered a potential risk when given in combination with other immunomodulating agents. Systemic immune activation should be included in the differential diagnosis for patients who, in the absence of an alternative etiology, develop a sepsis-like syndrome after administration of atezolizumab, and the initial evaluation should include the following:

- CBC with peripheral smear
- PT, PTT, fibrinogen, and D-dimer
- Ferritin
- Triglycerides
- AST, ALT, and total bilirubin
- LDH
- Complete neurologic and abdominal examination (assess for hepatosplenomegaly)

If systemic immune activation is still suspected after the initial evaluation, contact the Principal Investigator for additional recommendations.

6.1.2 Management of Specific AEs

Management of certain AEs of concern, including immune-related pneumonitis, hepatitis, colitis, endocrinopathies, pancreatitis, neuropathies, meningoencephalitis, and potential ocular toxicities are presented in the Atezolizumab Investigator's Brochure. See **Section 5.1.1** for guidelines for the management of Infusion Related Reactions and Anaphylaxis.

Pulmonary events

Dyspnea, cough, fatigue, hypoxia, pneumonitis, and pulmonary infiltrates have been associated with the administration of atezolizumab. Patients will be assessed for pulmonary signs and symptoms throughout the study and will also have CT scans of the chest performed at every tumor assessment.

All pulmonary events should be thoroughly evaluated for other commonly reported etiologies such as pneumonia or other infection, lymphangitic carcinomatosis, pulmonary embolism, heart failure, chronic obstructive pulmonary disease, or pulmonary hypertension. Management guidelines for pulmonary events are provided in the table below.

Endocrine disorders

Patients experiencing one or more unexplained AEs possibly indicative of endocrine dysfunction (including fatigue, myalgias, impotence, mental status changes, and constipation) should be investigated for the presence of thyroid, pituitary, or adrenal endocrinopathies. The patient should be referred to an endocrinologist if an endocrinopathy is suspected. Thyroid-stimulating hormone (TSH) and free T3 and T4 levels should be obtained to determine whether thyroid abnormalities are present. TSH, prolactin, and a morning cortisol level will help to differentiate primary adrenal insufficiency from primary pituitary insufficiency. The table below describes dose management guidelines for hyperthyroidism, hypothyroidism, symptomatic adrenal insufficiency, and hyperglycemia.

Meningoencephalitis

Immune-related meningoencephalitis is an identified risk associated with the administration of atezolizumab. Immune-related meningoencephalitis should be suspected in any patient presenting with signs or symptoms suggestive of meningitis or encephalitis, including, but not limited to, headache, neck pain, confusion, seizure, motor or sensory dysfunction, and altered or depressed level of consciousness. Encephalopathy from metabolic or electrolyte imbalances needs to be distinguished from potential meningoencephalitis resulting from infection (bacterial, viral, or fungal) or progression of malignancy, or secondary to a paraneoplastic process.

All patients being considered for meningoencephalitis should be urgently evaluated with a CT scan and/or MRI scan of the brain to evaluate for metastasis, inflammation, or edema. If deemed safe by the treating physician, a lumbar puncture should be performed and a neurologist should be consulted.

Patients with signs and symptoms of meningoencephalitis, in the absence of an identified alternate etiology, should be treated according to the guidelines.

Neurologic disorders

Myasthenia gravis and Guillain-Barré syndrome have been observed with single-agent atezolizumab. Patients may present with signs and symptoms of sensory and/or motor neuropathy. Diagnostic work-up is essential for an accurate characterization to differentiate between alternative etiologies.

The table below presents management and dose modification guidelines for specific AEs.

For recommendations to hold atezolizumab and begin corticosteroid treatment, use the following guidance for tapering the corticosteroid and resuming atezolizumab therapy after resolution of the event:

- Corticosteroids must be tapered over ≥1 month to ≤10 mg/day oral prednisone or equivalent before atezolizumab can be resumed.
- Atezolizumab may be held for a period of time beyond 12 weeks to allow for corticosteroids to be reduced to ≤10 mg/day oral prednisone or equivalent.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
Abdominal pain	Acute abdominal pain	Symptoms of abdominal pain associated with elevations of amylase and lipase, suggestive of pancreatitis, have been associated with administration of other immunomodulatory agents. The differential diagnosis of acute abdominal pain should include pancreatitis. Appropriate workup should include an evaluation for obstruction, as well as serum amylase and lipase tests. See the guidelines for "Amylase and/or lipase increase" and "Immune-related pancreatitis" elsewhere in this table, as needed. Right upper-quadrant abdominal pain and/or unexplained nausea or vomiting should be evaluated for potential hepatotoxicity (see the "Hepatotoxicity" guideline elsewhere in this table).

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
Adrenal insufficiency	Grade 2+ (symptomatic)	Hold atezolizumab. Refer patient to endocrinologist. Perform appropriate imaging. Initiate treatment with 1–2 mg/kg/day intravenous methylprednisolone or equivalent and convert to 1–2 mg/kg/day oral prednisone or equivalent upon improvement. If event resolves to grade 1 or better and patient is stable on replacement therapy (if required) within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better or patient is not stable on replacement therapy within 12 weeks.
Amylase and/or lipase increased	Grade 1	Continue atezolizumab. Monitor amylase and lipase prior to dosing.
	Grade 2	Continue atezolizumab. Monitor amylase and lipase weekly. For prolonged elevation (*e.g.*, >3 weeks), consider treatment with 10 mg/day oral prednisone or equivalent.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 3 or 4	Hold atezolizumab. Refer patient to gastrointestinal (GI) specialist. Monitor amylase and lipase every other day. If no improvement, consider treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks. For recurrent events, permanently discontinue atezolizumab. [c]
Dermatologic toxicity/rash (*e.g.*, maculopapular or purpura)	Grade 1	Continue atezolizumab. Consider topical steroids and/or other symptomatic therapy (*e.g.*, antihistamines).
	Grade 2	Continue atezolizumab. Consider dermatologist referral. Administer topical corticosteroids. Consider higher potency topical corticosteroids if event does not improve.
	Grade 3	Hold atezolizumab. Refer patient to dermatologist. Administer oral prednisone 10 mg or equivalent. If the event does not improve within 48–72 hours, increase dose to1–2 mg/kg/day or equivalent. Restart atezolizumab if event resolves to grade 1 or better within 12 weeks. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 4	Permanently discontinue atezolizumab. Patient may not resume treatment, regardless of benefit. Otherwise, manage as above.
	Persistent and/or severe rash or pruritus, any grade	A dermatologist should evaluate the event. A biopsy should be performed unless contraindicated.
Diarrhea or colitis	Any grade	Patients should be advised to inform the investigator if any diarrhea occurs, even if it is mild. All events of diarrhea or colitis should be thoroughly evaluated for other more common etiologies. For events of significant duration or magnitude or associated with signs of systemic inflammation or acute-phase reactants (*e.g.*, increased CRP, platelet count, or bandemia): Perform sigmoidoscopy (or colonoscopy, if appropriate) with colonic biopsy, with three to five specimens for standard paraffin block to check for inflammation and lymphocytic infiltrates to confirm colitis diagnosis.
	Grade 1	Continue atezolizumab. Initiate symptomatic treatment. Endoscopy is recommended if symptoms persist for >7 days. Monitor closely.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 2	Hold atezolizumab. Initiate symptomatic treatment. Patient referral to GI specialist is recommended. For recurrent events or events that persist >5 days, initiate treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks. Resumption of atezolizumab may be considered, after consultation with the trial PI, in patients who are deriving benefit and have fully recovered from the immune-related event.
	Grade 3	Hold atezolizumab. Refer patient to GI specialist for evaluation and confirmatory biopsy. Initiate treatment with 1–2 mg/kg/day intravenous methylprednisolone or equivalent and convert to 1–2 mg/kg/day oral prednisone or equivalent upon improvement. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to Grade 1 or better within 12 weeks. Resumption of atezolizumab may be considered, after consultation with the trial PI, in patients who are deriving benefit and have fully recovered from the immune-related event.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 4	Permanently discontinue atezolizumab. <u>Patient may not resume treatment, regardless of benefit.</u> Refer patient to GI specialist for evaluation and confirmation biopsy. Initiate treatment with 1–2 mg/kg/day intravenous methylprednisolone or equivalent and convert to 1–2 mg/kg/day oral prednisone or equivalent upon improvement. If event does not improve within 48 hours after initiating corticosteroids, consider adding an immunosuppressive agent. If event resolves to grade 1 or better, taper corticosteroids over ≥1 month.
Hepatotoxicity	Right upper-quadrant abdominal pain and/or unexplained nausea or vomiting	Risk of immune-mediated hepatitis. LFTs should be performed immediately, and LFTs should be reviewed before administration of the next dose of study drug. For patients with elevated LFTs, concurrent medication, viral hepatitis, and toxic or neoplastic etiologies should be considered and addressed, as appropriate. Symptoms of abdominal pain associated with elevations of amylase and lipase, suggestive of pancreatitis, have been associated with the administration of atezolizumab. The differential diagnosis of acute abdominal pain should also include pancreatitis, as described below.
	Grade 1 hepatic event	Continue atezolizumab. Monitor LFTs until values resolve to within normal limits.
	Grade 2 hepatic event, ≤5 days	Continue atezolizumab. Monitor LFTs more frequently until values resolve to baseline values.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 2 hepatic event, >5 days	Hold atezolizumab. Initiate treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to Grade 1 or better within 12 weeks.
	Grade 3 or 4 hepatic event	Permanently discontinue atezolizumab. Consider patient referral to GI specialist for evaluation and liver biopsy to establish etiology of hepatic injury. Initiate treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event does not improve within 48 hours after initiating corticosteroids, consider adding an immunosuppressive agent. If event resolves to grade 1 or better, taper corticosteroids over ≥1 month.
Hyperglycemia	Grade 1 or 2	Continue atezolizumab. Initiate treatment with insulin if needed. Monitor for glucose control.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 3 or 4	Hold atezolizumab. Initiate treatment with insulin. Monitor for glucose control. Resume atezolizumab when symptoms resolve and glucose levels are stable.
Hyperthyroidism	Grade 1 (asymptomatic)	**TSH ≥ 0.1 mU/L and < 0.5 mU/L:** Continue atezolizumab. Monitor TSH every 4 weeks. **TSH < 0.1 mU/L:** Follow guidelines for symptomatic hyperthyroidism.
	Grade 2+ (symptomatic)	Hold atezolizumab. Initiate treatment with anti-thyroid drug such as methimazole or carbimazole as needed. Consider patient referral to endocrinologist. Resume atezolizumab when symptoms are controlled and thyroid function is improving. Permanently discontinue atezolizumab for life-threatening immune-related hyperthyroidism.
Hypothyroidism	Grade 1 (asymptomatic)	Continue atezolizumab. Start thyroid-replacement hormone. Monitor TSH weekly.
	Grades 2+ (symptomatic)	Hold atezolizumab. Start thyroid-replacement hormone. Consider referral to an endocrinologist. Monitor TSH weekly. Restart atezolizumab when symptoms are controlled and thyroid function is improving.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
Meningo-encepahlitis, immune-related (signs and symptoms in absence of an identified alternate etiology)	All grades	Permanently discontinue atezolizumab. Patient may not resume treatment, regardless of benefit. Refer patient to neurologist. Initiate treatment with 1−2 mg/kg/day IV methylprednisolone or equivalent and convert to 1−2 mg/kg/day oral prednisone or equivalent upon improvement. If event resolves to grade 1 or better, taper corticosteroids over ≥1 month. If event does not improve within 48 hours after initiating corticosteroids, consider adding an immunosuppressive agent.
Myasthenia gravis and Guillain-Barré syndrome	All grades	Permanently discontinue atezolizumab. Patient may not resume treatment, regardless of benefit. Refer patient to neurologist. Initiate treatment as per institutional guidelines. Consider initiation of 1−2 mg/kg/day oral or IV prednisone or equivalent.
Myocarditis	All grades	Permanently discontinue atezolizumab. Patient may not resume treatment, regardless of benefit.
Neuropathy, immune-related	Grade 1	Continue atezolizumab. Evaluate for alternative etiologies.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
(sensory and/or motor)	Grade 2	Hold atezolizumab. Evaluate for alternative etiologies. Initiate treatment as per institutional guidelines. Resume atezolizumab if event resolves to grade 1 or better within 12 weeks. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks.
	Grade 3 or 4	Permanently discontinue atezolizumab. Initiate treatment as per institutional guidelines.
Ocular event (*e.g.*, uveitis, retinal events)	Grade 1	Continue atezolizumab. Patient referral to ophthalmologist is strongly recommended. Initiate treatment with topical corticosteroid eye drops and topical immunosuppressive therapy. If symptoms persist, treat as a grade 2 event.
	Grade 2	Withhold atezolizumab. Patient referral to ophthalmologist is strongly recommended. Initiate treatment with topical corticosteroid eye drops and topical immunosuppressive therapy. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 3 or 4	Permanently discontinue atezolizumab. Refer patient to ophthalmologist. Initiate treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event resolves to grade 1 or better, taper corticosteroids over ≥1 month. For grade 3 AEs, patient may only resume treatment after consultation with the trial PI; for grade 4, patient cannot resume treatment, regardless of benefit.
Pancreatitis, immune related	Grade 2 or 3	Hold atezolizumab. Refer patient to GI specialist. Initiate treatment with 1–2 mg/kg/day intravenous methylprednisolone or equivalent and convert to 1–2 mg/kg/day oral prednisone or equivalent upon improvement. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks. Patient may only resume treatment after consultation with the trial PI. For recurrent events, permanently discontinue atezolizumab. <u>Patient may not resume treatment, regardless of benefit.</u>

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 4	Permanently discontinue atezolizumab. <u>Patient may not resume treatment, regardless of benefit.</u> Refer patient to GI specialist. Initiate treatment with 1–2 mg/kg/day intravenous methylprednisolone or equivalent and convert to 1–2 mg/kg/day oral prednisone or equivalent upon improvement. If event does not improve within 48 hours after initiating corticosteroids, consider adding an immunosuppressive agent. If event resolves to grade 1 or better, taper corticosteroids over ≥1 month.
Pulmonary toxicity	All pulmonary events	Evaluate thoroughly for other commonly reported etiologies such as pneumonia/infection, lymphangitic carcinomatosis, pulmonary embolism, heart failure, chronic obstructive pulmonary disease (COPD), or pulmonary hypertension.
	Grade 1	Continue atezolizumab and monitor closely. Re-evaluate on serial imaging. Consider patient referral to a pulmonary specialist. For recurrent pneumonitis, treat as a grade 3 or 4 event.

AE Management and Dose Interruption Guidelines for Specific Toxicities		
Toxicity	**Severity/ Duration**	**Management**
	Grade 2	Hold atezolizumab. Refer patient to pulmonary and infectious disease specialists and consider bronchoscopy or bronchoscopic alveolar lavage (BAL). Initiate treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event resolves to grade 1 or better within 12 weeks, taper corticosteroids and resume atezolizumab according to the guidelines above. Permanently discontinue atezolizumab if event does not resolve to grade 1 or better within 12 weeks. For recurrent events, treat as a Grade 3 or 4 event.
	Grade 3 or 4	Permanently discontinue atezolizumab. Bronchoscopy or BAL is recommended. Initiate treatment with 1–2 mg/kg/day oral prednisone or equivalent. If event does not improve within 48 hours after initiating corticosteroids, consider adding an immunosuppressive agent. If event resolves to grade 1 or better, taper corticosteroids over ≥1 month. For grade 3 AEs, patient may only resume treatment after consultation with the trial PI; for grade 4, patient cannot resume treatment, regardless of benefit.

6.2 SGI-110

SGI-110 study therapy is intended to be administered for a total of 6 cycles (each cycle being 28 days). No dose modification should be made for the first 2 cycles to ensure initial dose intensity.

Timing of cycle initiation and dose level of SGI-110 starting Cycle 3 will be guided by peripheral blood (PB) neutrophil/platelet counts after the prior cycle, as indicated in the Table below. If Absolute Neutrophil Count (ANC) is ≥ 1500/mcL and platelets are ≥ 75,000/mcL at the start of the cycle, then full dose of SGI-110 should be administered as scheduled.

Patients will be withdrawn from the study if they fail to recover to ANC ≥ 1500/mcL and platelets ≥ 75,000/mcL, from a treatment-related toxicity within 4 weeks of holding treatment. For patients that recover in < 4 weeks, SGI-110 may be restarted as per table below.

If SGI-110 administration is delayed due to count recovery, reinitiation of therapy should be timed to lead into the next dose of atezolizumab similarly to the initial schedule. (Days 1-5 with the atezolizumab dose on Day 8 or Days 15-19 with atezolizumab on Day 22.)

SGI-110 Dosing Adjustment Guideline Based on Peripheral Blood Counts (starting with Cycle 3)

Neutrophils Platelets Threshold	Recovery within <2 weeks	Recovery in 2-4 weeks
ANC < 1500/mcL or platelets <75,000/mcL	Delay SGI-110 until ANC ≥ 1500/mcL and platelets ≥ 75,000/mcL, then administer full dose.	Delay study drug until ANC ≥ 1500/mcL and platelets ≥ 75,000/mcL, then reduce 1 dose level.

*If ANC is ≥ 1500/mcL and platelets are ≥ 75,000/mcL at the start of the cycle, administer full dose on schedule.
**Due to known myelosuppression of SGI-110, all patients receiving SGI-110 will receive G-CSF prophylactically, filgrastim (neupogen) or pegfilgrastim (neulasta) or biosimilar. The recommended dose of neupogen is 5 mcg/kg subcutaneously per day, rounding off the dose to the nearest vial size, until post-nadir ANC recovery to normal or near normal levels (≥ 1,500/mcL). Start filgrastim the next day, no later than 3 days after completion of SGI110, and treat through post-nadir recovery. The recommended dose of pegfilgrastim is 6 mg subcutaneously once per cycle, to be given within 24 hours after completing SGI110 or as per institutional guidelines".

Dose Level	***SGI-110*** **Dose**
1	30 mg/m^2, Day 1-5 every 28 days for 6 cycles
2	45 mg/m^2, Day 1-5 every 28 days for 6 cycles
-1	20 mg/m^2, Day 1-5 every 28 days for 6 cycles

7. ADVERSE EVENTS: LIST AND REPORTING REQUIREMENTS

Adverse event (AE) monitoring and reporting is a routine part of every clinical trial. The following list of AEs (Section 7.1) and the characteristics of an observed AE (Section 7.2) will determine whether the event requires expedited reporting via the CTEP Adverse Event Reporting System (CTEP-AERS) **in addition** to routine reporting.

7.1 Comprehensive Adverse Events and Potential Risks List(s) (CAEPRs)

The Comprehensive Adverse Event and Potential Risks list (CAEPR) provides a single list of reported and/or potential adverse events (AE) associated with an agent using a uniform presentation of events by body system. In addition to the comprehensive list, a subset, the Specific Protocol Exceptions to Expedited Reporting (SPEER), appears in a separate column and is identified with ***bold*** and ***italicized*** text. This subset of AEs (SPEER) is a list of events that are protocol-specific exceptions to expedited reporting to NCI (except as noted below). Refer to the 'CTEP, NCI Guidelines: Adverse Event Reporting Requirements' http://ctep.cancer.gov/protocolDevelopment/electronic_applications/docs/aeguidelines.pdf for further clarification.

NOTE: Report AEs on the SPEER **<u>ONLY IF</u>** they exceed the grade noted in parentheses next to the AE in the SPEER. If this CAEPR is part of a combination protocol using multiple investigational agents and has an AE listed on different SPEERs, use the lower of the grades to determine if expedited reporting is required.

7.1.1 CAEPRs for CTEP IND Agent(s)

7.1.1.1 CAEPR for Atezolizumab

Frequency is provided based on 3097 patients. Below is the CAEPR for Atezolizumab (MPDL3280A). **Version 2.2, July 23, 2018**[1]

Adverse Events with Possible Relationship to Atezolizumab (MPDL3280A) (CTCAE 5.0 Term) [n= 3097]			Specific Protocol Exceptions to Expedited Reporting (SPEER)
Likely (>20%)	**Less Likely (<=20%)**	**Rare but Serious (<3%)**	
BLOOD AND LYMPHATIC SYSTEM DISORDERS			
	Anemia		
CARDIAC DISORDERS			
		Heart failure(79)	
		Myocarditis(79)	
		Pericardial effusion(79)	
		Pericardial tamponade(79)	
		Pericarditis(79)	
ENDOCRINE DISORDERS			
		Adrenal insufficiency(79)	
		Endocrine disorders - Other (diabetes)(79)	
	Hyperthyroidism(79)		
		Hypophysitis(79)	
	Hypothyroidism(79)		
EYE DISORDERS			

Adverse Events with Possible Relationship to Atezolizumab (MPDL3280A) (CTCAE 5.0 Term) [n= 3097]			Specific Protocol Exceptions to Expedited Reporting (SPEER)
Likely (>20%)	**Less Likely (<=20%)**	**Rare but Serious (<3%)**	
		Eye disorders - Other (ocular inflammatory toxicity)(79)	
		Uveitis(79)	
GASTROINTESTINAL DISORDERS			
	Abdominal pain		***Abdominal pain (Gr 2)***
		Colitis(79)	
	Diarrhea		***Diarrhea (Gr 2)***
	Dysphagia		
	Nausea		***Nausea (Gr 2)***
		Pancreatitis(79)	
	Vomiting		***Vomiting (Gr 2)***
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS			
Fatigue			***Fatigue (Gr 2)***
	Fever[3]		
	Flu like symptoms[3]		
HEPATOBILIARY DISORDERS			
		Hepatic failure(79)	
		Hepatobiliary disorders - Other (hepatitis)(79)	
IMMUNE SYSTEM DISORDERS			
	Allergic reaction[3]		
		Anaphylaxis[3]	
		Cytokine release syndrome[3]	
		Immune system disorders - Other (systemic immune activation)(79)	
INFECTIONS AND INFESTATIONS			
Infection[4]			
INJURY, POISONING AND PROCEDURAL COMPLICATIONS			
	Infusion related reaction[3]		
INVESTIGATIONS			
	Alanine aminotransferase increased(79)		
	Alkaline phosphatase increased(79)		
	Aspartate aminotransferase increased(79)		
	Blood bilirubin increased(79)		
	GGT increased(79)		
	Lipase increased*		
		Platelet count decreased	
	Serum amylase increased*		
METABOLISM AND NUTRITION DISORDERS			
	Anorexia		***Anorexia (Gr 2)***
		Hyperglycemia(79)	

Adverse Events with Possible Relationship to Atezolizumab (MPDL3280A) (CTCAE 5.0 Term) [n= 3097]			Specific Protocol Exceptions to Expedited Reporting (SPEER)
Likely (>20%)	**Less Likely (<=20%)**	**Rare but Serious (<3%)**	
	Hypokalemia		
	Hyponatremia		
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS			
	Arthralgia(79)		
	Back pain		
		Generalized muscle weakness	
	Myalgia		
		Myositis(79)	
NERVOUS SYSTEM DISORDERS			
		Ataxia(79)	
		Encephalopathy(79)	
		Nervous system disorders - Other (encephalitis non-infective)(79) Guillain-Barre syndrome(79)	
		Nervous system disorders - Other (meningitis non-infective)(79)	
		Myasthenia gravis(79)	
		Paresthesia(79)	
		Peripheral motor neuropathy(79)	
		Peripheral sensory neuropathy(79)	
RENAL AND URINARY DISORDERS			
		Renal and urinary disorders - Other (nephritis)(79)	
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS			
	Cough		*Cough (Gr 2)*
	Dyspnea		
	Hypoxia		
	Nasal congestion		*Nasal congestion (Gr 2)*
		Pleural effusion(79)	
		Pneumonitis[2]	
SKIN AND SUBCUTANEOUS TISSUE DISORDERS			
		Bullous dermatitis(79)	
	Pruritus Rash acneiform		
	Rash maculo-papular		
	Skin and subcutaneous tissue disorders - Other (lichen planus)(79)		

Denotes adverse events that are <3%.

[1]This table will be updated as the toxicity profile of the agent is revised. Updates will be distributed to all Principal Investigators at the time of revision. The current version can be obtained by contacting PIO@CTEP.NCI.NIH.GOV. Your name, the name of the investigator, the protocol and the agent should be included in the e-mail.

[2]Atezolizumab, being a member of a class of agents involved in the inhibition of "immune checkpoints," may result in severe and possibly fatal immune-mediated adverse events probably due to T-cell activation and proliferation. Immune-mediated adverse reactions have been reported in patients receiving atezolizumab. Adverse events potentially related to atezolizumab may be manifestations of immune-mediated adverse events. In clinical trials, most immune-mediated adverse reactions were reversible and managed with interruptions of atezolizumab, administration of corticosteroids and supportive care.

[3]Infusion reactions, including high-grade hypersensitivity reactions, anaphylaxis, and cytokine release syndrome, which have been observed following administration of atezolizumab, may manifest as fever, chills, shakes, itching, rash, hypertension or hypotension, or difficulty breathing during and immediately after administration of atezolizumab.

[4]Infection includes all 75 sites of infection under the INFECTIONS AND INFESTATIONS SOC.

Adverse events reported on atezolizumab (MPDL3280A) trials, but for which there is insufficient evidence to suggest that there was a reasonable possibility that atezolizumab (MPDL3280A) caused the adverse event:

BLOOD AND LYMPHATIC SYSTEM DISORDERS - Blood and lymphatic system disorders - Other (pancytopenia); Febrile neutropenia
CARDIAC DISORDERS - Cardiac arrest; Ventricular tachycardia
GASTROINTESTINAL DISORDERS - Constipation; Dry mouth; Ileus
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS - Chills; Edema limbs; Malaise; Multi-organ failure
HEPATOBILIARY DISORDERS - Portal vein thrombosis
INVESTIGATIONS - Lymphocyte count decreased; Neutrophil count decreased; Weight loss; White blood cell decreased
METABOLISM AND NUTRITION DISORDERS - Hypophosphatemia; Tumor lysis syndrome
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS - Bone pain; Muscle cramp: Pain in extremity
NERVOUS SYSTEM DISORDERS - Headache
PSYCHIATRIC DISORDERS - Confusion; Insomnia; Suicide attempt
RENAL AND URINARY DISORDERS - Acute kidney injury
REPRODUCTIVE SYSTEM AND BREAST DISORDERS - Breast pain
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS - Bronchopulmonary hemorrhage; Pulmonary hypertension; Respiratory failure
SKIN AND SUBCUTANEOUS TISSUE DISORDERS - Alopecia; Dry skin[2]; Hyperhidrosis
VASCULAR DISORDERS - Hypertension; Hypotension; Thromboembolic event

Note: Atezolizumab (MPDL3280A) in combination with other agents could cause an exacerbation of any adverse event currently known to be caused by the other agent, or the combination may result in events never previously associated with either agent.

7.1.1.2 CAEPR for SGI-110

Frequency is provided based on 457 patients. Below is the CAEPR for SGI-110.

Version 2.0, March 4, 2016[1]

Adverse Events with Possible Relationship to SGI-110 (CTCAE 4.0 Term) [n= 457]			Specific Protocol Exceptions to Expedited Reporting (SPEER)
Likely (>20%)	**Less Likely (<=20%)**	**Rare but Serious (<3%)**	
BLOOD AND LYMPHATIC SYSTEM DISORDERS			
Anemia			***Anemia (Gr 2)***
	Febrile neutropenia		***Febrile neutropenia (Gr 2)***
GASTROINTESTINAL DISORDERS			
	Constipation		***Constipation (Gr 2)***
	Diarrhea		***Diarrhea (Gr 2)***
	Mucositis oral		***Mucositis oral (Gr 2)***
Nausea			***Nausea (Gr 2)***
	Vomiting		***Vomiting (Gr 2)***
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS			
Fatigue			***Fatigue (Gr 2)***
	Fever		
Injection site reaction			***Injection site reaction (Gr 2)***
	Pain		
INFECTIONS AND INFESTATIONS			
	Infection[2]		
INJURY, POISONING AND PROCEDURAL COMPLICATIONS			
	Bruising		***Bruising (Gr 1)***
INVESTIGATIONS			
	Lymphocyte count decreased		***Lymphocyte count decreased (Gr 4)***
Neutrophil count decreased			***Neutrophil count decreased (Gr 2)***
Platelet count decreased			***Platelet count decreased (Gr 2)***
	White blood cell decreased		***White blood cell decreased (Gr 2)***
METABOLISM AND NUTRITION DISORDERS			
	Anorexia		***Anorexia (Gr 2)***
NERVOUS SYSTEM DISORDERS			
	Dizziness		
	Headache		
PSYCHIATRIC DISORDERS			
	Insomnia		
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS			
	Cough		
	Dyspnea		
	Epistaxis		
SKIN AND SUBCUTANEOUS TISSUE DISORDERS			
	Rash maculo-papular		
	Skin and subcutaneous tissue disorders - Other (petechiae)		

[1]This table will be updated as the toxicity profile of the agent is revised. Updates will be distributed to all Principal Investigators at the time of revision. The current version can be obtained by contacting PIO@CTEP.NCI.NIH.GOV. Your name, the name of the investigator, the protocol and the agent should be included in the e-mail.

[2]Infection may include any of the 75 infection sites under the INFECTIONS AND INFESTATIONS SOC.

Adverse events reported on SGI-110 trials, but for which there is insufficient evidence to suggest that there was a reasonable possibility that SGI-110 caused the adverse event:

BLOOD AND LYMPHATIC SYSTEM DISORDERS - Bone marrow hypocellular
CARDIAC DISORDERS - Atrial fibrillation; Left ventricular systolic dysfunction
GASTROINTESTINAL DISORDERS - Abdominal distension; Abdominal pain; Colitis; Dry mouth; Dysphagia; Enterocolitis; Esophagitis; Gastritis; Small intestinal obstruction
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS - Chills; Edema limbs; Multi-organ failure; Non-cardiac chest pain
INVESTIGATIONS - Alanine aminotransferase increased; Alkaline phosphatase increased; Aspartate aminotransferase increased; Blood bilirubin increased; Ejection fraction decreased
METABOLISM AND NUTRITION DISORDERS - Hyperuricemia; Hypokalemia; Hypomagnesemia; Metabolism and nutrition disorders - Other (failure to thrive)
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS - Arthralgia; Back pain; Musculoskeletal and connective tissue disorder - Other (musculoskeletal stiffness); Myalgia; Pain in extremity
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS - Hypoxia; Pharyngolaryngeal pain; Pleural effusion; Respiratory failure
VASCULAR DISORDERS - Hypotension

Note: SGI-110 in combination with other agents could cause an exacerbation of any adverse event currently known to be caused by the other agent, or the combination may result in events never previously associated with either agent.

7.2 Adverse Event Characteristics

- **CTCAE term (AE description) and grade:** The descriptions and grading scales found in the revised NCI Common Terminology Criteria for Adverse Events (CTCAE) version 5.0 will be utilized for AE reporting. All appropriate treatment areas should have access to a copy of the CTCAE version 5.0. A copy of the CTCAE version 5.0 can be downloaded from the CTEP web site http://ctep.cancer.gov/protocolDevelopment/electronic_applications/ctc.htm.

- **For expedited reporting purposes only:**
  - AEs for the agent that are ***bold and italicized*** in the CAEPR (*i.e.*, those listed in the SPEER column, Section 7.1.1) should be reported through CTEP-AERS only if the grade is above the grade provided in the SPEER.
  - Other AEs for the protocol that do not require expedited reporting are outlined in section 7.3.4.

- **Attribution** of the AE:
  - Definite – The AE *is clearly related* to the study treatment.

- Probable – The AE *is likely related* to the study treatment.
- Possible – The AE *may be related* to the study treatment.
- Unlikely – The AE *is doubtfully related* to the study treatment.
- Unrelated – The AE *is clearly NOT related* to the study treatment.

7.3 Expedited Adverse Event Reporting

7.3.1 Expedited AE reporting for this study must use CTEP-AERS (CTEP Adverse Event Reporting System), accessed via the CTEP Web site (https://eapps-ctep.nci.nih.gov/ctepaers). Expedited reporting is necessary within 24 hours and full report is necessary in 5 calendar days. The reporting procedures to be followed are presented in the "NCI Guidelines for Investigators: Adverse Event Reporting Requirements for DCTD (CTEP and CIP) and DCP INDs and IDEs" which can be downloaded from the CTEP Web site (http://ctep.cancer.gov/protocolDevelopment/electronic_applications/adverse_events.htm) These requirements are briefly outlined in the tables below (Section 7.3.3).

In the rare occurrence when Internet connectivity is lost, a 24-hour notification is to be made to CTEP by telephone at 301-897-7497. Once Internet connectivity is restored, the 24-hour notification phoned in must be entered electronically into CTEP-AERS by the original submitter at the site.

7.3.2 Distribution of Adverse Event Reports

CTEP-AERS is programmed for automatic electronic distribution of reports to the following individuals: Principal Investigator and Adverse Event Coordinator(s) (if applicable) of the Corresponding Organization or Lead Organization, the local treating physician, and the Reporter and Submitter. CTEP-AERS provides a copy feature for other e-mail recipients.

7.3.3 Expedited Reporting Guidelines

Use the NCI protocol number and the protocol-specific patient ID assigned during trial registration on all reports.

Note: A death on study requires both routine and expedited reporting, regardless of causality. Attribution to treatment or other cause must be provided.

Death due to progressive disease should be reported as **Grade 5 "Disease progression" in the system organ class (SOC) "General disorders and administration site conditions."** Evidence that the death was a manifestation of underlying disease (*e.g.*, radiological changes suggesting tumor growth or progression: clinical deterioration associated with a disease process) should be submitted.

Late Phase 2 and Phase 3 Studies: Expedited Reporting Requirements for Adverse Events that Occur on Studies under an IND/IDE within 30 Days of the Last Administration of the Investigational Agent/Intervention[1, 2]

FDA REPORTING REQUIREMENTS FOR SERIOUS ADVERSE EVENTS (21 CFR Part 312)

NOTE: Investigators **MUST** immediately report to the sponsor (NCI) **ANY** Serious Adverse Events, whether or not they are considered related to the investigational agent(s)/intervention (21 CFR 312.64)

An adverse event is considered serious if it results in **ANY** of the following outcomes:

1) Death
2) A life-threatening adverse event
3) An adverse event that results in inpatient hospitalization or prolongation of existing hospitalization for ≥ 24 hours
4) A persistent or significant incapacity or substantial disruption of the ability to conduct normal life functions
5) A congenital anomaly/birth defect.
6) Important Medical Events (IME) that may not result in death, be life threatening, or require hospitalization may be considered serious when, based upon medical judgment, they may jeopardize the patient or subject and may require medical or surgical intervention to prevent one of the outcomes listed in this definition. (FDA, 21 CFR 312.32; ICH E2A and ICH E6).

ALL SERIOUS adverse events that meet the above criteria **MUST** be immediately reported to the NCI via electronic submission within the timeframes detailed in the table below.

<table>
<tr><th>Hospitalization</th><th>Grade 1 Timeframes</th><th>Grade 2 Timeframes</th><th>Grade 3 Timeframes</th><th>Grade 4 & 5 Timeframes</th></tr>
<tr><td>Resulting in Hospitalization ≥ 24 hrs</td><td colspan="3">10 Calendar Days</td><td rowspan="2">24-Hour 5 Calendar Days</td></tr>
<tr><td>Not resulting in Hospitalization ≥ 24 hrs</td><td colspan="2">Not required</td><td>10 Calendar Days</td></tr>
</table>

NOTE: Protocol specific exceptions to expedited reporting of serious adverse events are found in the Specific Protocol Exceptions to Expedited Reporting (SPEER) portion of the CAEPR

Expedited AE reporting timelines are defined as:

- "24-Hour; 5 Calendar Days" - The AE must initially be submitted electronically within 24 hours of learning of the AE, followed by a complete expedited report within 5 calendar days of the initial 24-hour report.
- "10 Calendar Days" - A complete expedited report on the AE must be submitted electronically within 10 calendar days of learning of the AE.

[1]Serious adverse events that occur more than 30 days after the last administration of investigational agent/intervention and have an attribution of possible, probable, or definite require reporting as follows:

Expedited 24-hour notification followed by complete report within 5 calendar days for:

- All Grade 4, and Grade 5 AEs

Expedited 10 calendar day reports for:

- Grade 2 adverse events resulting in hospitalization or prolongation of hospitalization
- Grade 3 adverse events

[2]For studies using PET or SPECT IND agents, the AE reporting period is limited to 10 radioactive half-lives, rounded UP to the nearest whole day, after the agent/intervention was last administered. Footnote "1" above applies after this reporting period.

Effective Date: May 5, 2011

7.3.4 Adverse Events of Special Interest in Atezolizumab Studies

The following AEs are considered of special interest in patients receiving atezolizumab and must be reported expeditiously through CTEP-AERS, irrespective of regulatory seriousness criteria:

- Pneumonitis
- Colitis
- Endocrinopathies: diabetes mellitus, pancreatitis, hyperthyroidism, hypophysitis, and adrenal insufficiency
- Hepatitis, including ALT >10 × ULN
- Systemic lupus erythematosus
- Neurological disorders: Guillain-Barré syndrome, myasthenic syndrome or myasthenia gravis, and meningoencephalitis
- Nephritis
- Events suggestive of hypersensitivity, cytokine release, influenza-like illness, systemic inflammatory response syndrome, systemic immune activation, or infusion-related reactions
- Ocular toxicities (*e.g.*, uveitis, retinitis)
- Myositis
- Myopathies, including rhabdomyolysis
- Grade ≥2 cardiac disorders (*e.g.*, atrial fibrillation, myocarditis, pericarditis)
- Vasculitis
- Autoimmune hemolytic anemia
- Severe cutaneous reactions (e.g. Stevens-Johnson Syndrome, dermatitis bullous, toxic epidermal necrolysis)

7.4 Routine Adverse Event Reporting

All Adverse Events **must** be reported in routine study data submissions. **AEs reported expeditiously through CTEP-AERS must also be reported in routine study data submissions.**

7.5 Secondary Malignancy

A *secondary malignancy* is a cancer caused by treatment for a previous malignancy (*e.g.*, treatment with investigational agent/intervention, radiation or chemotherapy). A secondary malignancy is not considered a metastasis of the initial neoplasm.

CTEP requires all secondary malignancies that occur following treatment with an agent under an NCI IND/IDE be reported expeditiously via CTEP-AERS. Three options are available to describe the event:

- Leukemia secondary to oncology chemotherapy (*e.g.*, acute myelocytic leukemia [AML])
- Myelodysplastic syndrome (MDS)
- Treatment-related secondary malignancy

Any malignancy possibly related to cancer treatment (including AML/MDS) should also be reported via the routine reporting mechanisms outlined in each protocol.

7.6 Second Malignancy

A second malignancy is one unrelated to the treatment of a prior malignancy (and is **NOT** a metastasis from the initial malignancy). Second malignancies require **ONLY** routine AE reporting unless otherwise specified.

8. PHARMACEUTICAL INFORMATION

A list of the adverse events and potential risks associated with the investigational agents administered in this study can be found in Section 7.1.

8.1 CTEP IND Agent(s)

8.1.1 Atezolizumab (NSC 783608)

Other Names: MPDL3280A

Classification: monoclonal antibody

M.W.: 150 KD

Mode of Action: anti-PD-L1

Description:
Atezolizumab is a humanized IgG1 monoclonal antibody consisting of two heavy chains (448 amino acids) and two light chains (214 amino acids). Atezolizumab targets human PD-L1 and inhibits its interaction with its receptor PD-1. Atezolizumab also blocks the binding of PD-L1 to B7.1, an interaction that is reported to provide additional inhibitory signals to T cells (80).

How Supplied:
Atezolizumab is provided by Genentech/F.Hoffmann-La Roche LTD and distributed by the Pharmaceutical Management Branch, CTEP, NCI. The agent is supplied in a single-use, 20-mL glass vial as a colorless-to-slightly-yellow, sterile, preservative-free clear liquid solution intended for IV administration. Atezolizumab is formulated as 60 mg/mL atezolizumab in 20 mM histidine acetate, 120 mM sucrose, 0.04% polysorbate 20, at a pH of 5.8. The vial is designed to deliver 20 mL (1200 mg) of atezolizumab solution but may contain more than the stated volume to enable

delivery of the entire 20 mL volume.

Preparation:
The prescribed dose of atezolizumab should be diluted in 250 mL 0.9% NaCl and infused through a 0.2 micrometer in-line filter. The IV bag may be constructed of PVC or PO; the IV infusion line may be constructed of PVC or PE; and the 0.2 micrometer in-line filter may be constructed of PES. The prepared solution may be stored at 2°C-8°C for up to 8 hours or room temperature for up to 4 hours.

Storage: 2°C-8°C (36°F-46°F) Vial contents should not be frozen or shaken and should be protected from direct sunlight.

If a storage temperature excursion is identified, promptly return atezolizumab to 2°C-8°C (36°F-46°F) and quarantine the supplies. Provide a detailed report of the excursion (including documentation of temperature monitoring and duration of the excursion) to PMBAfterHours@mail.nih.gov for determination of suitability.

Stability: Stability studies are ongoing.

CAUTION: No preservative is used in atezolizumab; therefore, the vial is intended for single use only. Discard any unused portion of drug remaining in a vial.

Route of Administration: IV infusion

Method of Administration:
Atezolizumab is administered as an intravenous infusion over 60 minutes. If the first infusion is tolerated, all subsequent infusions may be delivered over 30 minutes. Do not administer atezolizumab as an intravenous push or bolus. No premedication is indicated for administration of Cycle 1 of atezolizumab. Patients who experience an infusion related reaction with Cycle 1 of atezolizumab may receive premedication with antihistamines or antipyretics/analgesics (e.g. acetaminophen) for subsequent infusions.

Potential Drug Interactions:
Cytochrome P450 enzymes as well as conjugation/glucuronidation reactions are not involved in the metabolism of atezolizumab. No drug interaction studies for atezolizumab have been conducted or are planned. There are no known interactions with other medicinal products or other form of interactions.

Patient Care Implications:
Female patients of childbearing potential should utilize contraception and take active measures to avoid pregnancy while undergoing treatment on this study and for at least 5 months (150 days) after the last dose of study agent.

Availability

Atezolizumab is an investigational agent supplied to investigators by the Division of Cancer Treatment and Diagnosis (DCTD), NCI.

Atezolizumab is provided to the NCI under a Collaborative Agreement between the Pharmaceutical Collaborator and the DCTD, NCI (see Section 12.3).

8.1.2 SGI-110 (NSC 780463)

Chemical Name or Amino Acid Sequence:
Sodium (2*R*,3*S*,5*R*)-5-(4-amino-2-oxo-1,3,5-triazin-1(2*H*)-yl)-2-(hydroxymethyl) tetrahydrofuran-3-yl ((2*R*,3*S*,5*R*)-5-(2-amino-6-oxo-1*H*-purin-9(6*H*)-yl)-3-hydroxytetrahydrofuran-2-yl)methyl phosphate

Other Names: Guadecitabine

Classification: DNA methylation inhibitor

Molecular Formula: $C_{18}H_{23}N_9NaO_{10}P$

M.W.: 579.39 Da

Approximate Solubility: It is water soluble at about 30 mg/mL over a pH range of 6.0 to 7.0. It is unstable in aqueous solutions and is relatively more stable at neutral pH. SGI-110 is soluble in common organic solvent systems such as methanol, dimethylamine (DMA), and dimethyl sulfoxide (DMSO).

Mode of Action: Guadecitabine (SGI-110) is a potent inhibitor of DNA methylation. Guadecitabine is a dinucleotide of decitabine and deoxyguanosine linked with a phosphodiester bond. Decitabine is the active metabolite. Guadecitabine is a new chemical entity that was designed to enhance pharmacokinetic (PK) properties compared with decitabine, with potential to improve pharmacodynamics (PD), clinical efficacy, and safety.

How Supplied: SGI-110 for Injection is supplied by Astex Pharmaceutical and distributed by the Pharmaceutical Management Branch, CTEP/DCTD/NCI as follow:

- **SGI-110 for Injection**: 100 mg dry powder lyophile containing free acid equivalent in a single-use vial;
- **Diluent for Reconstitution**: 3 mL single-use vial containing non-aqueous diluent of propylene glycol, glycerin, and ethanol.

Both vials are packaged in clear glass vials with stoppers made of latex-free rubber stopper and capped with an aluminum flip-off seal.

Preparation:

1. Allow SGI-110 vial(s) and diluent vial(s) to reach room temperature approximately 60 minutes.
2. Next, loosen the lyophilized powder of SGI-110 by gently tapping and rotating the vial on a hard surface.
3. Use a 1 mL syringe and withdraw 0.9 mL of diluent and add it to the SGI-110 vial resulting in a final concentration of 100 mg/mL.
4. Vortex the diluted vial for approximately 5 to 10 minutes. Next, remove the vial and invert it 2 - 3 times, then vortex the diluted vial for another 10 minutes to ensure the powder is dissolved completely.
5. If the SGI-110 powder is not completely dissolved, repeat the vortex steps.
6. After the drug is completely dissolved, let the drug vial sit approximately 10 minutes until all bubbles dissipate.
7. The final solution of the diluted vial should be colorless to pale yellow and opalescent.
8. Withdraw the diluted solution into a syringe. If two or more syringes are needed to deliver the total prescribed dose, all dispensed syringes should have equivalent volumes for consistency and ease of administration.

Vortex: Sites that do not have access to a vortex may request it directly from Astex Pharmaceuticals, Inc.

Contact: [redacted]
Phone: [redacted]
Email: [redacted]

Storage: Store intact vials of SGI-110 refrigerated between 2^0 and 8^0 C (36^0 and 46^0 F). Store the diluents at 2^0 – 30^0 C (36^0 F – 86^0 F).

If a storage temperature excursion is identified, promptly return SGI-110 to 2^0 and 8^0 C and quarantine the supplies. Provide a detailed report of the excursion (including documentation of temperature monitoring and duration of the excursion) to PMBAfterHours@mail.nih.gov for determination of suitability.

Stability: Stability studies of SGI-110 intact vials are ongoing.

- Upon reconstitution, the diluted drug solution is good for 8 days refrigerated at 2 to 8 C

Route*(s)* **of Administration:** subcutaneous injection

Method of Administration: Administer by slow subcutaneous injection. Care must be taken to avoid intradermal injection. If injection site pain is reported upon injection, apply ice packs to the injection site both before and after injection for 5 – 10 minutes each; however, using icepack for the first dosing is not recommended. Pretreatment of topical or systemic analgesics can be considered.

Potential Drug Interactions*:* Drug-drug interaction studies have not been conducted with guadecitabine or decitabine. In vitro studies suggested that guadecitabine is unlikely to inhibit or induce major P450 enzymes and is not a CYPs substrate. Furthermore, guadecitabine nor the active metabolite decitabine inhibit major human drug transporters.

Special Handling: SGI-110 is a cytotoxic agent. Use PPE when preparing the drug. Direct contact of SGI-110 to skin must be washed immediately with soap and copious amount of water. The drug degrades rapidly in soap water. If SGI-110 contacts the mucous membranes, flush the affected area thoroughly with water.

Drug spilling can be inactivated by either using a 2 N sodium hydroxide solution or water and kericide CR Biocide B which consists of a blend of stabilized chlorine dioxide and a quaternary ammonium compound.

8.1.3 Agent Ordering and Agent Accountability

8.1.3.1 NCI-supplied agents may be requested by the Principal Investigator (or their authorized designee) at each participating institution. Pharmaceutical Management Branch (PMB) policy requires that agent be shipped directly to the institution where the patient is to be treated. PMB does not permit the transfer of agents between institutions (unless prior approval from PMB is obtained). The CTEP-assigned protocol number must be used for ordering all CTEP-supplied investigational agents. The responsible investigator at each participating institution must be registered with CTEP, DCTD through an annual submission of FDA Form 1572 (Statement of Investigator), Biosketch/Curriculum Vitae, Supplemental Investigator Data Form (IDF), and Financial Disclosure Form (FDF). If there are several participating investigators at one institution, CTEP-supplied investigational agents for the study should be ordered under the name of one lead investigator at that institution.

Starter supplies are not available for this study. The study agents may only be ordered after a patient has been enrolled onto the study.

Active CTEP-registered investigators and investigator-designated shipping designees and ordering designees can submit agent requests through the PMB Online Agent Order Processing (OAOP) application. Access to OAOP requires the establishment of a CTEP Identity and Access Management (IAM) account and the maintenance of an "active" account status and a "current" password. For questions about drug orders, transfers, returns, or accountability, call or email PMB any time. Refer to the PMB's website for specific policies and guidelines related to agent management.

8.1.3.2 Agent Inventory Records – The investigator, or a responsible party designated by the investigator, must maintain a careful record of the receipt, dispensing and final disposition of all agents received from the PMB using the appropriate NCI Investigational Agent (Drug) Accountability Record (DARF) available on the CTEP forms page. Store and maintain separate NCI Investigational Agent Accountability Records for each agent, strength, formulation and ordering investigator on this protocol.

8.1.3.3 Useful Links and Contacts

- CTEP Forms, Templates, Documents: http://ctep.cancer.gov/forms/
- NCI CTEP Investigator Registration: PMBRegPend@ctep.nci.nih.gov
- PMB policies and guidelines: http://ctep.cancer.gov/branches/pmb/agent_management.htm
- PMB Online Agent Order Processing (OAOP) application: https://eapps-ctep.nci.nih.gov/OAOP/pages/login.jspx
- CTEP Identity and Access Management (IAM) account: https://eapps-60ctep.nci.nih.gov/iam/
- CTEP Associate Registration and IAM account help: ctepreghelp@ctep.nci.nih.gov
- PMB email: PMBAfterHours@mail.nih.gov
- PMB phone and hours of service: (240) 276-6575 Monday through Friday between 8:30 am and 4:30 pm (ET)
- **IB Coordinator:** IBCoordinator@mail.nih.gov

Investigator Brochure Availability
The current versions of the IBs for the agents will be accessible to site investigators and research staff through the PMB Online Agent Order Processing (OAOP) application. Access to OAOP requires the establishment of a CTEP Identity and Access Management (IAM) account and the maintenance of an "active" account status and a "current" password. Questions about IB access may be directed to the PMB IB coordinator via email.

9. BIOMARKER, CORRELATIVE, AND SPECIAL STUDIES

Special Instructions for Biopsy Sample
At least 4 core biopsies to be collected each time if clinically feasible (pre- and on treatment, optional biopsy at progression when feasible). The size of each core biopsy is expected to be similar (about 1 cm in length collected with an 18 gauge needle). However, the number of cores available may differ between cases and therefore the following prioritization will be followed:

1. First core for formalin (FFPE Block)
2. Second core in RNA later
3. Third core flash frozen
4. Fourth core in RPMI media+10%FBS
5. If fifth available, flash frozen

To consider a sample to be acceptable for DNA or RNA extraction, it should have at least 30% neoplastic (tumor) nuclei and less than 20% of the tumor cells necrotic. Inflammatory nuclei are not scored.

Mediastinal, open surgical or laparoscopic, gastrointestinal, peritoneal or bronchial endoscopic biopsies are permitted ONLY when obtained incidentally to a clinically necessary procedure and not for the sole purpose of the clinical trial.

No laparoscopic, or endoscopic or open surgical procedure will be performed solely to obtain a biopsy for this protocol.

9.1 Integrated Laboratory Studies

9.1.1.1 NY-ESO-1 IHC and qPCR

In order to identify NY-ESO-1 antigen expression in ovarian tumors, paraffin-embedded tissue samples from patients are used to test for NY-ESO-1 expression by immunohistochemistry (IHC) and detection of the NY-ESO-1 transcript by qRT-PCR. Neither test has 100% sensitivity and a dual approach by both IHC and qRT-PCR is the optimal solution to identify the most accurate and precise NY-ESO-1 expression. Full details are provided in Appendix E.

9.1.1.2 Collection of Specimen(s)

Pre-treatment and 4 weeks on treatment (and at progression when available) tumor samples from all 3 cohorts will be collected to check the impact of SGI-110 (epigenetic modification by DNMTi) on CTA expression in tumor tissue and comparison within and across different cohorts. SOP in Appendix E:

Specimen type: Formalin paraffin embedded (FFPE) biopsies, excisions, or resections of human tissue can be used.
Specimen rejection criteria: Specimens are rejected if the corresponding H&E slides have no evidence of tumor or less than 10 percent tumor nuclei.

9.1.1.3 Handling of Specimens(s)

Tissues submitted to Pathology that are formalin fixed and paraffin embedded are used for NYESO-1 IHC and qRT-PCR testing. Tissues should be fixed for 24 to 48 hours, but not longer than 48 hours. For IHC tissues sections are cut 5 um thick using a microtome and placed onto positively charged slides. Slides are then air-dried and placed in a 60°C oven for 1 hour. These are subsequently stained with the NY-ESO-1 antibody.

For qRT-PCR a portion of the same FFPE tissue is obtained in order to extract RNA. This RNA is used to perform a qRT-PCR with specific primers and probes which can detect the presence or absence of NY-ESO-1 transcript.

9.1.1.4 Shipping of Specimen(s)

Specimens need to be shipped Monday through Thursday only (not on a holiday or day before holiday). Include Tumor Biopsy Collection Form and Shipping manifest form (Ambient and/or Frozen). Ship blocks and slides with a cold pack in warm weather, and ship frozen tissue on dry ice.

For Roswell Park patients, send samples to:

Roswell Park Cancer Institute
Correlative Science Pathology Office, GBSB S-636
Attn: I 285416 Samples
Elm & Carlton Streets
Buffalo, NY 14263
(716) 845-8917

E-mail the site name, patient ID #, study #, and FedEx tracking # to:
CRSLabPathTeam@RoswellPark.org.

For Network Sites, ship samples to (Refer to lab manual):

The Central Lab IASR (Immune Analysis Shared Resource, previously known as Immune Analysis facility). Samples will be analyzed in RPCI's Core Pathology Laboratory (Pathology Resource Network).

Roswell Park Cancer Institute
Immune Analysis Shared Resource (IASR), CCC room [redacted]
Attn: [redacted] (I 285416 Samples)
Elm & Carlton Streets
Buffalo, NY 14263
[redacted]

When shipping, e-mail the completed Tumor Biopsy Collection Form and Shipping manifest form (Ambient and/or Frozen) to:
[redacted] and include a copy in the shipment.

9.1.1.5 Site(s) Performing Correlative Study
Division of Pathology and Laboratory Medicine at Roswell Park Cancer Institute
Center for Personalized Medicine (CPM)

9.2 Exploratory/Ancillary Correlative Studies

9.2.1 Gut Microbiota

9.2.1.1 Fecal microbiome collection

Stool samples will be collected at baseline (within 7 days prior to start of treatment) and once on treatment on Cycle 4 Day 1 (or at time of disease progression, whichever is first). The stool specimen will be self-collected by participants. Patients will be provided with a sample collection kit. Patients will be provided with written instructions (Appendix D), and study staff will explain how to use the kits to collect fecal samples in the privacy of their own homes. Samples collected at other sites outside Roswell will be shipped to Roswell (DBBR lab) as instructed in Section 9.2.1.3 at ambient temperature, to be reached within 72 hours of collection.

9.2.1.2 Handling of Specimen
We will instruct the participants to return the stool sample within 24 hours of collection. If patient is unable to provide the sample with home collection or enough sample for analysis, a digital rectal exam will be performed at the time of their study visit.

9.2.1.3 Shipping of Specimen
All microbiome samples will be labeled with the participant's MR number, participant's initials, participant's study number, clinical study number, protocol timepoint, and protocol day. Collected stool samples will be sent to DBBR for de-identification, storage and future DNA extraction. Specimens need to be shipped Monday through Thursday only (not on a holiday or day before holiday) for receipt on Monday through Friday.

Roswell Park Cancer Institute
DBBR Laboratory
GBSB Bldg. 7th Floor, Rm. [redacted]
Attn: Study Number-I 285416
Elm and Carlton Streets
Buffalo, NY 14263
Tel: 716-845-1036
Fax: 716-845-1350
[redacted]

When shipping, e-mail the completed DBBR Microbiome Sample Shipping manifest (Refer to lab manual) to: [redacted]. and include a copy in the shipment.

9.2.2 AIM gene signatures
The NanoString nCounter® PanCancer Immune Profiling will be utilized for this purpose to evaluate the therapeutic efficacy on immune cell signature due to combination of PDL1 blockade, epigenetic modification, and vaccination. This panel is a novel new gene expression panel that enables developing profiles of the human immune response in all cancer types.

9.2.2.1 Collection of Specimen(s)
Fresh frozen tumor samples, approximately 10 mg, are required for optimal signal detection. The

NanoString technology is capable of providing an immune signature on FFPE samples if high quality RNA is isolated. In total, 100 – 300 ng of RNA is required from either fresh frozen or FFPE tumor samples. To detect a change in the immune profile, pre-treatment tumor biopsy and 4 weeks on-treatment tumor biopsy (and at progession when available) and PBMC samples will be used for NanoString analysis.

9.2.2.2 Handling of Specimens(s)

Specimens will be shipped to and collected at the Immune Analysis Shared resource (IASR) and provided to the Genomics Shared Resource (GSR) for the assay in batches.

9.2.2.3 Shipping of Specimen(s)

Specimens need to be shipped Monday through Thursday only (not on a holiday or day before holiday) for receipt on Monday through Friday. Include Biopsy collection specimen shipment log (refer to Lab manual). Ship frozen tissue on dry ice to:

Roswell Park Cancer Institute
Immune Analysis Shared Resource (IASR), CCC room [redacted]
Attn: [redacted] (I 285416 Samples)
Elm & Carlton Streets
Buffalo, NY 14263
[redacted]

When shipping, e-mail the completed Tumor Biopsy Collection Form and Shipping manifest form (refer to Lab manual) to:
[redacted] and include a copy in the shipment.

9.2.2.4 Site(s) Performing Correlative Study

Genomics Shared Resource (GSR) at the Roswell Park Cancer Institute (RPCI)

9.2.3 NY-ESO-1-specific immune responses

Anti-NY-ESO-1 antibody in serum will be measured by **ELISA**. Briefly serially diluted serum will be added to NY-ESO-1 protein-coated 96-well assay plate (Corning #3690). The antibodies bound to the plates will be detected using goat anti-human IgG AP-labeled antibody and Attophos substrate (Promega). Relative fluorescence unit will be measured by BioTek Synergy HT microplate reader. The fluorescence signal of the colored end product is proportional to the amount of anti-NY-ESO-1 antibody present in the serum. The antibody reciprocal titer will be determined based on the cutoff value of negative control serum.

Cellular immune responses against NY-ESO-1 will be assessed after *in vitro* stimulation with a pool of synthetic overlapping 20- to 25-mer NY-ESO-1 peptides covering the entire NY-ESO-1 protein sequence. Purified CD8+ T cells or CD4+ T cells isolated from PBMC and TILs (when available) will be presensitized with peptide-pulsed irradiated autologous CD8+ and CD4+ cell depleted population of PBMC. After 14 days of culture, NY-ESO-1-specific T-cell responses will be tested against PHA-stimulated CD4+ T cells (T-APC). The frequency of NY-ESO-1–specific

CD8+ T cells or CD4+ T cells will be determined by **ELISPOT** assay, tetramer staining or intracellular cytokine staining. For ELISPOT assays, the assay will be performed using Millipore MAHAS4510 plate and human IFN-γ ELISPOT kit (Mabtech). The plate will be enumerated by CTL S6 core analyzer. In a subset of patients with relevant HLA alleles, the frequencies of CD8+ T cells will be analyzed using human leukocyte antigen (HLA) multimers HLA-A2/NY-ESO-1157– 165, HLA-B35/NY-ESO-194–102, HLA-B35/NY-ESO-194–104, HLA-B35/NY-ESO-192–100, HLA- Cw3/NY-ESO-192-100 or HLA-Cw3/NYESO-196–104. HLA multimers will be obtained from TCMetrix or MHC Tetramer Production Facility at Baylor College of Medicine. NY-ESO-1-specific IFN-γ production from CD4+ T cells and CD8+ T cells will be determined by intracellular cytokine staining and flow cytometry. CD4+ T cells or CD8+ T cells will be stimulated with autologous T-APC for 6 hours at 37°C in the presence of monensin and/or brefeldin A. Cells will be fixed with 2% formaldehyde, followed by staining with IFN-γ (clone B27) in the presence of normal mouse IgG and permeabilization buffer (Invitrogen-Caltag). The data will be acquired by BD LSRII, BD Fortessa or FACSCalibur and analyzed using FCSExpress or FlowJo software.

9.2.3.1 Collection of Specimen(s)

Blood and tumor biopsy will be processed for isolating mononuclear cells/TILs in 6 hours upon arrival at IASR. Mononuclear cells in peripheral blood will be sorted by density gradient using lymphocyte separation medium. Serum will be isolated from serum separation tube by centrifugation.

9.2.3.2 Handling of Specimens(s)

Isolated peripheral blood mononuclear cells (PBMC), tumor infiltrating lymphocytes (TILs) and serum will be frozen down in liquid nitrogen freezer or −80°C freezer until use. All sample location in the repository will be tracked using Laboratory Information Management System (LIMS). Tumor biopsies at pre-treatment and 4 weeks on-treatment will be analyzed for frequency of NY-ESO-1-specific immune responses. Frequency of NY-ESO-1-specific responses in peripheral blood will be performed on samples collected every 4 weeks. Detailed SOPs are in Appendix E.

9.2.3.3 Shipping of Specimen(s)

Refer to lab manual. Specimens need to be shipped Monday through Thursday only (not on a holiday or day before holiday) for receipt on Monday through Friday. Ship samples to:

Roswell Park Cancer Institute
Immune Analysis Shared Resource (IASR), CCC room [redacted]
Attn: [redacted] (I 285416 Samples)
Elm & Carlton Streets
Buffalo, NY 14263
[redacted]

When shipping, e-mail the completed Tumor Biopsy Collection Form, Whole Blood collection form and Shipping manifest forms (Ambient/frozen) to:
[redacted] and include a copy of the forms in the shipment.

9.2.3.4 Site(s) Performing Correlative Study
Immune Analysis Shared Resource (IASR) at the Center for Immunotherapy, Roswell Park Cancer Institute

9.2.4 <u>PD-L1 expression</u>

PD-L1 expression will be checked for pre-treatment samples of all cohorts at a Genentech-designated external laboratory. Clone SP142 will be used for assessment of PD-L1 expression on tumor cells and immune cells using a predefined algorithm.

9.2.4.1 Collection of Specimen(s)
The same pre-treatment biopsy samples collected for NY-ESO-1 expression will be utilized for IHC for PD-L1 expression.

9.2.4.2 Handling of Specimens(s)
Samples will be shipped to a Genentech-designated external laboratory in batches.

9.2.4.3 Shipping of Specimen(s)
Samples stored at the Correlative Science Pathology Office (CSPO) or Immune Analysis Shared Resource (IASR) referred to in Section 9.1.1.4 will be batch-shipped to a Genentech-designated external laboratory. Slides should not be older than 60 days before shipment to the PD-L1 testing laboratory and a minimum of 5 unstained slides (4-5 µm) mounted on positively charged glass slides are required for testing.

Lab address for PD-L1 testing:
HistoGeneX
1331 W. 75th Street Suite 401
Naperville, IL 60540

9.2.4.4 Site(s) Performing Correlative Study
Genentech-designated external laboratory

9.2.5 <u>Neo-antigen and mutational antigen</u>

For neo-antigen load analysis, additional exploratory analysis may be done using whole exome sequencing (WES) and RNAseq on pre- and on treatment tumor (4 weeks and at progression when available)) and PBMC (every 4 weeks) samples for all cohorts.

9.2.5.1 Collection of Specimen(s)
Pre-treatment & on-treatment (4 weeks and at progression when available) biopsy samples and cryopreserved pre-, during and post-treatment PBMC samples collected monthly will be utilized for this.

9.2.5.2 Handling of Specimens(s)
These cryopreserved tumor and PBMC samples will be stored at IAF facility and handed to Genomics Shared Resource (GSR) in batches.

9.2.5.3 Shipping of Specimen(s)
Biopsy and blood specimens already received/processed and stored at the Immune Analysis Shared Resource (IASR) will be provided to the Genomics Shared Resource (GSR) in batches.

Site(s) Performing Correlative Study
Genomics Shared Resource (GSR), Roswell Park cancer Institute

9.2.6 T cell receptor (TCR) repertoire
TCRseq for Vβ will be studied to evaluate the therapeutic efficacy on TCR Vβ diversity due to combination of PDL1 blockade, epigenetic modification, and vaccination.

9.2.6.1 Collection of Specimen(s)
Pre- and on treatment tumor (4 weeks and at progression when available)) and PBMC (every 4 weeks) samples for all cohorts.

9.2.6.2 Handling of Specimens(s)
All samples will be shipped to and stored at the Immune Analysis Facility at the Center for Immnunotherapy.

9.2.6.3 Shipping of Specimen(s)
Biopsy and blood specimens already received/processed and stored at the Immune Analysis Shared Resource (IASR), will be utilized for this.

Site(s) Performing Correlative Study
Center for Immunotherapy, Roswell Park cancer Institute

9.3 Special Studies

9.3.1 *Assessing quality of life and emotional well-being and lifestyle using the Stool sample Collection Questionnaire and the Microbiome collection questionnaire – Special Correlative Study #1*

9.3.1.1 Outcome Measure:
Nutrition and basic lifestyle habits that have been shown to have an impact on human microbiome composition. This data will compliment the data from the gut microbiome data.

9.3.1.2 Assessment

9.3.1.2.1 Method of Assessment
The Stool sample Collection Questionnaire Appendix B and the Microbiome Collection Questionnaire (Appendix C) were specifically designed for patients enrolling in this study

in order to broaden the information gathered on nutrition and basic lifestyle habits that have been shown to have an impact on human microbiome composition.

9.3.1.2.2 Timing of Assessment

The Stool sample Collection Questionnaire will be completed only once at baseline, within 7 days prior to starting treatment. The Microbiome Collection Questionnaire will be obtained at baseline (within 7 days prior to start of treatment) and once on treatment on Cycle 4 Day 1 (or at time of disease progression, whichever is first).

9.3.1.3 Data Recording

9.3.1.3.1 Method of Recording

Paper questionnaires will be collected by the study nurse on the day of the patients visit and will be kept in a secured, locked up study folder in clinic. Data from collected questionnaires throughout the trial will be migrated to and stored in a secure REDCap data base.

9.3.1.3.2 Timing of Recording

The questionnaire timing will be coordinated with the stool sample collection timing, at baseline, on treatment (cycle 4, Day 1) and at the end of treatment.

10. STUDY CALENDAR

In the event that the patient's condition is deteriorating, laboratory evaluations should be repeated within 48 hours prior to initiation of the next cycle of therapy. Patients receiving atezolizumab will be closely assessed for adverse events related to pulmonary signs and symptoms throughout the study.

Table 1 Phase I, (SGI-110 + atezolizumab)

Evaluation	Baseline[1]	Cycle 1-6 (+ 5 day window)					Cycle 7-24 (+ 5 day window)			End of Treatment[16] (±7 day window)	Follow-Up[2] (±7 day window)	Long Term Follow-Up (every 2 months)[3]
Cycle Day		**D1**	**D2-5**	**D8**	**D15**	**D22**	**D1**	**D8**	**D22**			
Pre-Existing Conditions	X											
Concomitant Medications	X[9]	X --					X			X	X	X
Adverse Events		X --					X			X	X	X
Clinical Assessments												
Medical History	X											
Physical Examination[4]	X	X		X[19]		X[19]		X		X		
Vital Signs[5]	X	X	X	X		X		X	X	X		
Height/Weight[6]	X	X		X		X		X	X	X		
ECOG Performance Status	X											

Evaluation	Baseline[1]	Cycle 1-6 (+ 5 day window)					Cycle 7-24 (+5 day window)			End of Treatment[16] (± 7 day window)	Follow-Up[2] (± 7 day window)	Long Term Follow-Up (every 2 months)[3]
Cycle Day		**D1**	**D2-5**	**D8**	**D15**	**D22**	**D1**	**D8**	**D22**			
Stool sample Collection Questionnaire	X[21]											
Microbiome Collection Questionnaire[18]	X	X										
Laboratory Procedures												
Hematology[7]	X	X		X		X		X	X	X		
Chemistry[8]	X	X		X		X		X	X	X		
CA-125 level	X	X						X		X		
HLA typing[17]	X											
PT/INR, aPTT	X[15]	X[15]										
Blood for Correlative Studies[23]	X	X[24]						X[22]		X	X	X
Thyroid studies[20]	X											
Pregnancy Test Urine	X[14]											
Stool microbiome collection[18]	X	X										
Imaging Procedures												
CT guided biopsy	X	X[12]										
Electrocardiogram	X											
CT scan/MRI of chest, abdomen and pelvis	X	X[13]					X[13]				X	

Evaluation	Baseline[1]	Cycle 1-6 (+ 5 day window)					Cycle 7-24 (+5 day window)			End of Treatment[16] (± 7 day window)	Follow-Up[2] (± 7 day window)	Long Term Follow-Up (every 2 months)[3]
Cycle Day		**D1**	**D2-5**	**D8**	**D15**	**D22**	**D1**	**D8**	**D22**			
Study Drug Administration												
SGI-110[10]		X	X									
atezolizumab[11]				X		X		X	X			

1 Performed within 28 days prior to treatment start (unless otherwise noted).
2 Follow-up safety evaluations will occur 30 days (± 7 days) after last dose of study drug. (telephone contact is acceptable for AE's and conmeds)
3 Telephone contact is acceptable for AE's and conmeds
4 Research/study-specific assessments should be performed no later than 28 days prior to the start of treatment. A physical exam within 28 days of consent is acceptable.
5 Vital signs: temperature, heart rate, respiratory rate, blood pressure.
6 Height collected at baseline only. SGI-110 and atezolizumab doses will only be adjusted mid-cycle if weight changes $\geq$ 10%.
7 Hematology: Complete blood count (CBC) with automated differentials
8 Chemistry: Complete metabolic panel (CMP): chloride, CO_2, potassium, sodium, BUN, glucose, calcium, creatinine, total protein, albumin, total bilirubin, alkaline phosphatase, AST, ALT, A/G ratio, BUN/creatinine ratio, osmolality (Calc), anion gap, LDH
9 Medications ongoing, or discontinued, within 1 week prior to first dose of study drug.
10 SGI-110 to be given subcutaneously on days 1-5 of each 28 day cycle for a total of 6 cycles.
11 Atezolizumab to be given IV starting on day 8 and every 2 weeks thereafter. (on day 8 and day 22 for a total of 24 cycles)
12 An on treatment biopsy will occur 4 weeks after start of treatment, optional biopsy at progression when feasible.
13 Scan for reassessment imaging will commence 8 weeks (-7 days) after start of treatment and continue every other cycle (-7 days). Every other cycle is 8 weeks or may be more in the case of treatment delays. In the event that treatment is held, the patient will be closely monitored while on treatment hold and CT scan will be performed at 8 weeks to monitor disease status and hold of treatment may continue up to 12 weeks if patient is still stable in their disease.
14 Within 7 days of starting treatment for patients of child-bearing potential.
15 PT/INR, aPTT to be done on same day as CT guided biopsy prior to procedure **at baseline and an on-treatment biopsy at 4 weeks after start of treatment and optional biopsy at progression when feasible.**
16 At time decision is made to discontinue treatment.
17 Historic HLA typing is permitted, or 10 ml lavender top tube (to be sent to IASR at Roswell Park).

18	Stool sample and Microbiome Collection Questionnaire to be collected at baseline (within 7 days of C1D1), and once on treatment on Cycle 4 Day 1 (or at time of disease progression, whichever is first). At screening visit, the patient will be given a stool collection kit with instructions to bring in the stool sample prior to dosing on Day 1 (within 7 days). Stool sample will be collected by patient at home and returned to the local study site ASAP (within 24 hours of collection). If not possible, stool may be collected via digital rectal exam. Sites outside of Roswell will need to ship stool sample to Roswell Park DBBR lab to arrive within 72 hours of collection. Refer to lab manual for details.
19	Cycle 1 only.
20	A TSH level will be collected at baseline and as clinically indicated throughout the study. Free T3 and Free T4 will only be measured if the TSH level is abnormal. They should also be measured if there is clinical suspicion of an adverse event related to the endocrine system.
21	Within 7 days of starting treatment.
22	Obtain blood on Cycle 7 Day 8, then every 3 cycles on day 8.
23	Blood for Immune monitoring (about 60 ml in 6 x 10 ml green tops, plus 1 x 10 ml red top), 1 PAXgene RNA tube (2.5 ml)
24	Cycle 1 Day 1 sample to be collected as Baseline before administration of study drug only if a baseline sample hasn't been collected yet.

Table 2 Phase 2, Cohort 1 (atezolizumab only)

Evaluation	Baseline[1]	Cycle 1 Day 1	Cycle 1 Day 15 (+ 5 day window)	Cycle 2 Day 1 (+ 5 day window)	Cycle 2 Day 15 (+ 5 day window)	Cycle 3-24 Day 1 and 15 (+ 5 day window)	End of Treatment[14] (± 7 day window)	Follow-Up[2] (± 7 day window)	Long Term Follow-Up (every 2 months)[3]
Pre-Existing Conditions	X								
Concomitant Medications	X[9]	X -- X					X	X	X
Adverse Events		X -- X					X	X	X
Clinical Assessments									
Medical History	X								
Physical Examination[4]	X	X	X	X		X[22]	X		
Vital Signs[5]	X	X	X	X	X	X	X		
Height/Weight[6]	X	X	X	X	X	X	X		
ECOG Performance Status	X								
Stool sample Collection Questionnaire	X[19]								
Microbiome Collection Questionnaire[17]	X					X			
Laboratory Procedures									
Hematology[7]	X	X	X	X	X	X	X		
Chemistry[8]	X	X	X	X	X	X	X		

CA-125 level	X	X		X		X[22]	X		
Evaluation	Baseline[1]	Cycle 1 Day 1	Cycle 1 Day 15 (+ 5 day window)	Cycle 2 Day 1 (+ 5 day window)	Cycle 2 Day 15 (+ 5 day window)	Cycle 3-24 Day 1 and 15 (+ 5 day window)	End of Treatment[14] (± 7 day window)	Follow-Up[2] (± 7 day window)	Long Term Follow-Up (every 2 months)[3]
Thyroid studies[18]	X								
HLA typing	X[15]								
Blood for Correlative Studies[20]	X	X[23]		X		X[16]	X	X	X
PT/INR, aPTT	X[13]			X[13]					
Pregnancy Test Urine	X[12]								
Stool microbiome collection[17]	X					X			
Imaging Procedures									
CT guided biopsy	X			X[21]					
Electrocardiogram	X								
CT scan/MRI of chest, abdomen and pelvis	X					X[11]		X	
Study Drug Administration									
atezolizumab[10]		X	X	X	X	X			

1 Performed within 28 days prior to treatment start (unless otherwise noted).
2 Follow-up safety evaluations will occur 30 days (± 7 days) after last dose of study drug. (telephone contact is acceptable for AE's and con-meds)
3 Telephone contact is acceptable is acceptable for AE's and con-meds
4 Research/study-specific assessments should be performed no later than 28 days prior to the start of treatment. A physical exam within 28 days of consent is acceptable.
5 Vital signs: temperature, heart rate, respiratory rate, blood pressure
6 Height collected at baseline only. Weight will be obtained on day 1 and day 15 of each cycle. Atezolizumab dose will only be adjusted mid-cycle if

	weight changes ≥ 10%.
7	Hematology: Complete blood count (CBC) with automated differentials.
8	Chemistry: Complete metabolic panel (CMP): chloride, CO_2, potassium, sodium, BUN, glucose, calcium, creatinine, total protein, albumin, total bilirubin, alkaline phosphatase, AST, ALT, A/G ratio, BUN/creatinine ratio, osmolality (Calc), anion gap, LDH.
9	Medications ongoing, or discontinued, within 1 week prior to first dose of study drug.
10	Atezolizumab to be given IV on days 1 and 15 of each 28 days cycle for a total of 24 cycles.
11	Scan for reassessment imaging will commence 8 weeks (-7 days) after start of treatment and continue every other cycle (-7 days). Every other cycle is 8 weeks or may be more in the case of treatment delays. In the event that treatment is held, the patient will be closely monitored while on treatment hold and CT scan will be performed at 8 weeks to monitor disease status and hold of treatment may continue up to 12 weeks if patient is still stable in their disease.
12	Within 7 days of starting treatment for patients of child-bearing potential.
13	PT/INR, aPTT to be done on same day as CT guided biopsy prior to procedure **at baseline and an on-treatment biopsy at 4 weeks after start of treatment and optional biopsy at progression when feasible.**
14	At time decision is made to discontinue treatment.
15	Historic HLA typing is permitted, or 10 ml Lavendar top tube (to be sent to IASR at Roswell Park).
16	Obtain blood on Cycle 3 Day 1, Cycle 4 Day 1, Cycle 5 Day 1, Cycle 6 Day 1, Cycle 7 Day 1, then every 3 cycles on day 1.
17	Stool sample and Microbiome Collection Questionnaire to be collected at baseline (within 7 days of C1D1) and once on treatment on Cycle 4 Day 1 (or at time of disease progression, whichever is first). At screening visit, the patient will be given a stool collection kit with instructions to bring in the stool sample prior to dosing on Day 1 (within 7 days). Stool sample will be collected by patient at home and returned to the local study site ASAP (within 24 hours of collection). If not possible, stool may be collected via digital rectal exam. Sites outside of Roswell Park will need to ship stool sample to Roswell Park DBBR lab to arrive within 72 hours of collection. Refer to lab manual for details.
18	A TSH level will be collected at baseline and as clinically indicated throughout the study. Free T3 and Free T4 will only be measured if the TSH level is abnormal. They should also be measured if there is clinical suspicion of an adverse event related to the endocrine system.
19	Within 7 days of starting treatment.
20	Blood for Immune monitoring (about 60 ml in 6 x 10 ml green tops, plus 1x 10 ml red top), 1 PAXgene RNA tube (2.5 ml)
21	An on treatment biopsy will occur 4 weeks after start of treatment, optional biopsy at progression when feasible.
22	Day 1 only
23	Cycle 1 Day 1 sample to be collected as Baseline before administration of study drug only if a baseline sample hasn't been collected yet.

Table3 Phase 2, Cohort 2 (SGI-110 + atezolizumab)

Evaluation	Baseline[1]	Cycle 1-6 (+ 5 day window)					Cycle 7-24 (+ 5 day window)			End of Treatment[16] (±7 day window)	Follow-Up[2] (±7 day window)	Long Term Follow-Up (every 2 months)[3]
Cycle Day		**D1**	**D2-5**	**D8**	**D15**	**D22**	**D1**	**D8**	**D22**			
Pre-Existing Conditions	X											
Concomitant Medications	X[9]	X -- X								X	X	X
Adverse Events		X --X								X	X	X
Clinical Assessments												
Medical History	X											
Physical Examination[4]	X	X		X[19]		X[19]		X		X		
Vital Signs[5]	X	X	X	X		X		X	X	X		
Height/Weight[6]	X	X		X		X		X	X	X		
ECOG Performance Status	X											
Stool sample Collection Questionnaire	X[21]											
Microbiome Collection Questionnaire[18]	X	X										
Laboratory Procedures												
Hematology[7]	X	X		X		X		X	X	X		
Chemistry[8]	X	X		X		X		X	X	X		

Evaluation	Baseline[1]	Cycle 1-6 (+ 5 day window)					Cycle 7-24 (+ 5 day window)			End of Treatment[16] (±7 day window)	Follow-Up[2] (±7 day window)	Long Term Follow-Up (every 2 months)[3]
Cycle Day		D1	D2-5	D8	D15	D22	D1	D8	D22			
CA-125 level	X	X						X		X		
HLA typing[17]	X											
PT/INR, aPTT	X[15]	X[15]										
Blood for Correlative Studies[23]	X	X[24]						X[22]		X	X	X
Thyroid studies[20]	X											
Pregnancy Test Urine	X[14]											
Stool microbiome collection[18]	X	X										
Imaging Procedures												
CT guided biopsy	X	X[12]										
Electrocardiogram	X											
CT scan/MRI of chest, abdomen and pelvis	X	X[13]					X[13]				X	
Study Drug Administration												
SGI-110[10]		X	X									
atezolizumab[11]				X		X		X	X			

1 Performed within 28 days prior to treatment start (unless otherwise noted).
2 Follow-up safety evaluations will occur 30 days (± 7 days) after last dose of study drug. (telephone contact is acceptable for AE's and conmeds)
3 Telephone contact is acceptable for AE's and conmeds
4 Research/study-specific assessments should be performed no later than 28 days prior to the start of treatment. A physical exam within 28 days of consent is acceptable.
5 Vital signs: temperature, heart rate, respiratory rate, blood pressure.

6	Height collected at baseline only. SGI-110 and atezolizumab doses will only be adjusted mid-cycle if weight changes ≥ 10%.
7	Hematology: Complete blood count (CBC) with automated differentials
8	Chemistry: Complete metabolic panel (CMP): chloride, CO_2, potassium, sodium, BUN, glucose, calcium, creatinine, total protein, albumin, total bilirubin, alkaline phosphatase, AST, ALT, A/G ratio, BUN/creatinine ratio, osmolality (Calc), anion gap, LDH
9	Medications ongoing, or discontinued, within 1 week prior to first dose of study drug.
10	SGI-110 to be given subcutaneously on days 1-5 of each 28 day cycle for a total of 6 cycles.
11	Atezolizumab to be given IV starting on day 8 and every 2 weeks thereafter. (on day 8 and day 22 for a total of 24 cycles)
12	An on treatment biopsy will occur 4 weeks after start of treatment, optional biopsy at progression when feasible.
13	Scan for reassessment imaging will commence 8 weeks (-7 days) after start of treatment and continue every other cycle (-7 days). Every other cycle is 8 weeks or may be more in the case of treatment delays. In the event that treatment is held, the patient will be closely monitored while on treatment hold and CT scan will be performed at 8 weeks to monitor disease status and hold of treatment may continue up to 12 weeks if patient is still stable in their disease.
14	Within 7 days of starting treatment for patients of child-bearing potential.
15	PT/INR, aPTT to be done on same day as CT guided biopsy prior to procedure **at baseline and an on-treatment biopsy at 4 weeks after start of treatment and optional biopsy at progression when feasible.**
16	At time decision is made to discontinue treatment.
17	Historic HLA typing is permitted, or 10 ml Lavendar top tube (to be sent to IASR at Roswell Park).
18	Stool sample and Microbiome Collection Questionnaire to be collected at baseline (within 7 days of C1D1) and once on-treatment on Cycle 4 Day 1 (or at time of disease progression, whichever is first). At screening visit, the patient will be given a stool collection kit with instructions to bring in the stool sample prior to dosing on Day 1 (within 7 days). Stool sample will be collected by patient at home and returned to RPCI within 24 hours of collection. If not possible, stool may be collected via digital rectal exam. Sites outside of Roswell Park will need to ship stool sample to Roswell Park DBBR lab to arrive within 72 hours of collection. Refer to lab manual for details.
19	Cycle 1 only.
20	A TSH level will be collected at baseline and as clinically indicated throughout the study. Free T3 and Free T4 will only be measured if the TSH level is abnormal. They should also be measured if there is clinical suspicion of an adverse event related to the endocrine system.
21	Within 7 days of starting treatment.
22	Obtain blood on Cycle 7 Day 8, then every 3 cycles on day 8.
23	Blood for Immune monitoring (about 60 ml in 6 x 10 ml green tops, plus 1 x 10 ml red top), 1 PAXgene RNA tube (2.5 ml)
24	Cycle 1 Day 1 sample to be collected as Baseline before administration of study drug only if a baseline sample hasn't been collected yet.

MEASUREMENT OF EFFECT

Although the clinical benefit of these drug(s) has not yet been established, the intent of offering this treatment is to provide a possible therapeutic benefit, and thus the patients will be carefully monitored for tumor response and symptom relief in addition to safety and tolerability. Patients with measurable disease will be assessed by standard criteria. For the purposes of this study, patients should be re-evaluated every 8 weeks. In addition to a baseline scan, confirmatory scans will also be obtained 8 weeks following initial documentation of an objective response. Follow up safety evaluation will be done at 30 days after the last dose of study drug.

10.1 Antitumor Effect – Solid Tumors

Tumor response will be assessed using the Response Evaluation Criteria in Solid Tumors (RECIST 1.1) but Immune-Related response Criteria (irRECIST) as described by Nishino et al, 2013 (81) and Bohnsack et al, 2014 (82) will also be noted.

Baseline

Measurable Lesion Definitions and Target Lesion Selection

- All measurable lesions up to a maximum of 2 lesions per organ and 5 lesions in total, representative of all involved organs, will be identified as target lesions and recorded and measured at baseline
- Measurable lesions must be accurately measured in at least one dimension with a minimum size of:
    - 10 mm in the longest diameter by CT or MRI scan (or no less than double the slice thickness) for non- nodal lesions and ≥ 15 mm in short axis for nodal lesions.
    - 10 mm caliper measurement by clinical exam (lesions which cannot be accurately measured with calipers should be recorded as non-measurable).
    - 20 mm by chest X-ray
- A sum of the longest diameter (short axis for lymph nodes) of all target lesions will be calculated and reported as the baseline sum diameters. This will be used as reference to further characterize the objective tumor response of the measurable dimension of the disease.

Non-measurable Lesion Definitions

- Non-target lesions will include:
    - Measurable lesions not selected as target lesions

- All sites of non-measurable disease, such as neoplastic masses that are too small to measure because their longest uninterrupted diameter is < 10 mm (or < two times the axial slice thickness), i.e., the longest perpendicular diameter is ≥ 10 and < 15 mm.
- Other types of lesions that are confidently felt to represent neoplastic tissue, but are difficult to measure in a reproducible manner. These include bone metastases, leptomeningeal metastases, malignant ascites, pleural or pericardial effusions, ascites, inflammatory breast disease, lymphangitis cutis/pulmonis, cystic lesions, ill-defined abdominal masses, skin lesions, etc.

- All lesions or sites of disease not recorded as target lesions (e.g., small lesions and non-measurable lesions) should be identified as non-target lesions and indicated as present in the source documents at baseline. There is no limit to the number of non-target lesions that can be recorded at baseline. The general location will also be documented on the images, drawing a regularly-shaped Region of Interest.
- Measurements of the non-target lesions will not be performed, but the presence or absence of each should be noted throughout follow-up and evaluation.

Target and Non-Target Lymph Node Lesion Definitions

- To be considered pathologically enlarged and measurable, a lymph node must be ≥ 15 mm in short axis when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline and in follow-up, only the short axis will be measured and followed.

Non-Target Lesion Selection

- All lesions or sites of disease not recorded as target lesions should be recorded as non-target lesions at baseline. There is no limit to the number on non-target lesions that can be recorded at baseline.

Bone Lesions

- Bone scan, PET scan or plain films are not considered adequate imaging techniques to measure bone lesions. However, these techniques can be used to confirm the presence or disappearance of bone lesions.
- Regardless of the imaging modality blastic bone lesions will not be selected as target lesions. Lytic or mixed lytic-blastic lesions with a measurable soft tissue component ≥ 10 mm can be selected as target lesions.

Brain Lesions

- Brain Lesions detected on brain scans can be considered as both target or non-target lesions.

Cystic and Necrotic Lesions as Target Lesions

- Lesions that meet the criteria for radiographically defined simple cysts should not be considered as malignant lesions (neither measurable nor non-measurable) since they are, by definition, simple cysts.
- Lesions that are partially cystic or necrotic can be selected as target lesions.
- The longest diameter of such a lesion will be added to the ***Total Measured Tumor Burden*** (TMTB) of all target lesions at baseline.
- If other lesions with a non-liquid/non-necrotic component are present, those should be preferred.

Lesions with Prior Local Treatment

- During target lesion selection the radiologist will consider information on the anatomical sites of previous intervention (e.g. previous irradiation, RF-ablation, TACE, surgery, etc.).
- Lesions undergoing prior intervention will not be selected as target lesions unless there has been a demonstration of progress in the lesion.

No Disease at Baseline

- If a patient has no measurable and no non-measurable disease at baseline the radiologist will assign '***No Disease***' (ND) as the overall tumor assessment for any available follow-up time-points unless new measurable lesions are identified and contribute to the TMTB.

Follow-Up

Recording of Target and New Measurable Lesion Measurements

- The longest diameters of non-nodal target and new non-nodal measurable lesions, and short axes of nodal target and new nodal measurable lesions will be recorded. Together they determine the ***Total Measured Tumor Burden (TMTB)*** at follow-up.
- $\mathbf{TMTB\ \%\ change} = \frac{\mathbf{Baseline\ Tumor\ Burden - Current\ Tumor\ Burden}}{\mathbf{Baseline\ Tumor\ Burden}} \times \mathbf{100}$

Definition of New Measurable Lesions

- In order to be selected as new measurable lesions (≤ 2 lesions per organ, ≤ 5 lesions total, per time-point), new lesions must meet criteria as defined for baseline target lesion selection and meet the same minimum size requirements of 10 mm in long diameter and minimum 15 mm in short axis for new measurable lymph nodes. New measurable lesions shall be prioritized according to size, and the largest lesions shall be selected as new measured lesions.

Non-Target Lesion Assessment

- The RECIST 1.1 definitions for the assessment of non-target lesions apply to irRECIST as well (i.e., measurements of the non-target lesions will not be performed, but the presence or absence of each should be noted throughout follow-up and evaluation).
- The response of non-target lesions primarily contributes to the overall response assessments of CR and Non-CR/Non-PD (NN).
- Non-target lesions do not affect PR and SD assessments.
- Only a massive and unequivocal worsening of non-target lesions alone, even without progress in the TMTB is indicative of irPD.

New Non-Measurable Lesions Definition and Assessment

- All new lesions not selected as new measurable lesions are considered new non-measurable lesions and are followed qualitatively.
- Only a massive and unequivocal progression of new non-measurable lesions leads to an overall assessment of irPD for the time-point.
- Persisting new non- measurable lesions prevent CR.

RECIST 1.1 Overall Tumor Assessments

The following RECIST 1.1 criteria will be the primary method utilized in this study for the assessment and reporting of tumor response data. irRECIST will also be noted during the trial. The RECIST 1.1 will be used for tumor response assessment at time of continuing review and will be used for IRB reporting and formal analysis. Time point response assessments will be performed every 8 weeks and 30 days after the last dose of the study drug.

- **Complete Response (CR)**: Disappearance of all evidence of cancer in target and non-target lesions. CR must be confirmed by repeat assessments performed no less than 4 weeks after the criteria for response are first met to qualify as CR. This definition includes a pathological CR when hyperpigmented lesions visible by exam are biopsied and demonstrate no viable measurable cancer.
- **Partial Response (PR)**: At least a 30% decrease in the sum of the longest diameter (LD) of target lesions taking as reference the baseline sum LD. Non-target lesions may persist provided there is no unequivocal progression in these lesions. PR must be confirmed by repeat assessments performed no less than 4 weeks after the criteria for response are first met to qualify as PR;
- **Progressive Disease (PD)**: At least a 20% increase in the sum LD of the target lesions from the smallest sum LD recorded since the beginning of therapy or the appearance of one or more new lesions or unequivocal progression of existing non-target lesions; and
- **Stable Disease (SD)**: Measurements demonstrating neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify as PD after the start of treatment taking as

reference the smallest sum LD since the treatment started. During this time, non-target lesions may persist provided there is no unequivocal progression in these lesions.

Time Point Response Criteria (+/- non-target disease)

Target Lesions	Non-Target Lesions	New Lesions	Overall Response
CR	CR	No	CR
CR	Non-CR/Non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

Time Point Response Criteria (non-target disease only)

Non-Target Lesions	New Lesions	Overall Response
CR	No	CR
Non-CR/non-PD	No	Non-CR/non-PD[1]
Not all evaluated	No	NE
Unequivocal PD	Yes or No	PD
Any	Yes	PD

1 Non-CR/non-PD is preferred over SD for non-target disease since SD is used as endpoint for assessment of efficacy in trials so to assign this category when no lesions can be measured is not advised.

irRECIST Overall Tumor Assessments

The irRECIST overall tumor assessment is based on the ***TMTB*** (total measured tumor burden) of measured target and new lesions, non-target lesion assessment and new non-measurable lesions.

Time point response assessments will be performed every 8 weeks and 30 days after the last dose of the study drug. If "immune-related progression" is felt to occur, reassessment should be within 4 weeks such that the patient is not compromised from further therapy due to the delay.

- **irCR:** Complete disappearance of all measurable and non-measurable lesions. Lymph nodes must decrease to < 10 mm in short axis. Confirmation of response is not mandatory.
- **irPR:** Decrease of ≥ 30% in TMTB relative to baseline, non-target lesions are irNN, and no unequivocal progression of new non-measurable lesions.

- **irSD:** Failure to meet criteria for irCR or irPR in the absence of irPD.
- **irNN:** No target disease was identified at baseline and at follow-up the patient fails to meet criteria for irCR or irPD.
- **irPD:** Minimum 20% increase and minimum 5 mm absolute increase in TMTB compared to nadir, or irPD for non-target or new non-measurable lesions. Confirmation of progression is recommended minimum 4 weeks after the first irPD assessment.
- **irNE:** Used in exceptional cases where insufficient data exists.
- **irND:** In adjuvant setting when no disease is detected.

New Lesions: the presence of new lesion(s) does not define progression. The measurements of the new lesion(s) are included in the sum of the measurements (the sum of the measurements = the sum of the longest diameters of all target lesions and new lesions, if any).

Confirmation Measurement

A confirmatory assessment is required no less than 6 weeks after a PR or CR is deemed by RECIST or irRECIST

Guidelines for Evaluation of Measurable Disease

All measurements should be taken and recorded in metric notation. All baseline evaluations should be performed as closely as possible to the beginning of treatment and never more than 4 weeks before the beginning of the treatment.

The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Imaging-based evaluation is preferred to evaluation by clinical examination unless the lesion(s) being followed cannot be imaged but are assessable by clinical exam.

- **Clinical Lesions:** Clinical lesions will only be considered measurable when they are superficial (e.g., skin nodules and palpable lymph nodes) and ≥ 10 mm diameter as assessed using calipers (e.g., skin nodules). In the case of skin lesions, documentation by color photography, including a ruler to estimate the size of the lesion, is recommended.
- **Chest x-ray:** Lesions on chest x-ray are acceptable as measurable lesions when they are clearly defined and surrounded by aerated lung. However, CT is preferable.
- **Conventional CT and MRI:** This guideline has defined measurability of lesions on CT scan based on the assumption that CT slice thickness is 5 mm or less. If CT scans have slice thickness greater than 5 mm, the minimum size for a measurable lesion should be twice the slice thickness. MRI is also acceptable in certain situations (e.g., for body scans) however, conventional CT scan is the preferred modality to determine response to treatment.

Use of MRI remains a complex issue. MRI has excellent contrast, spatial, and temporal resolution; however, there are many image acquisition variables involved in MRI, which greatly impact image quality, lesion conspicuity, and measurement. Furthermore, the availability of MRI is variable globally. As with CT, if an MRI is performed, the technical specifications of the scanning sequences used should be optimized for the evaluation of the type and site of disease. Furthermore, as with CT, the modality used at follow-up should be the same as was used at baseline and the lesions should be measured/assessed on the same pulse sequence. It is beyond the scope of the irRECIST or RECIST guidelines to prescribe specific MRI pulse sequence parameters for all scanners, body parts, and diseases. Ideally, the same type of scanner should be used and the image acquisition protocol should be followed as closely as possible to prior scans. Body scans should be performed with breath-hold scanning techniques, if possible.

- **Ultrasound:** Ultrasound is not useful in assessment of lesion size and should not be used as a method of measurement. Ultrasound examinations cannot be reproduced in their entirety for independent review at a later date and, because they are operator dependent, it cannot be guaranteed that the same technique and measurements will be taken from one assessment to the next. If new lesions are identified by ultrasound in the course of the study, confirmation by CT or MRI is advised. If there is concern about radiation exposure at CT, MRI may be used instead of CT in selected instances.
- **Endoscopy, Laparoscopy:** The utilization of these techniques for objective tumor evaluation is not advised. However, such techniques may be useful to confirm complete pathological response when biopsies are obtained or to determine relapse in trials where recurrence following complete response (CR) or surgical resection is an endpoint.
- **Tumor Markers:** Tumor markers alone cannot be used to assess response. If markers are initially above the upper normal limit, they must normalize for a participant to be considered in complete clinical response.
- **Cytology, Histology:** These techniques can be used to differentiate between partial responses (PR) and complete responses (CR) in rare cases (e.g., residual lesions in tumor types, such as germ cell tumors, where known residual benign tumors can remain).

The cytological confirmation of the neoplastic origin of any effusion that appears or worsens during treatment when the measurable tumor has met criteria for response or stable disease is mandatory to differentiate between response or stable disease (an effusion may be a side effect of the treatment) and progressive disease.

11. STUDY OVERSIGHT AND DATA REPORTING/REGULATORY REQUIREMENTS

Adverse event lists, guidelines, and instructions for AE reporting can be found in Section 7.0 (Adverse Events: List and Reporting Requirements).

11.1 Study Oversight

This protocol is monitored at several levels, as described elsewhere in this section. The Protocol

Principal Investigator is responsible for monitoring the conduct and progress of the clinical trial, including the ongoing review of accrual, patient-specific clinical and laboratory data, and routine and serious adverse events; reporting of expedited adverse events; and accumulation of reported adverse events from other trials testing the same drug(s). The Protocol Principal Investigator and statistician have access to the data at all times through the CTMS web-based reporting portal.

For the Phase 1 portion of this study, all decisions regarding dose escalation/expansion/de-escalation require sign-off by the Protocol Principal Investigator through the CTMS/IWRS. In addition, for the Phase 1 portion, the Protocol Principal Investigator will have at least monthly, or more frequently, conference calls with the Study Investigators and the CTEP Medical Officer(s) to review accrual, progress, and adverse events and unanticipated problems.

For a Phase 1/2 trial, enrollment to the Phase 2 portion of the trial will not begin until a protocol amendment has been submitted which summarizes the Phase 1 results, the recommended Phase 2 dose, and the rationale for selecting it. The amendment must be reviewed and approved by CTEP before enrollment to the Phase 2 portion can begin.

All studies are also reviewed in accordance with the enrolling institution's data safety monitoring plan.

Data safety Monitoring:

Roswell Park as the primary oversight center for the conduct of the study NCI EDDOP #10017, "A Randomized Phase 2 Trial of Atezolizumab (MPDL3280A), SGI-110 and CDX-1401 Vaccine in Recurrent Ovarian Cancer" requires that in order to assure patient and research subject safety and integrity of research in clinical trials conducted by Roswell Park and participating centers, Roswell Park will serve as the reviewing entity of record for AE's, SAE's and DLT's. Data will be reviewed and discussed on a bi- weekly basis through a teleconference with participating sites.

Teleconference information needed:

- Diagnosis of patient
- Age
- Arm enrolled on
- Cycle/Day
- AE's/SAE's with grades and attributions

Any SAE's or possible DLT's will be e-mailed to Phase1DLTnetwork@roswellpark.org within 24 hours of site becoming aware of the event. This report will include the following information:

- Participant number
- Diagnosis
- Age

- Dose/Arm
- Cycle/Day
- Description of event to include: AE's/SAE with grading, Attributions (if available), interventions

*Please note that the adverse event/DLT information should also be entered into RAVE system as they occur.

During the Phase 2 portion of the study, the Protocol Principal Investigator will have, at a minimum, quarterly conference calls with the Study Investigators and the CTEP Medical Officer(s) to review accrual, progress, and pharmacovigilance. Decisions to proceed to the second stage of a Phase 2 trial will require sign-off by the Protocol Principal Investigator and the Protocol Statistician.

All Study Investigators at participating sites who register/enroll patients on a given protocol are responsible for timely submission of data via Medidata Rave and timely reporting of adverse events for that particular study. This includes timely review of data collected on the electronic CRFs submitted via Medidata Rave.

11.2 Data Reporting

Data collection for this study will be done exclusively through Medidata Rave. Access to the trial in Rave is granted through the iMedidata application to all persons with the appropriate roles assigned in the Regulatory Support System (RSS). To access Rave via iMedidata, the site user must have an active CTEP IAM account (https://ctepcore.nci.nih.gov/iam) and the appropriate Rave role (Rave CRA, Read-Only, CRA, Lab Admin, SLA, or Site Investigator) on either the LPO or Participating Organization roster at the enrolling site. To the hold Rave CRA role or CRA Lab Admin role, the user must hold a minimum of an AP registration type. To hold the Rave Site Investigator role, the individual must be registered as an NPIVR or IVR. Associates can hold read-only roles in Rave.

Upon initial site registration approval for the study in RSS, all persons with Rave roles assigned on the appropriate roster will be sent a study invitation e-mail from iMedidata. To accept the invitation, site users must log into the Select Login (https://login.imedidata.com/selectlogin) using their CTEP-IAM user name and password, and click on the "accept" link in the upper right-corner of the iMedidata page. Please note, site users will not be able to access the study in Rave until all required Medidata and study specific trainings are completed. Trainings will be in the form of electronic learnings (eLearnings), and can be accessed by clicking on the link in the upper right pane of the iMedidata screen.

Users that have not previously activated their iMedidata/Rave account at the time of initial site registration approval for the study in RSS will also receive a separate invitation from iMedidata to activate their account. Account activation instructions are located on the CTSU website, Rave tab under the Rave resource materials (Medidata Account Activation and Study Invitation Acceptance). Additional information on iMedidata/Rave is available on the CTSU members' website under the Rave tab or by contacting the CTSU Help Desk at 1-888-823-5923 or by e-mail at ctsucontact@westat.com.

11.2.1 Method

This study will be monitored by the Clinical Trials Monitoring Service (CTMS). Data will be submitted to CTMS at least once every two weeks via Medidata Rave (or other modality if approved by CTEP). Information on CTMS reporting is available at http://www.theradex.com/clinicalTechnologies/?National-Cancer-Institute-NCI-11. On-site audits will be conducted on an 18-36 month basis as part of routine cancer center site visits. More frequent audits may be conducted if warranted by accrual or due to concerns regarding data quality or timely submission.

For CTMS monitored studies, after users have activated their accounts, please contact the Theradex Help Desk at (609) 799-7580 or by email at CTMSSupport@theradex.com for additional support with Rave and completion of CRFs.

11.2.2 Responsibility for Data Submission

For ETCTN trials, it is the responsibility of the PI(s) at the site to ensure that all investigators at the ETCTN Sites understand the procedures for data submission for each ETCTN protocol and that protocol specified data are submitted accurately and in a timely manner to the CTMS via the electronic data capture system, Medidata Rave.

Data are to be submitted via Medidata Rave to CTMS on a real-time basis, but no less than once every 2 weeks. The timeliness of data submissions and timeliness in resolving data queries will be tracked by CTMS. Metrics for timeliness will be followed and assessed on a quarterly basis. For the purpose of Institutional Performance Monitoring, data will be considered delinquent if it is greater than 4 weeks past due.

Data from Medidata Rave and CTEP-AERS is reviewed by the CTMS on an ongoing basis as data is received. Queries will be issued by CTMS directly within Rave. The queries will appear on the Task Summary Tab within Rave for the CRA at the ETCTN to resolve. Monthly web-based reports are posted for review by the Drug Monitors in the IDB, CTEP. Onsite audits will be conducted by the CTMS to ensure compliance with regulatory requirements, GCP, and NCI policies and procedures with the overarching goal of ensuring the integrity of data generated from NCI-sponsored clinical trials, as described in the ETCTN Program Guidelines, which may be found on the CTEP (http://ctep.cancer.gov/protocolDevelopment/electronic_applications/adverse_events.htm) and CTSU websites.

An End of Study CRF is to be completed by the PI, and is to include a summary of study endpoints not otherwise captured in the database, such as (for phase I trials) the recommended phase 2 dose (RP2D), and a description of any dose-limiting toxicities (DLTs). CTMS will utilize a core set of eCRFs that are Cancer Data Standards Registry and Repository (caDSR) compliant (http://cbiit.nci.nih.gov/ncip/biomedical-informatics-resources/interoperability-and-semantics/metadata-and-models). Customized eCRFs will be included when appropriate to meet unique study requirements. The PI is encouraged to review the eCRFs, working closely with CTMS to ensure prospectively that all required items are appropriately captured in the eCRFs prior to study activation. CTMS will prepare the eCRFs with built-in edit checks to the extent possible to promote data integrity.

CDUS data submissions for ETCTN trials activated after March 1, 2014, will be carried out by the CTMS contractor, Theradex. CDUS submissions are performed by Theradex on a monthly basis. The trial's lead institution is responsible for timely submission to CTMS via Rave, as above.

Further information on data submission procedures can be found in the ETCTN Program Guidelines (http://ctep.cancer.gov/protocolDevelopment/electronic_applications/adverse_events.htm).

11.3 CTEP Multicenter Guidelines-N/A

11.4 Collaborative Agreements Language

The agent(s) supplied by CTEP, DCTD, NCI used in this protocol is/are provided to the NCI under a Collaborative Agreement (CRADA, CTA, CSA) between the Pharmaceutical Company(ies) (hereinafter referred to as "Collaborator(s)") and the NCI Division of Cancer Treatment and Diagnosis. Therefore, the following obligations/guidelines, in addition to the provisions in the "Intellectual Property Option to Collaborator" (http://ctep.cancer.gov/industryCollaborations2/intellectual_property.htm) contained within the terms of award, apply to the use of the Agent(s) in this study:

1. Agent(s) may not be used for any purpose outside the scope of this protocol, nor can Agent(s) be transferred or licensed to any party not participating in the clinical study. Collaborator(s) data for Agent(s) are confidential and proprietary to Collaborator(s) and shall be maintained as such by the investigators. The protocol documents for studies utilizing Agents contain confidential information and should not be shared or distributed without the permission of the NCI. If a copy of this protocol is requested by a patient or patient's family member participating on the study, the individual should sign a confidentiality agreement. A suitable model agreement can be downloaded from: http://ctep.cancer.gov.

2. For a clinical protocol where there is an investigational Agent used in combination with (an)other Agent(s), each the subject of different Collaborative Agreements, the access to and use of data by each Collaborator shall be as follows (data pertaining to such combination use shall hereinafter be referred to as "Multi-Party Data"):

   a. NCI will provide all Collaborators with prior written notice regarding the existence and nature of any agreements governing their collaboration with NCI, the design of the proposed combination protocol, and the existence of any obligations that would tend to restrict NCI's participation in the proposed combination protocol.

b. Each Collaborator shall agree to permit use of the Multi-Party Data from the clinical trial by any other Collaborator solely to the extent necessary to allow said other Collaborator to develop, obtain regulatory approval or commercialize its own Agent.

c. Any Collaborator having the right to use the Multi-Party Data from these trials must agree in writing prior to the commencement of the trials that it will use the Multi-Party Data solely for development, regulatory approval, and commercialization of its own Agent.

3. Clinical Trial Data and Results and Raw Data developed under a Collaborative Agreement will be made available to Collaborator(s), the NCI, and the FDA, as appropriate and unless additional disclosure is required by law or court order as described in the IP Option to Collaborator (http://ctep.cancer.gov/industryCollaborations2/intellectual_property.htm). Additionally, all Clinical Data and Results and Raw Data will be collected, used and disclosed consistent with all applicable federal statutes and regulations for the protection of human subjects, including, if applicable, the *Standards for Privacy of Individually Identifiable Health Information* set forth in 45 C.F.R. Part 164.

4. When a Collaborator wishes to initiate a data request, the request should first be sent to the NCI, who will then notify the appropriate investigators (Group Chair for Cooperative Group studies, or PI for other studies) of Collaborator's wish to contact them.

5. Any data provided to Collaborator(s) for Phase 3 studies must be in accordance with the guidelines and policies of the responsible Data Monitoring Committee (DMC), if there is a DMC for this clinical trial.

6. Any manuscripts reporting the results of this clinical trial must be provided to CTEP by the Group office for Cooperative Group studies or by the principal investigator for non-Cooperative Group studies for immediate delivery to Collaborator(s) for advisory review and comment prior to submission for publication. Collaborator(s) will have 30 days from the date of receipt for review. Collaborator shall have the right to request that publication be delayed for up to an additional 30 days in order to ensure that Collaborator’s confidential and proprietary data, in addition to Collaborator(s)’s intellectual property rights, are protected. Copies of abstracts must be provided to CTEP for forwarding to Collaborator(s) for courtesy review as soon as possible and preferably at least three (3) days prior to submission, but in any case, prior to presentation at the meeting or publication in the proceedings. Press releases and other media presentations must also be forwarded to CTEP prior to release. Copies of any manuscript, abstract and/or press release/ media presentation should be sent to:

Email: ncicteppubs@mail.nih.gov

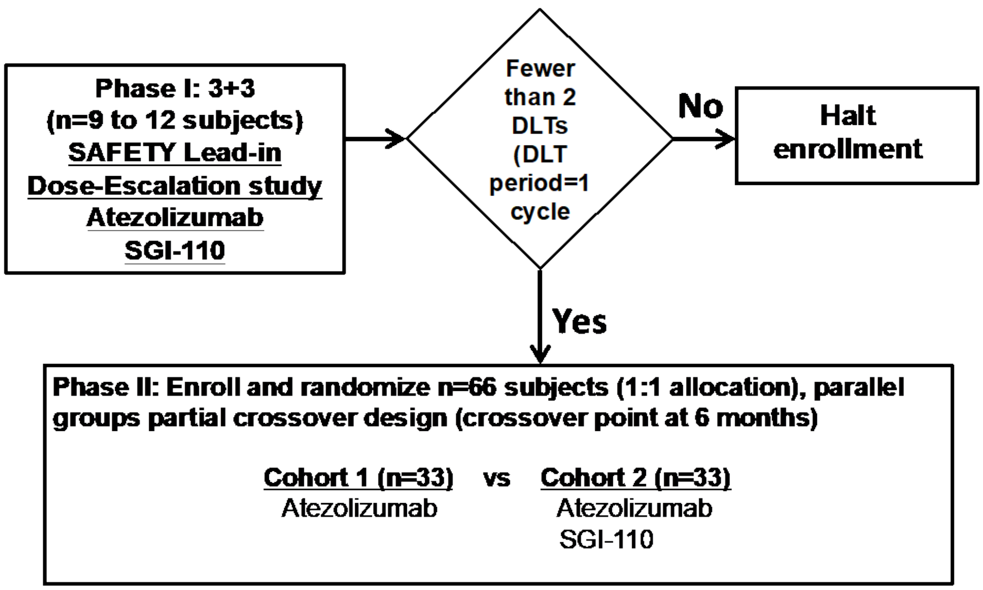
The Regulatory Affairs Branch will then distribute them to Collaborator(s). No publication, manuscript or other form of public disclosure shall contain any of Collaborator’s confidential/ proprietary information.

11.5 Genomic Data Sharing Plan- N/A

12. STATISTICAL CONSIDERATIONS

12.1 Study Design/Endpoints

STUDY SCHEMA



For the Phase I safety lead-in dose escalation component of the study, 3 eligible patients will receive SGI-110 (guadecitabine), on days 1-5 at an initial dose of 30mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles. If 0/3 dose limiting toxicities (DLTs) are observed, proceed to the next dose level of SGI-110 in combination with Atezolizumab (i.e. SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). If 1/3 DLTs are observed enroll 3 more patients at the same dose of SGI-110 (i.e. 30mg/m^2). If 1/6 DLTs are observed, proceed to the next dose level of SGI-110 in combination with Atezpolizumab (i.e. SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). At the next dose level of 45mg/m^2, if 0 or 1 out of 3 patients experiences a DLT, 3 more patients will be added. If 0 or 1 out of the 6 patients experiences a DLT, this will be declared as a safe dose for the Phase II. If $\geq$ 2 DLTs are experienced, the treatment will be considered too toxic and the lower dose-level will be considered the MTD (SGI-110 30mg/m^2). A total of 9-12 subjects will be enrolled into the Phase I component of the study. The DLT monitoring period is defined as 1 cycle of treatment.

Considering the number of subjects who experienced neutropenia on the study at the SGI-110 30mg/m^2 dose level, the FDA advised to de-escalate to SGI-110 20mg/m^2 and study 3 to 6 subjects as an expansion of the Phase 1 portion of the study (Dose Level -1 in the table below). If 1 out of 3 patients experience a DLT, 3 more patients will be added. If 0 out of 3 or 1 out of the 6 patients experience a DLT, this will be declared as a safe dose for the Phase II.

Phase I Dosing

Dose Level	Dose	
Dose Level 1	SGI-110: 30 mg/m^2 subcutaneously Day 1-5 every 28 days	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle.
Dose Level 2	SGI-110: 45 mg/m^2 subcutaneously Day 1-5 every 28 days	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle.
Dose Level -1	SGI-110: 20 mg/m^2 subcutaneously Day 1-5 every 28 days, total 6 cycles	Atezolizumab: 840 mg IV Day 8 and 22 of every 28 day cycle, total 24 cycles.

In patients receiving therapeutic benefit, dose reduction with continued treatment can be considered regardless of degree and type of toxicity at the discretion of the Principal Investigator.

Phase I Study Objectives

The primary endpoint of the Phase I portion of the study is establishing safety. Assessment of safety and tolerability will be performed by the internal data safety-monitoring panel on an ongoing basis, based on data review and regular conference calls with the investigators. All patients who receive any amount of the study drug will be evaluable for toxicity. AEs will be listed individually per patient according to CTCAE version 5.0, and the number of patients experiencing each AE will be summarized using descriptive statistics.

Phase I: The following **decision rules** will be followed: (i) If 0/3 dose limiting toxicities (DLTs) are observed, proceed to the next dose level of SGI-110 in combination with Atezolizumab (i.e. Dose Level 2: SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). (ii) If 1/3 DLTs are observed, enroll 3 more patients at the same dose of SGI-110 (i.e. Dose Level 1: 30mg/m^2). If 1/6 DLTs are observed, proceed to the next dose level of SGI-110 in combination with Atezpolizumab (i.e. Dose Level 2: SGI-110, on days 1-5 at a dose of 45mg/m^2 every 28 days subcutaneously for a total of 6 cycles; and Atezolizumab 840 mg IV on day 8 and 22 of every 28 day cycle for a total of 24 cycles). At the next dose level of 45mg/m^2, if 0 or 1 out of 3 patients experiences a DLT, 3 more patients will be added. If 0 or 1 out of the 6 patients experiences a DLT, this will be declared as a safe dose for the Phase II. If $\geq$ 2 DLTs are experienced, the treatment will be considered too toxic and the lower dose-level will be considered the MTD.

Phase IIb Study Objectives
The phase IIb component of this trial is designed as a randomized partial crossover design (83) with the crossover point following the primary measure of efficacy at 6 months.
The **primary objective** of this study is:
To compare the 6-Month Progression Free Survival (PFS) rate in patients with recurrent ovarian cancer between 2 arms using the RECIST criteria (see section 10.1). Eligible patients will be randomized to one of the following two treatment groups:
Cohort 1: Atezolizumab 840 mg IV every 2 weeks; on days 1 and 15 of each 28 day cycle for a total of 24 cycles.
Cohort 2: SGI-110 subcutaneous on days 1-5 of each cycle for a total of 6 cycles plus Atezolizumab 840 mg IV every 2 weeks, on days 8 and 22, for a total of 24 cycles.
(1 cycle = 28 days).
Subjects randomized to Cohort 1 will be crossed over at the six month time point and will be eligible to receive SGI-110 in addition to Atezolizumab.

Accrual into the randomized Phase IIb study (Cohort 2) will utilize the dose of SGI-110 that has been determined to be safe in the Phase I trial, which is 20 mg/m^2.

Randomized Phase IIb Dosing

Treatment Cohorts	# of Patients	SGI-110	Atezolizumab
Cohort 1	33	None	840 mg IV Day 1 and 15 of each 28 day cycle for a total of 24 cycles.
Cohort 2	33	20 mg/m^2 SC on days 1-5 of each 28 day cycle for a total of 6 cycles.	840 mg IV Day 8 and 22 of each 28 day cycle for a total of 24 cycles.

In patients receiving therapeutic benefit, dose reduction with continued treatment can be considered regardless of degree and type of toxicity at the discretion of the Principal Investigator.

The **secondary objectives** of this study are:
To compare 6 month Overall Survival (OS) rates between Cohort 1 and Cohort 2.

To compare overall PFS and OS between Cohort 1 and Cohort 2 using the time-varying covariate accounting for the crossover portion of the trial at 6 months.
The **correlative objective** of this study is:
We will examine potential differential effect of NY-ESO-1 expression on PFS and OS within the context of the time-varying covariate model.

Study Endpoints
Primary Endpoint:

- PFS-6 is defined as the percentage of evaluable patients that have not progressed 6 months after the date of diagnosis. All patients will be followed for a minimum of 6 months relative to evaluating the primary endpoint.

Secondary Endpoints:

- OS-6 is defined as percentage of evaluable patients that have survived 6 months after the date of diagnosis.

Correlative Endpoints:

- Anti-tumor immune response, the epigenetic modification of immune gene signatures and to assess toxicities associated with the combination of treatments in cohort 2.

Statistical Analysis Plan
The phase IIb component of this protocol is designed as a parallel groups partial crossover design with a 1:1 randomation allocation to Cohort 1 versus Cohort 2(described above). The primary endpoint is 6 month PFS rates. The sample size will be n_1=33 for Cohort 1 and n_2=33 for Cohort 2 for a total sample size of n_1+n_2=66. The primary statistical analysis will be carried forth using Fisher's exact test (one-sided). The hypotheses of interest are H_0: $\pi_{\text{cohort }1}=\pi_{\text{cohort2}}$ versus H_1: $\pi_{\text{cohort }1}<\pi_{\text{cohort2}}$, where π denotes the 6 month PFS rate.
Secondary analyses of time-to-event endpoints will be carried out using a Cox proportional hazards regression model with a time-varying covariate to account for the crossover point at 6 months. Immune response and gene signature differences between cohorts will be examined using Wilcoxon rank-sum tests.
All subjects meeting the eligibility criteria who signed a consent form and have been randomized to receive the Cohort 1 or Cohort 2 treatment assignments will be considered evaluable for efficacy analysis. Safety analysis will be performed on all patients who have signed a consent form and received at least one dose of study medication. Subjects will be followed for survival status after completion or discontinuation of the study drug for progression or discontinuation of the study drug for other reasons.

The primary efficacy analysis of the study will be on the intention to treat (ITT) population, which will include all patients who have signed a consent form and undergone randomization, according to the treatment group to which they were allocated.

All data collected will be summarized and presented. Continuous variables will be described as the mean, median, standard deviation and range of the observations. Categorical data will be described with contingency tables including frequency and percentage. Individual patient listings will be generated and presented.

Statistical descriptions and analyses will be carried out using SAS statistical analysis software version 9.4 or higher (SAS Institute, Inc., Cary, North Carolina, USA).

Randomization
A block randomization scheme will be employed with random block sizes as developed by the study biostatistician and stratified by clinical site.

12.2 Sample Size/Accrual Rate

A total of 9-12 patients are expected to enroll in the Phase I safety lead-in cohort followed by the randomized Phase IIb, an additional n=66 patients, allocated n=33 each for cohort 1 (Atezolizumab alone) and cohort 2 (Atezolizumab + SGI-110).
The sample size considerations were based on the test H_0: $\pi_{cohort\ 1}=\pi_{cohort2}$ versus H_1: $\pi_{cohort\ 1}<\pi_{cohort2}$ at $\alpha=0.10$ under the assumption that under the 6 month PFS rates will be $\pi_{cohort\ 1}=\pi_{cohort2}=0.25$ versus the alternative that the 6 month PFS rates will be $\pi_{cohort\ 1}=0.25$ and $\pi_{cohort2}=0.52$ at 0.80 power.

Study Duration and Compliance
All study drug administration and compliance data will be summarized.

Prior and Concomitant Medication
All relevant prior medication and all concomitant medications will be summarized by frequencies and percentages. All medications will be coded using the World Health Organization (WHO) drug dictionary.

Safety and Toxicity Analyses
Adverse events monitoring and clinical findings including physical examinations, vital signs, laboratory test results, concomitant medications, and withdrawals/terminations will be used to assess safety.

Adverse Events and Serious Adverse Events
AEs will be categorized by SOC and Preferred Terms using the MedDRA dictionary. The incidence of AEs as well as the severity and relationship to study drug will be presented by treatment group. The incidence of AEs leading to withdrawal from the study and serious AEs (SAEs) will be summarized by frequency and percentages.
As per NCI CTCAE Version 5.0, the term toxicity is defined as adverse events that are classified as either possibly, probably, or definitely related to study treatment. The maximum grade for each type of toxicity will be recorded for each patient and will be used for reporting. Frequency tables will be reviewed to determine toxicity patterns. In addition, all adverse event data graded as 3, 4, or 5 and classified as either "unrelated or unlikely to be related" to study treatment in the

event of an actual relationship developing will be reviewed.

Serious adverse events will be similarly summarized. Clinical findings will include evaluation of physical examinations, vital signs and laboratory test results, concomitant medications, and withdrawals/terminations. These findings will be summarized by treatment group.

Procedure for Accounting for Missing, Unused and Spurious Data
Missing data will be indicated in the listings but excluded from all descriptive analyses. All data will be listed, including otherwise unused data. Spurious data will be identified as such, wherever possible.

Demographics and Baseline Characteristics: Descriptive Analyses
Measured outcome variables will be summarized overall and by relevant demographic and baseline variables. Descriptive statistics such as frequencies and relative frequencies will be computed for all categorical variables. Numeric variables will be summarized using simple descriptive statistics such as the mean, standard deviation and range. A variety of graphical techniques will also be used to display data, ex. histograms, boxplots, scatterplots, etc. Descriptive statistics (as appropriate: number, percent, mean, median, min, max) will also be used to summarize by treatment group, demographics and baseline characteristics.

Safety Analysis
Safety and tolerability will be assessed by incidence, severity, and changes from baseline of all relevant parameters including adverse events (AEs), laboratory values and vital signs.
Vital sign results (systolic and diastolic blood pressure, pulse, respiration, and temperature) will be summarized descriptively for each scheduled and unscheduled protocol time point. Changes will be calculated relative to the assessments at baseline.
The changes in hematology, chemistry, and other laboratory values will be summarized descriptively for each scheduled and unscheduled protocol assessment time point. Changes will be calculated relative to the values collected at baseline. Data listings of all laboratory data collected during the study will be presented. Laboratory values outside normal limits will be identified in data listings and will include flags for high and low values.
The frequency of toxicities will be tabulated by grade. All participants who receive any study treatment will be considered evaluable for toxicity.

Adverse Event Analysis
AEs will be coded using the latest available version of the Medical Dictionary for Regulatory Activities (MedDRA). AEs will be summarized for each treatment arm by the number and percentage of patients who experienced the event, according to system organ class and preferred term. Additional summaries will also be provided by severity grade and relationship to study drug, and for SAEs and events resulting in the permanent discontinuation of therapy. A patient reporting multiple cases of the same AE will be counted once within each system organ class and similarly counted once within each preferred term, and adverse events will be graded by worst severity grade. Unless specified otherwise, the denominator for these calculations will be based on the number of patients in each treatment arm who receive at least one dose of study drug,

irrespective of the total number of doses administered. All participants who receive any study treatment will be considered evaluable for toxicity.

Interim Analysis and Criteria for Early Termination of the Study
There are no planned formal interim analyses. Informal interim analyses may be performed if:

1. Safety concerns arise, and
2. At the discretion of the data monitoring committee or the principal investigator

Based on previous studies we anticipate a less than 10% dropout rate. However, our overall sample size will be increased to n=33 per group as a caution against any unforeseen dropout behavior. The primary analysis will be an intention-to-treat analysis. Any subject who drops out of the study prior to a minimal 12 months follow-up will be considered a censored observation in the intent-to-treat context.
With 66 patients at multiple sites, accrual is estimated to take about 2 years but may be subject to an additional 6 months (30 months total) to account for an initial slow start of patient accrual.

PLANNED ENROLLMENT REPORT

Racial Categories	Ethnic Categories				Total (Phase I + Phase II
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	0	0	0	0	**0**
Asian	5	0	0	0	**4+1**
Native Hawaiian or Other Pacific Islander	0	0	0	0	**0**
Black or African American	9	0	2	0	**10+1**
White	52	0	7	0	**52 +7**
More Than One Race	0	0	0	0	**0**
Total	**66**	**0**	**9**	**0**	**66 +9**

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12.3 Reporting and Exclusions

12.3.1 Evaluation of Toxicity

All patients will be evaluable for toxicity from the time of their first treatment with Atezolizumab, and SGI-110.

12.3.2 Evaluation of Response

All patients included in the study must be assessed for response to treatment, even if there are major protocol treatment deviations or if they are ineligible. Each patient will be assigned one of the following categories: 1) complete response, 2) partial response, 3) stable disease, 4) progressive disease, 5) early death from malignant disease, 6) early death from toxicity, 7) early death because of other cause, or 9) unknown (not assessable, insufficient data). [Note: By arbitrary convention, category 9 usually designates the "unknown" status of any type of data in a clinical database.]

All of the patients who met the eligibility criteria (with the possible exception of those who received no study medication) should be included in the main analysis of the response rate. Patients in response categories 4-9 should be considered to have a treatment failure (disease progression). Thus, an incorrect treatment schedule or drug administration does not result in exclusion from the analysis of the response rate. Precise definitions for categories 4-9 will be protocol specific.

All conclusions should be based on all eligible patients. Subanalyses may then be performed on the basis of a subset of patients, excluding those for whom major protocol deviations have been identified (*e.g.*, early death due to other reasons, early discontinuation of treatment, major protocol violations, etc.). However, these subanalyses may not serve as the basis for drawing conclusions concerning treatment efficacy, and the reasons for excluding patients from the analysis should be clearly reported. The 95% confidence intervals will also be provided.

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Note: Appendices in separate files as PART A2, B and C due to file size

14. APPENDICES

14.1 Appendix A Performance Status Criteria

ECOG Performance Status Scale		**Karnofsky Performance Scale**	
Grade	Descriptions	Percent	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.	100	Normal, no complaints, no evidence of disease.
		90	Able to carry on normal activity; minor signs or symptoms of disease.
1	Symptoms, but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (*e.g.*, light housework, office work).	80	Normal activity with effort; some signs or symptoms of disease.
		70	Cares for self, unable to carry on normal activity or to do active work.
2	In bed <50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.	60	Requires occasional assistance, but is able to care for most of his/her needs.
		50	Requires considerable assistance and frequent medical care.
3	In bed >50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.	40	Disabled, requires special care and assistance.
		30	Severely disabled, hospitalization indicated. Death not imminent.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.	20	Very sick, hospitalization indicated. Death not imminent.
		10	Moribund, fatal processes progressing rapidly.
5	Dead.	0	Dead.

14.2 Appendix B Stool Sample Collection Questionnaire

Patient Name:
MRN:
Date:

Stool Sample Collection Questionnaire

1. **What is your race? (one or more categories may be selected)**
   - American Indian or Alaska Native
   - Black or African American
   - White
   - Asian, please specify:__________
   - Native Hawaiian/Other specific Islander:
   - Other Race, please specify:______________

2. **Are you Hispanic or Spanish origin?**
   - No
   - Yes

3. **Were you breastfed as an infant?**
   - No
   - Yes
   - I don't know

4. **Were you delivered via vaginal delivery or C-section?**
   - Vaginal delivery
   - C-section
   - I don't know

5. **Have you had an appendectomy?**
   - No
   - Yes
   - I don't know

6. **Have you ever had any bowel surgery/removal a portion of your bowel or stomach?**
   - No
   - Yes, please specify________________________
   - I don't know

7. Were you ever diagnosed with an autoimmune disease?

- o No _________
- o Yes, please specify______________________
- o I don't know ______

14.3 Appendix C Microbiome Collection Questionnaire

Patient Name:
MRN:
Date:

MICROBIOME COLLECTION QUESTIONNAIRE

1. **Have you had any fevers or infection in the past 2 weeks?**
   - No
   - Yes, please specify:____________________________

2. **Have you taken antibiotics or antifungal agents in the past 4 weeks?**
   - No
   - Yes, please tell us which medication(s):_________________________
     Why were you given antibiotics? ________________________________
     For how long did you take it? ___________________________________

3. **How many times a week do you eat red meat?**
   - I don't eat red meat
   - 1-3 times a week
   - More than 3 times a week

4. **How many times a week you eat cruciferous vegetables such as cabbage, broccoli, kale, and cauliflower?**
   - I don't eat cruciferous vegetables
   - 1-3 times a week
   - More than 3 times a week

5. **Do you eat yogurt?**
   - I don't eat yogurt
   - 1-3 times a week (please specify which kind)______________________________
   - More than 3 times a week (please specify which kind)____________________

6. **Do you take probiotics?**
   - No
   - Yes, please indicate how often and brand name if known:
     ____

7. Are there foods you must avoid?
   - No
   - Yes, please list the food(s):______________________________
   - Foods you avoid for religious, personal, or cultural reasons:______________________
   - Foods your Doctor told you to avoid:___________________________

8. If you are CURRENTLY on a special diet, please indicate below:
   - Not on a special diet
   - Weight loss
   - Weight gain
   - Vegetarian
   - Diet for diabetes
   - Diet for heart disease
   - Diet for kidney disease
   - Other:_____________

9. Are you currently taking prescribed or over-the-counter medications to lose weight or maintain your current weight?
   - No
   - Yes, I am on these weight loss medications:______________________

10. Are you taking laxatives to have a bowel movement on a regular base?
   - No
   - Yes, please describe which kind and how often:______________

11. How would you describe your bowel movements in the past 2 weeks:
   - Regular bowel movements every 1-2 days with no change
   - Frequent issues diarrhea, please describe____________________
   - Frequent issues with constipation, please describe_______________

12. Do you participate in regular physical activity?
   - No
   - Yes, please describe what exercise do you like to do?_______________
     How long? _________________
     How many times a week? _________________

13. Do you smoke?
   - No
   - Yes (please indicate how many cigarettes a day)_____________

Appendix D Stool Collection Instructions

STOOL SAMPLE COLLECTION INSTRUCTIONS

On the morning of your visit (or the day prior, but within 24 hours of your visit), please collect a small stool sample and fill out the Stool Sample Collection Questionnaire.

Before starting:

Please check to make sure that you have received the following:

1. A white stool collection container and toilet ring
2. Two (2) plastic tubes with a scoop attached to each lid.
3. A plastic zip-lock bag with an orange biohazard sign
4. A pair of disposable gloves

If something is missing, or if any of the following instructions are unclear, please call the research nurse.

Getting ready to collect:

- Line the white collection container with toilet paper. It is important that you do NOT use the bag with the orange biohazard sign on it, because that bag cannot be flushed down the toilet.
- Follow the instructions on the container lid for placing the container under the toilet seat. The toilet seat and collection kit should be set up like this:

- Sit on the toilet, relax, and collect a stool sample into the container.

Getting sample into the vials we provide:

1. Put on the gloves.
2. Remove the lid from the first tube provided. Using the scoop that is attached to the lid, place 2 pea-sized scoops into the tube. Please do not collect more than 2 pea-sized scoops.
3. Screw the lid onto the tube, and **make sure it is on properly and tightly**.
4. Repeat steps 2-3 with the other tube provided.

5. Place both tubes into the plastic zip-lock bag with the red biohazard sign on it, and **seal it tightly**.

Disposing of the leftover stool:

Dispose of the contents of the collection container (toilet paper, leftover stool) in the toilet.

Dispose of the white plastic collection container in the regular trash.

Remove the gloves, and throw them in the trash. **Do NOT** flush any part of the white collection container or the gloves down your toilet.

Completing the form:

When you are finished and have washed your hands, complete the "Stool Sample Collection Questionnaire" form (on the following page).

Storing before delivery:

Store the collection bag in a cool place, but not the refrigerator. Bring the tubes in the biohazard bag and the Stool Sample Collection Form with you on the morning of your visit, and give them to the research nurse.

Clinic Use Only
Study ID:
Date:
Collection:

STOOL SAMPLE COLLECTION QUESTIONNAIRE

Did you have any problems or concerns with the stool collection? (Please describe):

____

____

____

1) Date stool sample collected: ______-______-______ **(MM/DD/YYYY)**

2) Time of collection: _______: _____ **(Hr:Min)** ☐**AM** ☐**PM**

PLEASE BRING THIS QUESTIONNAIRE AND THE SAMPLE MATERIALS TO YOUR NEXT VISIT.

14.4 Appendix E Bioassay Templates

14.4.1 **IHC and qRT-PCR SOPs for NY-ESO-1 Expression**

NOTE: Submitted as a separate attachment due to the file size "P10017_O00UnofficalProtocol100416_PartB"

14.5.2 **NY-ESO-1-specific humoral and cellular immune response** (Document included in the initial application as part of the Study Checklist for Biomarker Assays)

NOTE: Submitted as a separate attachment due to the file size "P10017_O00UnofficalProtocol100416_PartC"

14.5 Appendix F Patient Wallet Card

ROSWELL PARK CANCER INSTITUTE

UNDERSTAND PREVENT & CURE CANCER

Research Patient
**Patient is on clinical trial # I 285416 NCI # 10017

Receiving **Atezolizumab, SGI-110, CDX-1401**

Study PI: Kunle Odunsi, MD, PhD

Please call: Mon-Fri 8am-5pm
Clinic (name/#) GYN; RPCI / [redacted]
Treating Dr. ___________________________
After 5pm/weekends/holidays: [redacted]

Roswell Park Cancer Institute
Elm & Carlton Streets
Buffalo, NY 14263
www.roswellpark.org